

**UNITED STATES
SECURITIES AND EXCHANGE COMMISSION**
WASHINGTON, D.C. 20549

FORM 10-Q

**QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE
SECURITIES EXCHANGE ACT OF 1934**

For the quarterly period ended September 30, 2018

Commission file number: 1-3285

3M COMPANY

(Exact name of registrant as specified in its charter)

DELAWARE

(State or other jurisdiction of
incorporation or organization)

41-0417775

(I.R.S. Employer
Identification No.)

3M Center, St. Paul, Minnesota
(Address of principal executive offices)

55144
(Zip Code)

(651) 733-1110

(Registrant's telephone number, including area code)

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.:

Large accelerated filer ☒

Accelerated filer ☐

Non-accelerated filer ☐

Smaller reporting company ☐

Emerging growth company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

Indicate the number of shares outstanding of each of the issuer's classes of common stock, as of the latest practicable date.

Class	Outstanding at September 30, 2018
Common Stock, \$0.01 par value per share	582,287,135 shares

3M COMPANY
Form 10-Q for the Quarterly Period Ended September 30, 2018
TABLE OF CONTENTS

	BEGINNING PAGE
<u>PART I</u>	
<u>FINANCIAL INFORMATION</u>	
<u>ITEM 1.</u>	
<u>Financial Statements</u>	
Index to Financial Statements:	
<u>Consolidated Statement of Income</u>	3
<u>Consolidated Statement of Comprehensive Income</u>	4
<u>Consolidated Balance Sheet</u>	5
<u>Consolidated Statement of Cash Flows</u>	6
<u>Notes to Consolidated Financial Statements</u>	
<u>Note 1. Significant Accounting Policies</u>	7
<u>Note 2. Revenue</u>	13
<u>Note 3. Acquisitions and Divestitures</u>	17
<u>Note 4. Goodwill and Intangible Assets</u>	18
<u>Note 5. Restructuring Actions and Exit Activities</u>	20
<u>Note 6. Supplemental Income Statement Information</u>	21
<u>Note 7. Supplemental Equity and Comprehensive Income Information</u>	21
<u>Note 8. Income Taxes</u>	25
<u>Note 9. Marketable Securities</u>	27
<u>Note 10. Long-Term Debt and Short-Term Borrowings</u>	27
<u>Note 11. Pension and Postretirement Benefit Plans</u>	28
<u>Note 12. Derivatives</u>	29
<u>Note 13. Fair Value Measurements</u>	35
<u>Note 14. Commitments and Contingencies</u>	38
<u>Note 15. Stock-Based Compensation</u>	48
<u>Note 16. Business Segments</u>	51
<u>Report of Independent Registered Public Accounting Firm</u>	53
<u>ITEM 2.</u>	
<u>Management's Discussion and Analysis of Financial Condition and Results of Operations</u>	
Index to Management's Discussion and Analysis:	
<u>Overview</u>	54
<u>Results of Operations</u>	61
<u>Performance by Business Segment</u>	66
<u>Financial Condition and Liquidity</u>	73
<u>Cautionary Note Concerning Factors That May Affect Future Results</u>	78
<u>ITEM 3.</u>	
<u>Quantitative and Qualitative Disclosures About Market Risk</u>	78
<u>ITEM 4.</u>	
<u>Controls and Procedures</u>	79
<u>PART II</u>	
<u>OTHER INFORMATION</u>	
<u>ITEM 1.</u>	
<u>Legal Proceedings</u>	80
<u>ITEM 1A.</u>	
<u>Risk Factors</u>	80
<u>ITEM 2.</u>	
<u>Unregistered Sales of Equity Securities and Use of Proceeds</u>	82
<u>ITEM 3.</u>	
<u>Defaults Upon Senior Securities</u>	83
<u>ITEM 4.</u>	
<u>Mine Safety Disclosures</u>	83
<u>ITEM 5.</u>	
<u>Other Information</u>	83
<u>ITEM 6.</u>	
<u>Exhibits</u>	83

3M COMPANY
FORM 10-Q
For the Quarterly Period Ended September 30, 2018
PART I. Financial Information

Item 1. Financial Statements.

3M Company and Subsidiaries
Consolidated Statement of Income
(Unaudited)

(Millions, except per share amounts)	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Net sales	\$ 8,152	\$ 8,172	\$ 24,820	\$ 23,667
Operating expenses				
Cost of sales	4,159	4,059	12,622	11,961
Selling, general and administrative expenses	1,547	1,637	5,920	4,871
Research, development and related expenses	430	468	1,384	1,422
Gain on sale of businesses	—	—	(530)	(490)
Total operating expenses	6,136	6,164	19,396	17,764
Operating income	2,016	2,008	5,424	5,903
Other expense (income), net	51	11	144	27
Income before income taxes	1,965	1,997	5,280	5,876
Provision for income taxes	419	564	1,266	1,532
Net income including noncontrolling interest	\$ 1,546	\$ 1,433	\$ 4,014	\$ 4,344
Less: Net income attributable to noncontrolling interest	3	4	12	9
Net income attributable to 3M	\$ 1,543	\$ 1,429	\$ 4,002	\$ 4,335
Weighted average 3M common shares outstanding — basic	585.6	597.6	591.1	597.9
Earnings per share attributable to 3M common shareholders — basic	\$ 2.64	\$ 2.39	\$ 6.77	\$ 7.25
Weighted average 3M common shares outstanding — diluted	598.4	612.7	605.1	612.5
Earnings per share attributable to 3M common shareholders — diluted	\$ 2.58	\$ 2.33	\$ 6.61	\$ 7.08
Cash dividends paid per 3M common share	\$ 1.36	\$ 1.175	\$ 4.08	\$ 3.525

The accompanying Notes to Consolidated Financial Statements are an integral part of this statement.

3M Company and Subsidiaries
Consolidated Statement of Comprehensive Income
(Unaudited)

(Millions)	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Net income including noncontrolling interest	\$ 1,546	\$ 1,433	\$ 4,014	\$ 4,344
Other comprehensive income (loss), net of tax:				
Cumulative translation adjustment	(112)	44	(441)	269
Defined benefit pension and postretirement plans adjustment	114	80	344	241
Cash flow hedging instruments, unrealized gain (loss)	46	(49)	147	(176)
Total other comprehensive income (loss), net of tax	48	75	50	334
Comprehensive income (loss) including noncontrolling interest	1,594	1,508	4,064	4,678
Comprehensive (income) loss attributable to noncontrolling interest	—	(3)	(4)	(11)
Comprehensive income (loss) attributable to 3M	\$ 1,594	\$ 1,505	\$ 4,060	\$ 4,667

The accompanying Notes to Consolidated Financial Statements are an integral part of this statement.

3M Company and Subsidiaries
Consolidated Balance Sheet
(Unaudited)

	September 30, 2018	December 31, 2017
(Dollars in millions, except per share amount)		
Assets		
Current assets		
Cash and cash equivalents	\$ 3,185	\$ 3,053
Marketable securities — current	338	1,076
Accounts receivable — net	5,329	4,911
Inventories		
Finished goods	2,125	1,915
Work in process	1,350	1,218
Raw materials and supplies	962	901
Total inventories	4,437	4,034
Prepays	745	937
Other current assets	385	266
Total current assets	14,419	14,277
Property, plant and equipment	24,765	24,914
Less: Accumulated depreciation	(16,135)	(16,048)
Property, plant and equipment — net	8,630	8,866
Goodwill	10,123	10,513
Intangible assets — net	2,726	2,936
Other assets	1,377	1,395
Total assets	\$ 37,275	\$ 37,987
Liabilities		
Current liabilities		
Short-term borrowings and current portion of long-term debt	\$ 1,307	\$ 1,853
Accounts payable	2,029	1,945
Accrued payroll	783	870
Accrued income taxes	186	310
Other current liabilities	3,031	2,709
Total current liabilities	7,336	7,687
Long-term debt	13,539	12,096
Pension and postretirement benefits	3,183	3,620
Other liabilities	2,906	2,962
Total liabilities	\$ 26,964	\$ 26,365
Commitments and contingencies (Note 14)		
Equity		
3M Company shareholders' equity:		
Common stock par value, \$.01 par value; 944,033,056 shares issued	\$ 9	\$ 9
Additional paid-in capital	5,597	5,352
Retained earnings	40,120	39,115
Treasury stock, at cost: 361,745,921 shares at September 30, 2018; 349,148,819 shares at December 31, 2017	(28,510)	(25,887)
Accumulated other comprehensive income (loss)	(6,968)	(7,026)
Total 3M Company shareholders' equity	10,248	11,563
Noncontrolling interest	63	59
Total equity	\$ 10,311	\$ 11,622
Total liabilities and equity	\$ 37,275	\$ 37,987

The accompanying Notes to Consolidated Financial Statements are an integral part of this statement.

3M Company and Subsidiaries
Consolidated Statement of Cash Flows
(Unaudited)

(Millions)	Nine months ended September 30,	
	2018	2017
Cash Flows from Operating Activities		
Net income including noncontrolling interest	\$ 4,014	\$ 4,344
Adjustments to reconcile net income including noncontrolling interest to net cash provided by operating activities		
Depreciation and amortization	1,117	1,195
Company pension and postretirement contributions	(303)	(314)
Company pension and postretirement expense	306	243
Stock-based compensation expense	258	266
Gain on sale of businesses	(530)	(490)
Deferred income taxes	(73)	(105)
Changes in assets and liabilities		
Accounts receivable	(596)	(595)
Inventories	(562)	(436)
Accounts payable	148	(25)
Accrued income taxes (current and long-term)	122	249
Other — net	280	48
Net cash provided by operating activities	4,181	4,380
Cash Flows from Investing Activities		
Purchases of property, plant and equipment (PP&E)	(1,046)	(914)
Proceeds from sale of PP&E and other assets	143	18
Acquisitions, net of cash acquired	13	(12)
Purchases of marketable securities and investments	(1,352)	(1,055)
Proceeds from maturities and sale of marketable securities and investments	2,066	745
Proceeds from sale of businesses, net of cash sold	806	862
Other — net	8	2
Net cash provided by (used in) investing activities	638	(354)
Cash Flows from Financing Activities		
Change in short-term debt — net	(698)	60
Repayment of debt (maturities greater than 90 days)	(456)	(650)
Proceeds from debt (maturities greater than 90 days)	2,247	—
Purchases of treasury stock	(3,601)	(1,564)
Proceeds from issuance of treasury stock pursuant to stock option and benefit plans	401	582
Dividends paid to shareholders	(2,406)	(2,104)
Other — net	(36)	(23)
Net cash used in financing activities	(4,549)	(3,699)
Effect of exchange rate changes on cash and cash equivalents	(138)	106
Net increase (decrease) in cash and cash equivalents	132	433
Cash and cash equivalents at beginning of year	3,053	2,398
Cash and cash equivalents at end of period	\$ 3,185	\$ 2,831

The accompanying Notes to Consolidated Financial Statements are an integral part of this statement.

3M Company and Subsidiaries

Notes to Consolidated Financial Statements

(Unaudited)

NOTE 1. Significant Accounting Policies

Basis of Presentation

The interim consolidated financial statements are unaudited but, in the opinion of management, reflect all adjustments necessary for a fair statement of the Company's consolidated financial position, results of operations and cash flows for the periods presented. These adjustments consist of normal, recurring items. The results of operations for any interim period are not necessarily indicative of results for the full year. The interim consolidated financial statements and notes are presented as permitted by the requirements for Quarterly Reports on Form 10-Q. This Quarterly Report on Form 10-Q should be read in conjunction with the Company's consolidated financial statements and notes included in its Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K).

As described in the "New Accounting Pronouncements" section, the Company adopted Accounting Standards Update (ASU) No. 2017-07, *Improving the Presentation of Net Periodic Pension Cost and Net Periodic Postretirement Benefit Cost*, effective January 1, 2018 on a retrospective basis. This ASU changed how 3M presents net periodic benefit cost within its consolidated statement of income, as reflected in the table that follows. The financial information presented herein reflects these impacts for all periods presented.

Three months ended September 30, 2017 (Millions)	Previously Reported	Revised	Change
Net Sales	\$ 8,172	\$ 8,172	\$ —
Operating expenses			
Cost of sales	4,045	4,059	14
Selling, general and administrative expenses	1,623	1,637	14
Research, development and related expenses	463	468	5
Gain on sale of businesses	—	—	—
Total operating expenses	6,131	6,164	33
Operating income	\$ 2,041	\$ 2,008	\$ (33)
Other expense (income), net	\$ 44	\$ 11	\$ (33)
Income before income taxes	\$ 1,997	\$ 1,997	\$ —
Nine months ended September 30, 2017 (Millions)	Previously Reported	Revised	Change
Net Sales	\$ 23,667	\$ 23,667	\$ —
Operating expenses			
Cost of sales	11,921	11,961	40
Selling, general and administrative expenses	4,830	4,871	41
Research, development and related expenses	1,407	1,422	15
Gain on sale of businesses	(490)	(490)	—
Total operating expenses	17,668	17,764	96
Operating income	\$ 5,999	\$ 5,903	\$ (96)
Other expense (income), net	\$ 123	\$ 27	\$ (96)
Income before income taxes	\$ 5,876	\$ 5,876	\$ —

In addition, as described in Note 16, effective in the first quarter of 2018, the Company changed its business segment reporting in its continuing effort to improve the alignment of businesses around markets and customers. These changes included the consolidation of customer account activity within international countries (expanding dual credit reporting) and the centralization of manufacturing and supply chain technology platforms. Segment information presented herein reflects the impact of these changes for all periods presented.

The Company updated its financial information and disclosures in its Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K) to reflect the retrospective application of ASU No. 2017-07 and the preceding business reporting changes.

Changes to Significant Accounting Policies

The following accounting policies have been updated since the Company's 2017 Annual Report on Form 10-K (which was updated by 3M's Current Report on Form 8-K dated May 8, 2018 in regards to the adoption of ASU 2017-07 and the business segment reporting changes discussed above).

Revenue (sales) recognition: As described in the "New Accounting Pronouncements" section, 3M adopted ASU No. 2014-09, *Revenue from Contracts with Customers*, and other related ASUs on January 1, 2018 using the modified retrospective transition approach. The Company's accounting policy with respect to revenue recognition and additional disclosure relative to this ASU are included in Note 2.

Investments: As described in the "New Accounting Pronouncements" section, 3M adopted ASU No. 2016-01, *Recognition and Measurement of Financial Assets and Financial Liabilities*, effective January 1, 2018. As a result, all equity securities that do not result in consolidation and are not accounted for under the equity method are measured at fair value with changes therein reflected in net income. 3M utilizes the measurement alternative for equity investments that do not have readily determinable fair values and measures these investments at cost less impairment plus or minus observable price changes in orderly transactions. Further, the change in balance of these securities for the three and nine months ended September 30, 2018 was not considered material for additional disclosure.

Foreign Currency Translation

Local currencies generally are considered the functional currencies outside the United States. Assets and liabilities for operations in local-currency environments are translated at month-end exchange rates of the period reported. Income and expense items are translated at month-end exchange rates of each applicable month. Cumulative translation adjustments are recorded as a component of accumulated other comprehensive income (loss) in shareholders' equity.

3M has a subsidiary in Venezuela, the financial statements of which are remeasured as if its functional currency were that of its parent because Venezuela's economic environment is considered highly inflationary. The operating income of this subsidiary is immaterial as a percent of 3M's consolidated operating income for 2017. The Venezuelan government sets official rates of exchange and conditions precedent to purchase foreign currency at these rates with local currency. The government has also operated various expanded secondary currency exchange mechanisms that have been eliminated and replaced from time to time. Such rates and conditions have been and continue to be subject to change. For the periods presented, the financial statements of 3M's Venezuelan subsidiary were remeasured utilizing the rate associated with the secondary auction mechanism, Tipo de Cambio Complementario (DICOM), or its predecessor. During the third quarter of 2018, the Venezuelan government effected a conversion of its currency to the Sovereign Bolivar (VES), essentially equating to its previous Venezuelan Bolivar divided by 100,000.

Note 1 in 3M's Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K) provides additional information the Company considers in determining the exchange rate used relative to its Venezuelan subsidiary as well as factors which could lead to its deconsolidation. The Company continues to monitor these circumstances. Changes in applicable exchange rates or exchange mechanisms may continue in the future. These changes could impact the rate of exchange applicable to remeasure the Company's net monetary assets (liabilities) denominated in VES. As of September 30, 2018, the Company had a balance of net monetary assets denominated in VES of less than 2 million VES and the DICOM exchange rate was approximately 62 VES per U.S. dollar. A need to deconsolidate the Company's Venezuelan subsidiary's operations may result from a lack of exchangeability of VES-denominated cash coupled with an acute degradation in the ability to make key operational decisions due to

government regulations in Venezuela. Based upon a review of factors as of September 30, 2018, the Company continues to consolidate its Venezuelan subsidiary. As of September 30, 2018, the balance of accumulated other comprehensive loss associated with this subsidiary was approximately \$145 million, and the amount of intercompany receivables due from this subsidiary and its equity balance were not significant.

3M has subsidiaries in Argentina, the operating income of which is less than one half of one percent of 3M's consolidated operating income for 2017. Based on various indices, Argentina's cumulative three-year inflation rate exceeded 100 percent in the second quarter of 2018, thus being considered highly inflationary. As a result, beginning in the third quarter of 2018, the financial statements of the Argentine subsidiaries were remeasured as if their functional currency were that of their parent. As of September 30, 2018, the Company had a balance of net monetary assets denominated in Argentine pesos (ARS) of approximately 300 million ARS and the exchange rate was approximately 41 ARS per U.S. dollar.

Earnings Per Share

The difference in the weighted average 3M shares outstanding for calculating basic and diluted earnings per share attributable to 3M common shareholders is a result of the dilution associated with the Company's stock-based compensation plans. Certain options outstanding under these stock-based compensation plans were not included in the computation of diluted earnings per share attributable to 3M common shareholders because they would not have had a dilutive effect (3.2 million average options for the three months ended September 30, 2018; 2.8 million average options for the nine months ended September 30, 2018; insignificant for the three months ended September 30, 2017, 1.1 million average options for the nine months ended September 30, 2017). The computations for basic and diluted earnings per share follow:

Earnings Per Share Computations

(Amounts in millions, except per share amounts)	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Numerator:				
Net income attributable to 3M	\$ 1,543	\$ 1,429	\$ 4,002	\$ 4,335
Denominator:				
Denominator for weighted average 3M common shares outstanding – basic	585.6	597.6	591.1	597.9
Dilution associated with the Company's stock-based compensation plans	12.8	15.1	14.0	14.6
Denominator for weighted average 3M common shares outstanding – diluted	598.4	612.7	605.1	612.5
Earnings per share attributable to 3M common shareholders – basic	\$ 2.64	\$ 2.39	\$ 6.77	\$ 7.25
Earnings per share attributable to 3M common shareholders – diluted	\$ 2.58	\$ 2.33	\$ 6.61	\$ 7.08

New Accounting Pronouncements

See the Company's Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K) for a more detailed discussion of the standards in the tables that follow, except for those pronouncements issued subsequent to the most recent Form 10-K filing date for which separate, more detailed discussion is provided below.

Standards Adopted During the Current Fiscal Year			
Standard	Relevant Description	Effective Date for 3M	Impact and Other Matters
ASU No. 2014-09, <i>Revenue from Contracts with Customers</i> (as amended by ASU Nos. 2015-14, 2016-08, 2016-10, 2016-12, and 2016-20) and related ASU No. 2017-10, <i>Determining the Customer of the Operation Services</i>	- Provides a single comprehensive model to be used in the accounting for revenue arising from contracts with customers and supersedes most previous revenue recognition guidance, including industry-specific guidance. - Core principle is that an entity should recognize revenue to depict the transfer of promised goods or services to customers in an amount that reflects the consideration to which the entity expects to be entitled in exchange for those goods or services. - Requires disclosures about the nature, amount, timing, and uncertainty of revenue and cash flows arising from contracts with customers. - Specifies the accounting for some costs to obtain or fulfill a contract with a customer.	January 1, 2018	- See Note 2 for detailed discussion and disclosures. - Adopted using a modified retrospective approach. January 1, 2018 balance of retained earnings was increased by less than \$2 million.
ASU No. 2016-01, <i>Recognition and Measurement of Financial Assets and Financial Liabilities</i>	- Requires investments in equity securities in an entity that do not result in consolidation and are not accounted for under the equity method to be measured at fair value with changes therein reflected in net income. - Simplifies the impairment assessment and allows for a fair value measurement alternative for equity investments without a readily determinable fair value. - Eliminates the previous cost method of accounting for certain equity securities that did not have readily determinable fair values.	January 1, 2018	- Measurement alternative adopted prospectively. - See the preceding "Changes to Significant Accounting Policies" section for impact.
ASU No. 2016-16, <i>Intra-Entity Transfers of Assets Other Than Inventory</i>	- Exempts income tax accounting that requires companies to defer the income tax effects of certain intercompany transactions only for intercompany inventory transactions. - The exception no longer applies to intercompany sales and transfers of other assets (e.g., intangible assets).	January 1, 2018	Adopted using a modified retrospective approach. January 1, 2018 balance of retained earnings was decreased by less than \$2 million.
ASU No. 2017-01, <i>Clarifying the Definition of a Business</i>	Narrows the previous definition of a business and provides a framework for evaluating whether a transaction should be accounted for as an acquisition (or disposal) of assets or a business.	January 1, 2018	- Adopted prospectively with no immediate impact. - Fewer sets of transferred assets and activities are expected to be considered businesses.
ASU No. 2017-05, <i>Clarifying the Scope of Asset Derecognition Guidance and Accounting for Partial Sales of Nonfinancial Assets</i>	- Largely impacts the sale of nonfinancial assets (such as real estate and intellectual property) that do not constitute a business, when the purchaser is not a customer. - Seller applies certain recognition and measurement principles of ASU No. 2014-09, <i>Revenue from Contracts with Customers</i>, even though the purchaser is not a customer.	January 1, 2018	Adopted coincident with the adoption of ASU No. 2014-09 with no material impact.

Standards Adopted During the Current Fiscal Year (continued)			
Standard	Relevant Description	Effective Date for 3M	Impact and Other Matters
ASU No. 2017-07, <i>Improving the Presentation of Net Periodic Pension Cost and Net Periodic Postretirement Benefit Cost</i>	- Changes previous classification of net periodic defined benefit pension and postretirement benefit costs within operating expenses. - Requires that only the service cost component of net periodic benefit cost be included in operating expenses and that only the service cost component is eligible for capitalization into assets such as inventory. - Specifies that other net periodic benefit costs components (such as interest, expected return on plan assets, prior service cost amortization and actuarial gain/loss amortization) would be reported outside of operating income.	January 1, 2018	- Adopted on a retrospective basis. - No impact on previously reported income before income taxes and net income attributable to 3M. However, non-service cost components of net periodic benefit costs in prior periods have been reclassified from operating expenses and are now reported outside of operating income within other expense (income), net. - See the “Basis of Presentation” section above for impact of this ASU’s adoption on prior period income statement amounts. - Prospective impact on costs capitalized into assets was not material.
ASU No. 2017-09, <i>Scope of Modification Accounting</i>	Provides that fewer changes to the terms of share-based payment awards will require accounting under the modification model (which generally would have required additional compensation cost).	January 1, 2018	- Adopted prospectively with no immediate impact. 3M does not typically make changes to the terms or conditions of its issued share-based payments.

Standards Issued and Not Yet Adopted			
Standard	Relevant Description	Effective Date for 3M	Impact and Other Matters
ASU No. 2016-02, <i>Leases</i> (as amended by ASU Nos. 2018-10 and 2018-11)	Introduces a lessee model that requires entities to recognize assets and liabilities for most leases, but recognize expenses on their income statements in a manner similar to current accounting. This ASU does not make fundamental changes to existing lessor accounting.	January 1, 2019	- As amended, provides for retrospective transition applied to earliest period presented or an adoption method by which entities would not need to recast the comparative periods presented. 3M does not plan on recasting prior periods as it adopts this ASU. 3M continues to evaluate the ASU’s impact on its consolidated results of operations and financial condition, has conducted initial analysis, executed project management relative to the process of adopting this ASU including implementing a new lease accounting system, conducted detailed contract reviews, and is finalizing its assessment of adoption impacts. - Note 15 in 3M’s Current Report on Form 8-K dated May 8, 2018 (which updated 3M’s 2017 Annual Report on Form 10-K) provides information regarding rent expense for operating leases and minimum lease payments for capital and operating leases under existing lease guidance. - See the “Relevant New Standards Issued Subsequent to Most Recent Annual Report” below for further discussion on ASU Nos. 2018-10 and 2018-11 issued in July 2018.

Standards Issued and Not Yet Adopted (continued)			
Standard	Relevant Description	Effective Date for 3M	Impact and Other Matters
ASU No. 2016-13, <i>Measurement of Credit Losses on Financial Instruments</i>	- Introduces an approach, based on expected losses, to estimate credit losses on certain types of financial instruments and modifies the impairment model for available-for-sale debt securities. - Amends the current other-than-temporary impairment model for available-for-sale debt securities. For such securities with unrealized losses, entities will still consider if a portion of any impairment is related only to credit losses and therefore recognized as a reduction in income.	January 1, 2020	- Required to make a cumulative-effect adjustment to retained earnings as of the beginning of the first reporting period in which the guidance is adopted. 3M is currently assessing this ASU's impact.
ASU No. 2017-08, <i>Premium Amortization on Purchased Callable Debt Securities</i>	Shortens the amortization period to the earliest call date for the premium related to certain callable debt securities that have explicit, noncontingent call features and are callable at a fixed price and preset date.	January 1, 2019	3M's marketable security portfolio includes limited instances of callable debt securities held at a premium. 3M does not expect this ASU to have a material impact.
ASU No. 2017-11, (Part I) <i>Accounting for Certain Financial Instruments with Down Round Features</i> , (Part II) <i>Replacement of the Indefinite Deferral for Mandatorily Redeemable Financial Instruments of Certain Nonpublic Entities and Certain Mandatorily Redeemable Noncontrolling Interests with a Scope Exception</i>	Amends (1) the classification of financial instruments with down-round features as liabilities or equity by revising certain guidance relative to evaluating if they must be accounted for as derivative instruments and (2) the guidance on recognition and measurement of freestanding equity-classified instruments.	January 1, 2019	- No financial instruments with down-round features have been issued. 3M does not expect this ASU to have a material impact.
ASU No. 2017-12, <i>Targeted Improvements to Accounting for Hedging Activities</i>	- Amends existing guidance to simplify application of hedge accounting in certain situations and allow companies to better align their hedge accounting with risk management activities. - Simplifies related accounting by eliminating requirement to separately measure and report hedge ineffectiveness. - Expands an entity's ability to hedge nonfinancial and financial risk components.	January 1, 2019	- Required to make a cumulative-effect adjustment to retained earnings as of the beginning of the first reporting period in which the guidance is adopted. 3M does not expect this ASU to have a material impact.

Relevant New Standards Issued Subsequent to Most Recent Annual Report

In February 2018, the FASB issued ASU No. 2018-02, *Reclassification of Certain Tax Effects from Accumulated Other Comprehensive Income*, which permits entities to reclassify, to retained earnings, the one-time income tax effects stranded in accumulated other comprehensive income (AOCI) arising from the change in the U.S. federal corporate tax rate as a result of the Tax Cuts and Jobs Act of 2017. An entity that elects to make this reclassification must consider all items in AOCI that have tax effects stranded as a result of the tax rate change, and must disclose the reclassification of these tax effects as well as the entity's policy for releasing income tax effects from AOCI. The ASU may be applied either retrospectively or as of the beginning of the period of adoption. For 3M, the ASU is effective January 1, 2019. While this ASU will have no impact on 3M's results of operations, the Company is currently assessing this standard's impact on its consolidated financial condition.

In June 2018, the FASB issued ASU No. 2018-07, *Improvements to Nonemployee Share-Based Payment Accounting*, which largely aligns the measurement and classification guidance for share-based payments to nonemployees with the guidance for share-based payments to employees. The ASU also clarifies that any share-based payment issued to a customer should be evaluated under ASC 606, *Revenue from Contracts with Customers*. The ASU requires a modified retrospective transition approach. For 3M, the ASU is effective as of January 1, 2019. Because the Company does not grant share-based payments to nonemployees or customers, this ASU will not have a material impact on its consolidated results of operations and financial condition.

In June 2018, the FASB issued ASU No. 2018-08, *Clarifying the Scope and the Accounting Guidance for Contributions Received and Contributions Made*. The ASU applies to entities that receive or make contributions, which primarily are not-for-profit entities but also affects business entities that make contributions. In the context of business entities that make contributions, the FASB clarified

that a contribution is conditional if the arrangement includes both a barrier for the recipient to be entitled to the assets transferred and a right of return for the assets transferred (or a right of release of the business entity's obligation to transfer assets). The recognition of contribution expense is deferred for conditional arrangements and is immediate for unconditional arrangements. The ASU requires modified prospective transition to arrangements that have not been completed as of the effective date or that are entered into after the effective date, but full retrospective application to each period presented is permitted. For 3M, the ASU is effective as of January 1, 2019. The Company does not expect this ASU to have a material impact on its consolidated results of operations and financial condition.

In July 2018, the FASB issued ASU No. 2018-10, *Codification Improvements to Topic 842, Leases*, which amends ASU No. 2016-02, *Leases*. The new ASU includes certain clarifications to address potential narrow-scope implementation issues which the Company is incorporating into its assessment and adoption of ASU No. 2016-02. This ASU has the same transition requirements and effective date as ASU No. 2016-02, which for 3M is January 1, 2019.

In July 2018, the FASB issued ASU No. 2018-11, *Leases (Topic 842): Targeted Improvements*, which amends ASU No. 2016-02, *Leases*. The new ASU offers an additional transition method by which entities may elect not to recast the comparative periods presented in financial statements in the period of adoption and allows lessors to elect a practical expedient to not separate lease and nonlease components when certain conditions are met. This ASU has the same transition requirements and effective date as ASU No. 2016-02, which for 3M is January 1, 2019.

In August 2018, the FASB issued ASU No. 2018-13, *Changes to the Disclosure Requirements for Fair Value Measurement*, which eliminates, amends, and adds disclosure requirements for fair value measurements. The amended and new disclosure requirements primarily relate to Level 3 fair value measurements. For 3M, the ASU is effective as of January 1, 2020. The removal and amendment of certain disclosures may be early adopted with retrospective application while the new disclosure requirements are to be applied prospectively. As this ASU relates only to disclosures, there will be no impact to the Company's consolidated results of operations and financial condition.

In August 2018, the FASB issued ASU No. 2018-14, *Changes to the Disclosure Requirements for Defined Benefit Plans*, which makes minor changes to the disclosure requirements related to defined benefit pension and other postretirement plans. The ASU requires a retrospective transition approach. For 3M, the ASU is effective as of January 1, 2021. As this ASU relates only to disclosures, there will be no impact to the Company's consolidated results of operations and financial condition.

In August 2018, the FASB issued ASU No. 2018-15, *Customer's Accounting for Implementation Costs Incurred in a Cloud Computing Arrangement That Is a Service Contract*, which aligns the accounting for implementation costs incurred in a cloud computing arrangement that is a service arrangement (i.e. hosting arrangement) with the guidance on capitalizing costs in ASC 350-40, *Internal-Use Software*. The ASU permits either a prospective or retrospective transition approach. For 3M, the ASU is effective as of January 1, 2020. The Company is currently assessing this standard's impact on its consolidated results of operations and financial condition.

NOTE 2. Revenue

The Company adopted ASU No. 2014-09 and related standards (collectively, Accounting Standards Codification (ASC) 606, *Revenue from Contracts with Customers*), as described in Note 1, on January 1, 2018 using the modified retrospective method of adoption. Prior periods have not been restated. Due to the cumulative net impact of adopting ASC 606, the January 1, 2018 balance of retained earnings was increased by less than \$2 million, primarily relating to the accelerated recognition for software installation service and training revenue. This cumulative impact reflects retrospective application of ASC 606 only to contracts that were not completed as of January 1, 2018. Further, the Company applied the practical expedient permitting the effect of all contract modifications that occurred before January 1, 2018 to be aggregated in the transition accounting. The impact of applying ASC 606 as compared with previous guidance applied to revenues and costs was not material for the three and nine months ended September 30, 2018.

Performance Obligations:

The Company sells a wide range of products to a diversified base of customers around the world and has no material concentration of credit risk or significant payment terms extended to customers. The vast majority of 3M's customer arrangements contain a single performance obligation to transfer manufactured goods as the promise to transfer the individual goods is not separately identifiable from other promises in the contracts and, therefore, not distinct. However, to a limited extent 3M also enters into customer

arrangements that involve intellectual property out-licensing, multiple performance obligations (such as equipment, installation and service), software with coterminous post-contract support, services and non-standard terms and conditions.

Revenue is recognized when control of goods has transferred to customers. For the majority of the Company's customer arrangements, control transfers to customers at a point-in-time when goods/services have been delivered as that is generally when legal title, physical possession and risks and rewards of goods/services transfers to the customer. In limited arrangements, control transfers over time as the customer simultaneously receives and consumes the benefits as 3M completes the performance obligation(s).

Revenue is recognized at the transaction price which the Company expects to be entitled. When determining the transaction price, 3M estimates variable consideration applying the portfolio approach practical expedient under ASC 606. The main sources of variable consideration for 3M are customer rebates, trade promotion funds, and cash discounts. These sales incentives are recorded as a reduction to revenue at the time of the initial sale using the most-likely amount estimation method. The most-likely amount method is based on the single most likely outcome from a range of possible consideration outcomes. The range of possible consideration outcomes are primarily derived from the following inputs: sales terms, historical experience, trend analysis, and projected market conditions in the various markets served. Because 3M serves numerous markets, the sales incentive programs offered vary across businesses, but the most common incentive relates to amounts paid or credited to customers for achieving defined volume levels or growth objectives. There are no material instances where variable consideration is constrained and not recorded at the initial time of sale. Free goods are accounted for as an expense and recorded in cost of sales. Product returns are recorded as a reduction to revenue based on anticipated sales returns that occur in the normal course of business. 3M primarily has assurance-type warranties that do not result in separate performance obligations. Sales, use, value-added, and other excise taxes are not recognized in revenue. The Company has elected to present revenue net of sales taxes and other similar taxes.

For substantially all arrangements recognized over time, the Company applies the "right to invoice" practical expedient. As a result, 3M recognizes revenue at the invoice amount when the entity has a right to invoice a customer at an amount that corresponds directly with the value to the customer of the Company's performance completed to date.

For contracts with multiple performance obligations, the Company allocates the contract's transaction price to each performance obligation using 3M's best estimate of the standalone selling price of each distinct good or service in the contract.

The Company did not recognize any material revenue in the current reporting period for performance obligations that were fully satisfied in previous periods.

Contract Balances:

Deferred revenue (current portion) as of September 30, 2018 and December 31, 2017 was \$614 million and \$513 million, respectively, and primarily relates to revenue that is recognized over time for one-year software license contracts, the changes in balance of which are related to the satisfaction or partial satisfaction of these contracts. The balance also contains a deferral for goods that are in-transit at period end for which control transfers to the customer upon delivery. Approximately \$70 million and \$460 million of the December 31, 2017 balance was recognized as revenue during the three and nine months ended September 30, 2018, respectively. The amount of noncurrent deferred revenue is not considered significant.

Exemptions and Practical Expedients Applied or Elected:

3M applies ASC 606 utilizing the following allowable exemptions or practical expedients:

- Exemption to not disclose the unfulfilled performance obligation balance for contracts with an original length of one year or less.
- Practical expedient relative to costs of obtaining a contract by expensing sales commissions when incurred because the amortization period would have been one year or less.
- Portfolio approach practical expedient relative to estimation of variable consideration.
- "Right to invoice" practical expedient based on 3M's right to invoice the customer at an amount that reasonably represents the value to the customer of 3M's performance completed to date.
- Election to present revenue net of sales taxes and other similar taxes.
- Sales-based royalty exemption permitting future intellectual property out-licensing royalty payments to be excluded from the otherwise required remaining performance obligations disclosure.

Disaggregated revenue information:

The Company views the following disaggregated disclosures as useful to understanding the composition of revenue recognized during the respective reporting periods:

Net Sales (Millions)	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Abrasives	\$ 427	\$ 441	\$ 1,375	\$ 1,308
Adhesives and Tapes	1,164	1,170	3,482	3,341
Advanced Materials	314	281	935	855
Automotive and Aerospace	499	490	1,577	1,487
Automotive Aftermarket	397	414	1,253	1,249
Separation and Purification	224	228	698	667
Other Industrial	(2)	(1)	(5)	(2)
Total Industrial Business Group	\$ 3,023	\$ 3,023	\$ 9,315	\$ 8,905
Commercial Solutions	\$ 439	\$ 444	\$ 1,426	\$ 1,344
Personal Safety	881	732	2,800	2,160
Roofing Granules	83	98	283	291
Transportation Safety	259	279	752	877
Other Safety and Graphics	(2)	(2)	(3)	(2)
Total Safety and Graphics Business Group	\$ 1,660	\$ 1,551	\$ 5,258	\$ 4,670
Drug Delivery	\$ 102	\$ 137	\$ 340	\$ 378
Food Safety	83	78	249	227
Health Information Systems	208	197	618	583
Medical Solutions	734	756	2,282	2,199
Oral Care	317	318	1,013	984
Other Health Care	1	(1)	(1)	(2)
Total Health Care Business Group	\$ 1,445	\$ 1,485	\$ 4,501	\$ 4,369
Electronics	\$ 1,103	\$ 1,095	\$ 2,959	\$ 2,856
Energy	339	419	1,170	1,236
Other Electronics and Energy	1	1	1	4
Total Electronics and Energy Business Group	\$ 1,443	\$ 1,515	\$ 4,130	\$ 4,096
Consumer Health Care	\$ 96	\$ 111	\$ 298	\$ 317
Home Care	249	263	773	777
Home Improvement	518	517	1,472	1,385
Stationery and Office	362	376	1,012	1,008
Other Consumer	10	12	30	34
Total Consumer Business Group	\$ 1,235	\$ 1,279	\$ 3,585	\$ 3,521
Corporate and Unallocated	\$ 35	\$ 3	\$ 47	\$ 6
Elimination of Dual Credit	(689)	(684)	(2,016)	(1,900)
Total Company	\$ 8,152	\$ 8,172	\$ 24,820	\$ 23,667

Three months ended September 30, 2018

Net Sales (Millions)	United States	Asia Pacific	Europe, Middle East and Africa	Latin America and Canada	Other Unallocated	Worldwide
Industrial	\$ 1,137	\$ 898	\$ 683	\$ 308	\$ (3)	\$ 3,023
Safety and Graphics	678	411	381	191	(1)	1,660
Health Care	687	282	340	137	(1)	1,445
Electronics and Energy	212	1,079	95	57	—	1,443
Consumer	779	221	129	107	(1)	1,235
Corporate and Unallocated	34	(1)	(2)	—	4	35
Elimination of Dual Credit	(262)	(269)	(99)	(59)	—	(689)
Total Company	\$ 3,265	\$ 2,621	\$ 1,527	\$ 741	\$ (2)	\$ 8,152

Nine months ended September 30, 2018

Net Sales (Millions)	United States	Asia Pacific	Europe, Middle East and Africa	Latin America and Canada	Other Unallocated	Worldwide
Industrial	\$ 3,419	\$ 2,712	\$ 2,245	\$ 942	\$ (3)	\$ 9,315
Safety and Graphics	2,077	1,327	1,260	595	(1)	5,258
Health Care	2,092	873	1,115	423	(2)	4,501
Electronics and Energy	685	2,889	371	186	(1)	4,130
Consumer	2,131	729	408	318	(1)	3,585
Corporate and Unallocated	42	(1)	(1)	3	4	47
Elimination of Dual Credit	(789)	(728)	(321)	(178)	—	(2,016)
Total Company	\$ 9,657	\$ 7,801	\$ 5,077	\$ 2,289	\$ (4)	\$ 24,820

Three months ended September 30, 2017

Net Sales (Millions)	United States	Asia Pacific	Europe, Middle East and Africa	Latin America and Canada	Other Unallocated	Worldwide
Industrial	\$ 1,111	\$ 883	\$ 700	\$ 331	\$ (2)	\$ 3,023
Safety and Graphics	612	393	352	195	(1)	1,551
Health Care	730	263	354	140	(2)	1,485
Electronics and Energy	233	1,069	142	71	—	1,515
Consumer	785	242	139	112	1	1,279
Corporate and Unallocated	3	(1)	—	1	—	3
Elimination of Dual Credit	(251)	(269)	(98)	(66)	—	(684)
Total Company	\$ 3,223	\$ 2,580	\$ 1,589	\$ 784	\$ (4)	\$ 8,172

Nine months ended September 30, 2017

Net Sales (Millions)	United States	Asia Pacific	Europe, Middle East and Africa	Latin America and Canada	Other Unallocated	Worldwide
Industrial	\$ 3,307	\$ 2,546	\$ 2,105	\$ 950	\$ (3)	\$ 8,905
Safety and Graphics	1,832	1,194	1,067	578	(1)	4,670
Health Care	2,117	784	1,060	410	(2)	4,369
Electronics and Energy	697	2,775	421	203	—	4,096
Consumer	2,066	732	404	319	—	3,521
Corporate and Unallocated	6	—	—	1	(1)	6
Elimination of Dual Credit	(734)	(696)	(286)	(184)	—	(1,900)
Total Company	\$ 9,291	\$ 7,335	\$ 4,771	\$ 2,277	\$ (7)	\$ 23,667

NOTE 3. Acquisitions and Divestitures

Acquisitions:

3M makes acquisitions of certain businesses from time to time that are aligned with its strategic intent with respect to, among other factors, growth markets and adjacent product lines or technologies. Goodwill resulting from business combinations is largely attributable to the existing workforce of the acquired businesses and synergies expected to arise after 3M's acquisition of these businesses.

There were no business combinations that closed during the nine months ended September 30, 2018.

As discussed in 3M's Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K), in October 2017, 3M completed the acquisition of Scott Safety for \$2.0 billion of cash, net of cash acquired. Adjustments in 2018 to the purchase price allocation were approximately \$7 million and related to identification of certain immaterial acquired assets, tax-related and contingent liabilities, and resolution of certain acquired working capital and other purchase price adjustments with the seller. The change to provisional amounts did not result in material impacts to results of operations in 2018 or any portion related to earlier quarters in the measurement period. The allocation of purchase consideration related to Scott Safety was completed in the third quarter of 2018.

Divestitures:

3M may divest certain businesses from time to time based upon reviews of the Company's portfolio considering, among other items, factors relative to the extent of strategic and technological alignment and optimization of capital deployment, in addition to considering if selling the businesses results in the greatest value creation for the Company and its shareholders.

2018 divestitures:

In February 2018, 3M closed on the sale of certain personal safety product offerings primarily focused on noise, environmental, and heat stress monitoring to TSI, Inc. This business has annual sales of approximately \$15 million. The transaction resulted in a pre-tax gain of less than \$20 million that was reported within the Company's Safety and Graphics business.

In addition, during the first quarter of 2018, 3M divested a polymer additives compounding business, formerly part of the Company's Industrial business, and reflected a gain on final closing adjustments from a prior divestiture which, in aggregate, were not material.

In May 2018, 3M divested an abrasives glass products business, formerly part of the Company's Industrial business, with annual sales of approximately \$10 million. The transaction resulted in a pre-tax gain of less than \$15 million.

In June 2018, 3M completed the sale of substantially all of its Communication Markets Division to Corning Incorporated. This business, with annual sales of approximately \$400 million, consists of optical fiber and copper passive connectivity solutions for the telecommunications industry including 3M's xDSL, FTTx, and structured cabling solutions and, in certain countries, telecommunications system integration services. 3M received cash proceeds of \$772 million and reflected a pre-tax gain of \$494 million as a result of this divestiture, which was reported within the Company's Electronics and Energy business. The sale of the telecommunications system integration services portion of the business based in Germany for approximately \$30 million remains pending and is expected to be completed by the end of 2018.

2017 divestitures:

During 2017, as described in Note 2 in 3M's Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K), the Company divested of a number of business including its: safety prescription eyewear; identity management; tolling and automated license/number plate recognition; electronic monitoring; and electrical marking/labeling businesses.

The aggregate operating income of these businesses was approximately \$25 million and \$20 million in the first nine months of 2018 and 2017, respectively. The approximate amounts of major assets and liabilities associated with disposal groups classified as held-for-sale as of September 30, 2018 were not material and as of December 31, 2017 included the following:

(Millions)	December 31, 2017
Accounts receivable	\$ 25
Property, plant and equipment (net)	20

In addition, an immaterial amount and approximately \$275 million of goodwill was estimated to be attributable to disposal groups classified as held-for-sale as of September 30, 2018 and December 31, 2017, respectively, based upon relative fair value. The amounts above have not been segregated and are classified within the existing corresponding line items on the Company's consolidated balance sheet.

Refer to Note 2 in 3M's Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K) for more information on 3M's acquisitions and divestitures.

NOTE 4. Goodwill and Intangible Assets

There were no acquisitions that closed during the first nine months of 2018. The acquisition activity in the following table relates to the net impact of adjustments to the preliminary allocation of purchase price within the one year measurement period following prior acquisitions, which increased goodwill by \$7 million during the nine months ended September 30, 2018. Divestiture activity within the Electronics and Energy business segment relates to the sale of substantially all of the Communication Markets Division. The amounts in the "Translation and other" column in the following table primarily relate to changes in foreign currency exchange rates. The goodwill balances by business segment as of December 31, 2017 and September 30, 2018, follow:

Goodwill

(Millions)	December 31, 2017 Balance	Acquisition activity	Divestiture activity	Translation and other	September 30, 2018 Balance
Industrial	\$ 2,678	\$ —	\$ (4)	\$ (50)	\$ 2,624
Safety and Graphics	4,419	7	(8)	(53)	4,365
Health Care	1,682	—	—	(20)	1,662
Electronics and Energy	1,524	—	(247)	(11)	1,266
Consumer	210	—	—	(4)	206
Total Company	\$ 10,513	\$ 7	\$ (259)	\$ (138)	\$ 10,123

Accounting standards require that goodwill be tested for impairment annually and between annual tests in certain circumstances such as a change in reporting units or the testing of recoverability of a significant asset group within a reporting unit. At 3M, reporting units generally correspond to a division.

As described in Note 16, effective in the first quarter of 2018, the Company changed its business segment reporting in its continuing effort to improve the alignment of its businesses around markets and customers. In addition, certain shared film manufacturing and supply technology platform resources formerly reflected within the Electronics and Energy business segment were combined with other shared and centrally managed material resource centers of expertise within Corporate and Unallocated. For any product changes that resulted in reporting unit changes, the Company applied the relative fair value method to determine the impact on goodwill of the associated reporting units. During the first quarter of 2018, the Company completed its assessment of any potential goodwill impairment for reporting units impacted by this new structure and determined that no impairment existed.

Acquired Intangible Assets

The carrying amount and accumulated amortization of acquired finite-lived intangible assets, in addition to the balance of non-amortizable intangible assets, as of September 30, 2018, and December 31, 2017, follow:

(Millions)	September 30, 2018	December 31, 2017
Customer related intangible assets	\$ 2,301	\$ 2,332
Patents	544	561
Other technology-based intangible assets	578	583
Definite-lived tradenames	664	678
Other amortizable intangible assets	125	207
Total gross carrying amount	\$ 4,212	\$ 4,361
Accumulated amortization — customer related	(966)	(874)
Accumulated amortization — patents	(486)	(489)
Accumulated amortization — other technology based	(323)	(292)
Accumulated amortization — definite-lived tradenames	(268)	(256)
Accumulated amortization — other	(86)	(162)
Total accumulated amortization	\$ (2,129)	\$ (2,073)
Total finite-lived intangible assets — net	\$ 2,083	\$ 2,288
Non-amortizable intangible assets (primarily tradenames)	643	648
Total intangible assets — net	\$ 2,726	\$ 2,936

3M does not amortize certain acquired tradenames because they have been in existence for over 55 years, have a history of leading-market share positions, are intended to be continuously renewed, and the associated products are expected to generate cash flows for 3M for an indefinite period of time.

Amortization expense for acquired intangible assets for the three and nine months ended September 30, 2018 and 2017 follows:

(Millions)	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Amortization expense	\$ 61	\$ 51	\$ 188	\$ 162

Expected amortization expense for acquired amortizable intangible assets recorded as of September 30, 2018:

(Millions)	Remainder of 2018	2019	2020	2021	2022	2023	After 2023
Amortization expense	\$ 62	\$ 241	\$ 229	\$ 220	\$ 206	\$ 175	\$ 950

The preceding expected amortization expense is an estimate. Actual amounts of amortization expense may differ from estimated amounts due to additional intangible asset acquisitions, changes in foreign currency exchange rates, impairment of intangible assets, accelerated amortization of intangible assets and other events. 3M expenses the costs incurred to renew or extend the term of intangible assets.

NOTE 5. Restructuring Actions and Exit Activities

2018 Restructuring Actions:

During the second quarter of 2018, management approved and committed to undertake certain restructuring actions related to addressing corporate functional costs following the Communication Markets Division divestiture. These actions affected approximately 900 positions worldwide and resulted in a second quarter 2018 pre-tax charge of \$105 million, essentially all within Corporate and Unallocated.

The restructuring charges were recorded in the income statement as follows:

(Millions)	Second Quarter 2018
Cost of sales	\$ 12
Selling, general and administrative expenses	89
Research, development and related expenses	4
Total	\$ 105

Components of these restructuring actions, follow:

(Millions)	Employee-Related
Expense incurred in the second quarter of 2018	\$ 105
Cash payments	(12)
Adjustments	(7)
Accrued restructuring action balances as of September 30, 2018	\$ 86

Remaining activities related to this restructuring are expected to be largely completed through the first half of 2019.

2017 Restructuring Actions:

During the second quarter of 2017, management approved and committed to undertake certain restructuring actions primarily focused on portfolio and footprint optimization. These actions affected approximately 1,300 positions worldwide and resulted in a second quarter 2017 pre-tax charge of \$99 million.

Components of these restructuring actions, including cash and non-cash impacts, follow:

(Millions)	Employee-Related
Expense incurred in the second quarter of 2017	\$ 99
Cash payments	(8)
Adjustments	(3)
Accrued restructuring action balances as of December 31, 2017	\$ 88
Cash payments	(17)
Adjustments	(27)
Accrued restructuring action balances as of September 30, 2018	\$ 44

Remaining activities related to this restructuring are expected to be largely completed by the end of 2018. A portion of the adjustments detailed above include certain severance accruals taken in 2017, the obligation for which was relieved and reflected as part of the gain on divestiture when that business was sold in 2018.

2017 Exit Activities:

During the first quarter of 2017, the Company recorded net pre-tax charges of \$24 million related to exit activities. These charges related to employee reductions, primarily in Western Europe.

NOTE 6. Supplemental Income Statement Information

Other expense (income), net consists of the following:

(Millions)	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Interest expense	\$ 85	\$ 57	\$ 255	\$ 156
Interest income	(15)	(13)	(52)	(33)
Pension and postretirement net periodic benefit cost (benefit)	(19)	(33)	(59)	(96)
Total	\$ 51	\$ 11	\$ 144	\$ 27

Pension and postretirement net periodic benefit costs described in the table above include all components of defined benefit plan net periodic benefit costs except service cost, which is reported in various operating expense lines. Refer to Note 11 for additional details on the components of pension and postretirement net periodic benefit costs.

NOTE 7. Supplemental Equity and Comprehensive Income Information

Consolidated Changes in Equity

Three months ended September 30, 2018

(Millions)	Total	3M Company Shareholders				
		Common Stock and Additional Paid-in Capital	Retained Earnings	Treasury Stock	Accumulated Other Comprehensive Income (Loss)	Non-controlling Interest
Balance at June 30, 2018	\$ 10,428	\$ 5,559	\$ 39,442	\$ (27,617)	\$ (7,019)	\$ 63
Net income	1,546		1,543			3
Other comprehensive income (loss), net of tax:						
Cumulative translation adjustment	(112)				(109)	(3)
Defined benefit pension and post-retirement plans adjustment	114				114	—
Cash flow hedging instruments - unrealized gain (loss)	46				46	—
Total other comprehensive income (loss), net of tax	48					
Dividends declared	(794)		(794)			
Stock-based compensation	47	47				
Reacquired stock	(1,058)			(1,058)		
Issuances pursuant to stock option and benefit plans	94		(71)	165		
Balance at September 30, 2018	\$ 10,311	\$ 5,606	\$ 40,120	\$ (28,510)	\$ (6,968)	\$ 63

Nine months ended September 30, 2018

(Millions)	Total	3M Company Shareholders				Non-controlling Interest
		Common Stock and Additional Paid-in Capital	Retained Earnings	Treasury Stock	Accumulated Other Comprehensive Income (Loss)	
Balance at December 31, 2017	\$ 11,622	\$ 5,361	\$ 39,115	\$ (25,887)	\$ (7,026)	\$ 59
Net income	4,014		4,002			12
Other comprehensive income (loss), net of tax:						
Cumulative translation adjustment	(441)				(433)	(8)
Defined benefit pension and post-retirement plans adjustment	344				344	—
Cash flow hedging instruments - unrealized gain (loss)	147				147	—
Total other comprehensive income (loss), net of tax	50					
Dividends declared	(2,406)		(2,406)			
Stock-based compensation	245	245				
Reacquired stock	(3,621)			(3,621)		
Issuances pursuant to stock option and benefit plans	407		(591)	998		
Balance at September 30, 2018	\$ 10,311	\$ 5,606	\$ 40,120	\$ (28,510)	\$ (6,968)	\$ 63

Three months ended September 30, 2017

(Millions)	Total	3M Company Shareholders				Non-controlling Interest
		Common Stock and Additional Paid-in Capital	Retained Earnings	Treasury Stock	Accumulated Other Comprehensive Income (Loss)	
Balance at June 30, 2017	\$ 11,644	\$ 5,253	\$ 38,793	\$ (25,466)	\$ (6,989)	\$ 53
Net income	1,433		1,429			4
Other comprehensive income (loss), net of tax:						
Cumulative translation adjustment	44				45	(1)
Defined benefit pension and post-retirement plans adjustment	80				80	—
Cash flow hedging instruments - unrealized gain (loss)	(49)				(49)	—
Total other comprehensive income (loss), net of tax	75					
Dividends declared	(701)		(701)			
Stock-based compensation	58	58				
Reacquired stock	(394)			(394)		
Issuances pursuant to stock option and benefit plans	87		(67)	154		
Balance at September 30, 2017	\$ 12,202	\$ 5,311	\$ 39,454	\$ (25,706)	\$ (6,913)	\$ 56

Nine months ended September 30, 2017

(Millions)	3M Company Shareholders					Non-controlling Interest
	Total	Common Stock and Additional Paid-in Capital	Retained Earnings	Treasury Stock	Accumulated Other Comprehensive Income (Loss)	
Balance at December 31, 2016	\$ 10,343	\$ 5,070	\$ 37,907	\$ (25,434)	\$ (7,245)	\$ 45
Net income	4,344		4,335			9
Other comprehensive income (loss), net of tax:						
Cumulative translation adjustment	269				267	2
Defined benefit pension and post-retirement plans adjustment	241				241	—
Cash flow hedging instruments - unrealized gain (loss)	(176)				(176)	—
Total other comprehensive income (loss), net of tax	334					
Dividends declared	(2,104)		(2,104)			
Stock-based compensation	241	241				
Reacquired stock	(1,547)			(1,547)		
Issuances pursuant to stock option and benefit plans	591		(684)	1,275		
Balance at September 30, 2017	\$ 12,202	\$ 5,311	\$ 39,454	\$ (25,706)	\$ (6,913)	\$ 56

Changes in Accumulated Other Comprehensive Income (Loss) Attributable to 3M by Component

Three months ended September 30, 2018

(Millions)	Cumulative Translation Adjustment	Defined Benefit Pension and Postretirement Plans Adjustment	Cash Flow Hedging Instruments, Unrealized Gain (Loss)	Total Accumulated Other Comprehensive Income (Loss)
Balance at June 30, 2018, net of tax:	\$ (1,962)	\$ (5,046)	\$ (11)	\$ (7,019)
Other comprehensive income (loss), before tax:				
Amounts before reclassifications	(110)	—	22	(88)
Amounts reclassified out	—	150	37	187
Total other comprehensive income (loss), before tax	(110)	150	59	99
Tax effect	1	(36)	(13)	(48)
Total other comprehensive income (loss), net of tax	(109)	114	46	51
Balance at September 30, 2018, net of tax:	\$ (2,071)	\$ (4,932)	\$ 35	\$ (6,968)

Nine months ended September 30, 2018

(Millions)	Cumulative Translation Adjustment	Defined Benefit Pension and Postretirement Plans Adjustment	Cash Flow Hedging Instruments, Unrealized Gain (Loss)	Total Accumulated Other Comprehensive Income (Loss)
Balance at December 31, 2017, net of tax:	\$ (1,638)	\$ (5,276)	\$ (112)	\$ (7,026)
Other comprehensive income (loss), before tax:				
Amounts before reclassifications	(392)	—	122	(270)
Amounts reclassified out	—	452	99	551
Total other comprehensive income (loss), before tax	(392)	452	221	281
Tax effect	(41)	(108)	(74)	(223)
Total other comprehensive income (loss), net of tax	(433)	344	147	58
Balance at September 30, 2018, net of tax:	\$ (2,071)	\$ (4,932)	\$ 35	\$ (6,968)

Three months ended September 30, 2017

(Millions)	Cumulative Translation Adjustment	Defined Benefit Pension and Postretirement Plans Adjustment	Cash Flow Hedging Instruments, Unrealized Gain (Loss)	Total Accumulated Other Comprehensive Income (Loss)
Balance at June 30, 2017, net of tax:	\$ (1,786)	\$ (5,167)	\$ (36)	\$ (6,989)
Other comprehensive income (loss), before tax:				
Amounts before reclassifications	(29)	—	(80)	(109)
Amounts reclassified out	—	119	2	121
Total other comprehensive income (loss), before tax	(29)	119	(78)	12
Tax effect	74	(39)	29	64
Total other comprehensive income (loss), net of tax	45	80	(49)	76
Balance at September 30, 2017, net of tax:	\$ (1,741)	\$ (5,087)	\$ (85)	\$ (6,913)

Nine months ended September 30, 2017

(Millions)	Cumulative Translation Adjustment	Defined Benefit Pension and Postretirement Plans Adjustment	Cash Flow Hedging Instruments, Unrealized Gain (Loss)	Total Accumulated Other Comprehensive Income (Loss)
Balance at December 31, 2016, net of tax:	\$ (2,008)	\$ (5,328)	\$ 91	\$ (7,245)
Other comprehensive income (loss), before tax:				
Amounts before reclassifications	30	—	(255)	(225)
Amounts reclassified out	—	357	(21)	336
Total other comprehensive income (loss), before tax	30	357	(276)	111
Tax effect	237	(116)	100	221
Total other comprehensive income (loss), net of tax	267	241	(176)	332
Balance at September 30, 2017, net of tax	\$ (1,741)	\$ (5,087)	\$ (85)	\$ (6,913)

Income taxes are not provided for foreign translation relating to permanent investments in international subsidiaries, but tax effects within cumulative translation do include impacts from items such as net investment hedge transactions. Reclassification adjustments are made to avoid double counting in comprehensive income items that are also recorded as part of net income.

Reclassifications out of Accumulated Other Comprehensive Income Attributable to 3M

Details about Accumulated Other Comprehensive Income Components (Millions)	Amount Reclassified from Accumulated Other Comprehensive Income				Location on Income Statement
	Three months ended September 30,		Nine months ended September 30,		
	2018	2017	2018	2017	
Gains (losses) associated with defined benefit pension and postretirement plans amortization					
Prior service benefit	\$ 20	\$ 21	\$ 58	\$ 65	See Note 11
Net actuarial loss	(170)	(140)	(510)	(422)	See Note 11
Total before tax	(150)	(119)	(452)	(357)	
Tax effect	36	39	108	116	Provision for income taxes
Net of tax	\$ (114)	\$ (80)	\$ (344)	\$ (241)	
Cash flow hedging instruments gains (losses)					
Foreign currency forward/option contracts	\$ (37)	\$ (2)	\$ (99)	\$ 21	Cost of sales
Interest rate swap contracts	—	—	—	—	Interest expense
Total before tax	(37)	(2)	(99)	21	
Tax effect	8	—	22	(8)	Provision for income taxes
Net of tax	\$ (29)	\$ (2)	\$ (77)	\$ 13	
Total reclassifications for the period, net of tax	\$ (143)	\$ (82)	\$ (421)	\$ (228)	

NOTE 8. Income Taxes

The Company files income tax returns in the U.S. federal jurisdiction as well as various state, local, and foreign jurisdictions. With few exceptions, the Company is no longer subject to income tax examinations by tax authorities for years before 2005.

The IRS has completed its field examination of the Company's U.S. federal income tax returns for the years 2005 through 2014, and 2016. The Company protested certain IRS positions within these tax years and entered into the administrative appeals process with the IRS. Currently, the Company is under examination by the IRS for its U.S. federal income tax returns for the years 2015 and 2017. As of September 30, 2018, the IRS has not proposed any significant adjustments to the Company's tax positions for which the Company is not adequately reserved.

Payments relating to other proposed assessments arising from the 2005 through 2017 examinations may not be made until a final agreement is reached between the Company and the IRS on such assessments or upon a final resolution resulting from the administrative appeals process or judicial action. In addition to the U.S. federal examination, there is also audit activity in several U.S. state and foreign jurisdictions.

The Company applies a more-likely-than-not threshold to the recognition and derecognition of uncertain tax positions. Accordingly, the Company recognizes the amount of tax benefit that has a greater than 50 percent likelihood of being ultimately realized upon settlement. The total amounts of unrecognized tax benefits that, if recognized, would affect the effective tax rate as of September 30, 2018 and December 31, 2017 are \$584 million and \$526 million, respectively.

The Company recognizes interest and penalties accrued related to unrecognized tax benefits in tax expense. The Company recognized in the consolidated statement of income on a gross basis approximately \$5 million and \$8 million of expense for the three months ended September 30, 2018 and September 30, 2017, respectively, and approximately \$8 million and \$16 million of expense for the nine months ended September 30, 2018 and September 30, 2017, respectively. At September 30, 2018 and December 31, 2017, accrued interest and penalties in the consolidated balance sheet on a gross basis were \$65 million and \$68 million, respectively. Included in these interest and penalty amounts are interest and penalties related to tax positions for which the ultimate deductibility is highly certain but for which there is uncertainty about the timing of such deductibility.

The provision for income taxes is determined using the asset and liability approach. Under this approach, a valuation allowance is recorded to reduce its deferred tax assets when uncertainty regarding their realizability exists. As of September 30, 2018 and December 31, 2017, the Company had valuation allowances of \$73 million and \$81 million on its deferred tax assets, respectively.

The effective tax rate for the third quarter of 2018 was 21.3 percent, compared to 28.3 percent in the third quarter of 2017, a decrease of 7.0 percentage points. Primary factors that decreased the Company's effective tax rate included favorable aspects of the 2017 Tax Cuts and Jobs Act (TCJA) such as the decrease in the U.S. income tax rate and provisions incentivizing foreign-derived intangible income (FDII), in addition to impacts associated with composition of geographic mix of income before taxes. These decreases were partially offset by multiple factors that increased the Company's effective tax rate. These increases were primarily driven by impacts of the TCJA such as the elimination of the domestic manufacturing deduction and the global intangible low-taxed income (GILTI) provision; and lower excess tax benefits related to employee share-based payments.

The effective tax rate for the first nine months of 2018 was 24.0 percent, compared to 26.1 percent in the first nine months of 2017, a decrease of 2.1 percentage points. Primary factors that decreased the Company's effective tax rate included favorable aspects of the TCJA such as the decrease in the U.S. income tax rate and FDII, in addition to impacts associated with composition of geographic mix of income before taxes. These decreases were partially offset by impacts related to the TCJA such as the first quarter 2018 measurement period adjustment to the provisional accounting of the TCJA's enactment (discussed further below), the elimination of the domestic manufacturing deduction and the GILTI provision; and lower excess tax benefits related to employee share-based payments.

The TCJA was enacted in December 2017. Among other things, the TCJA reduces the U.S. federal corporate tax rate from 35 percent to 21 percent beginning in 2018, requires companies to pay a one-time transition tax on previously unremitted earnings of non-U.S. subsidiaries that were previously tax deferred and creates new taxes on certain foreign sourced earnings. The SEC staff issued Staff Accounting Bulletin (SAB) 118, which provides guidance on accounting for enactment effects of the TCJA. SAB 118 provides a

measurement period of up to one year from the TCJA's enactment date for companies to complete their accounting under ASC 740. In accordance with SAB 118, to the extent that a company's accounting for certain income tax effects of the TCJA is incomplete but it is able to determine a reasonable estimate, it must record a provisional estimate in its financial statements. If a company cannot determine a provisional estimate to be included in its financial statements, it should continue to apply ASC 740 on the basis of the provisions of the tax laws that were in effect immediately before the enactment of the TCJA. In connection with 3M's initial analysis of the impact of the enactment of the TCJA, the Company recorded a net tax expense of \$762 million in the fourth quarter of 2017. As further discussed below, during the first quarter of 2018, 3M recognized a measurement period adjustment resulting in an additional tax expense of \$217 million to this provisional accounting. The Company made no additional adjustments in the second and third quarters of 2018.

Transition tax: 3M was able to make a reasonable estimate of the transition tax and recorded a provisional obligation and related income tax expense of \$745 million in the fourth quarter of 2017, with an additional \$132 million in the first quarter of 2018 as a result of subsequent U.S. Internal Revenue Service guidance. The Company made no additional adjustments in the second and third quarters of 2018. The Company continues to gather additional information and will consider further guidance to more precisely compute the transition tax. The provisional amount may change when 3M finalizes the calculation of post-1986 foreign earnings and profit previously deferred from U.S. federal taxation and finalizes the amounts held in cash or other specified assets. The TCJA's transition tax is payable over eight years beginning in 2018. As of September 30, 2018, 3M reflected \$74 million and \$733 million in current accrued income taxes and long-term income taxes payable, respectively, associated with the transition tax.

Remeasurement of deferred tax assets/liabilities and other impacts: 3M remeasured certain deferred tax assets and liabilities based on the rates at which they are expected to reverse in the future, which is generally 21 percent under the TCJA. In the fourth quarter of 2017, 3M recorded a net additional income tax expense of \$17 million related to remeasurement of deferred tax assets/liabilities and other impacts. In the first quarter of 2018, 3M recorded an additional tax expense of \$85 million as a measurement period adjustment as a result of subsequent U.S. Internal Revenue Service guidance and other impacts related to TCJA. The Company made no additional adjustments in the second and third quarters of 2018. 3M is still analyzing certain aspects of the TCJA, considering additional technical guidance, and refining its calculations, which could potentially affect the measurement of these balances or potentially give rise to new deferred tax amounts. This includes the potential impacts of the GILTI provision within the TCJA on deferred tax assets/liabilities. 3M also is considering other impacts of the 2017 enactment of the TCJA including, but not limited to, effects on the Company's indefinite reinvestment assertion. 3M previously has not provided deferred taxes on unremitted earnings attributable to international companies that have been considered to be reinvested indefinitely. The full effects of underlying tax rates of the TCJA causes some reassessment of previous indefinite reinvestment assertions with respect to certain jurisdictions. While 3M was able to make a reasonable estimate of these impacts, it may be affected by other analyses related to the TCJA, including, but not limited to, the calculation of the transition tax on deferred foreign income.

While 3M has recorded current tax on GILTI relative to 2018 operations, the Company has not yet elected a policy as to whether it will recognize deferred taxes for basis differences expected to reverse as GILTI or whether 3M will account for GILTI as period costs if and when incurred.

NOTE 9. Marketable Securities

The following is a summary of amounts recorded on the Consolidated Balance Sheet for marketable securities (current and non-current).

(Millions)	September 30, 2018	December 31, 2017
Corporate debt securities	\$ —	\$ 14
Commercial paper	304	899
Certificates of deposit/time deposits	31	76
U.S. municipal securities	3	3
Asset-backed securities:		
Automobile loan related	—	16
Credit card related	—	68
Asset-backed securities total	—	84
Current marketable securities	\$ 338	\$ 1,076
U.S. municipal securities	\$ 27	\$ 27
Non-current marketable securities	\$ 27	\$ 27
Total marketable securities	\$ 365	\$ 1,103

At September 30, 2018 and December 31, 2017, gross unrealized, gross realized, and net realized gains and/or losses (pre-tax) were not material.

The balances at September 30, 2018 for marketable securities by contractual maturity are shown below. Actual maturities may differ from contractual maturities because the issuers of the securities may have the right to prepay obligations without prepayment penalties.

(Millions)	September 30, 2018
Due in one year or less	\$ 338
Due after one year through five years	13
Due after five years through ten years	14
Total marketable securities	\$ 365

3M does not currently expect risk related to its holding in asset-backed securities to materially impact its financial condition or liquidity.

NOTE 10. Long-Term Debt and Short-Term Borrowings

In September 2018, 3M issued \$400 million aggregate principal amount of 3-year fixed rate medium-term notes due 2021 with a coupon rate of 3.00%, \$300 million aggregate principal amount of 5.5-year fixed rate medium-term notes due 2024 with a coupon rate of 3.25%, \$300 million aggregate principal amount of 5.5-year floating rate medium-term notes due 2024 with a rate based on a floating three-month LIBOR index, \$600 million aggregate principal amount of 10-year fixed rate medium-term notes due 2028 with a coupon rate of 3.625%, and \$650 million aggregate principal amount of 30-year fixed rate medium-term notes due 2048 with a coupon rate of 4.00%. Upon debt issuance, the Company entered into a fixed-to-floating interest rate swap on \$200 million aggregate principal amount of the 3-year fixed rate medium-term notes issued with an interest rate based on a three-month LIBOR index.

As of September 30, 2018, the Company had zero commercial paper outstanding. As of December 31, 2017, the Company had approximately \$745 million in commercial paper outstanding.

In August 2018, 3M repaid \$450 million aggregate principal amount of fixed rate medium-term notes that matured.

NOTE 11. Pension and Postretirement Benefit Plans

The service cost component of defined benefit net periodic benefit cost is recorded in cost of sales, selling, general and administrative expenses, and research, development and related expenses. The other components of net periodic benefit cost are reflected in other expense (income), net. Components of net periodic benefit cost and other supplemental information for the three and nine months ended September 30, 2018 and 2017 follow:

Benefit Plan Information

(Millions)	Three months ended September 30,					
	Qualified and Non-qualified Pension Benefits				Postretirement Benefits	
	United States		International			
	2018	2017	2018	2017	2018	2017
Net periodic benefit cost (benefit)						
Operating expense						
Service cost	\$ 72	\$ 67	\$ 37	\$ 33	\$ 13	\$ 13
Non-operating expense						
Interest cost	\$ 141	\$ 141	\$ 40	\$ 38	\$ 20	\$ 20
Expected return on plan assets	(272)	(259)	(78)	(71)	(21)	(21)
Amortization of prior service benefit	(6)	(5)	(4)	(3)	(10)	(13)
Amortization of net actuarial loss	126	97	29	29	15	14
Total non-operating expense (benefit)	(11)	(26)	(13)	(7)	4	—
Total net periodic benefit cost (benefit)	\$ 61	\$ 41	\$ 24	\$ 26	\$ 17	\$ 13
(Millions)	Nine months ended September 30,					
	Qualified and Non-qualified Pension Benefits				Postretirement Benefits	
	United States		International			
	2018	2017	2018	2017	2018	2017
Net periodic benefit cost (benefit)						
Operating expense						
Service cost	\$ 216	\$ 201	\$ 110	\$ 100	\$ 39	\$ 38
Non-operating expense						
Interest cost	\$ 423	\$ 425	\$ 120	\$ 112	\$ 60	\$ 59
Expected return on plan assets	(816)	(777)	(235)	(209)	(63)	(63)
Amortization of prior service benefit	(18)	(17)	(10)	(9)	(30)	(39)
Amortization of net actuarial loss	378	291	87	89	45	42
Total non-operating expense (benefit)	(33)	(78)	(38)	(17)	12	(1)
Total net periodic benefit cost (benefit)	\$ 183	\$ 123	\$ 72	\$ 83	\$ 51	\$ 37

For the nine months ended September 30, 2018, contributions totaling \$300 million were made to the Company's U.S. and international pension plans and \$3 million to its postretirement plans. For total year 2018, the Company expects to contribute up to \$400 million of cash to its global defined benefit pension and postretirement plans. The Company does not have a required minimum cash pension contribution obligation for its U.S. plans in 2018. Future contributions will depend on market conditions, interest rates and other factors. 3M's annual measurement date for pension and postretirement assets and liabilities is December 31 each year, which is also the date used for the related annual measurement assumptions.

The Company adopted ASU No. 2017-07, *Improving the Presentation of Net Periodic Pension Cost and Net Periodic Postretirement Benefit Cost*, effective January 1, 2018 on a retrospective basis. This ASU changed how employers that sponsor defined benefit pension and/or other postretirement benefit plans present the net periodic benefit cost in the income statement. Under the new standard, only the service cost component of net periodic benefit cost is included in operating expenses and only the service cost component is eligible for capitalization into assets such as inventory. All other net periodic benefit costs components (such as interest, expected return on plan assets, prior service cost amortization and actuarial gain/loss amortization) are reported outside of operating income. See Note 1 for additional details.

NOTE 12. Derivatives

The Company uses interest rate swaps, currency swaps, commodity price swaps, and forward and option contracts to manage risks generally associated with foreign exchange rate, interest rate and commodity price fluctuations. The information that follows explains the various types of derivatives and financial instruments used by 3M, how and why 3M uses such instruments, how such instruments are accounted for, and how such instruments impact 3M's financial position and performance.

Additional information with respect to derivatives is included elsewhere as follows:

- Impact on other comprehensive income of nonderivative hedging and derivative instruments is included in Note 7.
- Fair value of derivative instruments is included in Note 13.
- Derivatives and/or hedging instruments associated with the Company's long-term debt are described in Note 11 in 3M's Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K).

*Types of Derivatives/Hedging Instruments and Inclusion in Income/Other Comprehensive Income*Cash Flow Hedges:

For derivative instruments that are designated and qualify as cash flow hedges, the effective portion of the gain or loss on the derivative is reported as a component of other comprehensive income and reclassified into earnings in the same period during which the hedged transaction affects earnings. Gains and losses on the derivative representing either hedge ineffectiveness or hedge components excluded from the assessment of effectiveness are recognized in current earnings.

Cash Flow Hedging - Foreign Currency Forward and Option Contracts: The Company enters into foreign exchange forward and option contracts to hedge against the effect of exchange rate fluctuations on cash flows denominated in foreign currencies. These transactions are designated as cash flow hedges. The settlement or extension of these derivatives will result in reclassifications (from accumulated other comprehensive income) to earnings in the period during which the hedged transactions affect earnings. 3M may dedesignate these cash flow hedge relationships in advance of the occurrence of the forecasted transaction. The portion of gains or losses on the derivative instrument previously included in accumulated other comprehensive income for dedesignated hedges remains in accumulated other comprehensive income until the forecasted transaction occurs or becomes probable of not occurring. Changes in the value of derivative instruments after dedesignation are recorded in earnings and are included in the Derivatives Not Designated as Hedging Instruments section below. The maximum length of time over which 3M hedges its exposure to the variability in future cash flows of the forecasted transactions is 36 months.

Cash Flow Hedging — Interest Rate Contracts: The Company may use forward starting interest rate contracts to hedge exposure to variability in cash flows from interest payments on forecasted debt issuances. The amortization of gains and losses on forward starting interest rate swaps is included in the tables below as part of the gain/(loss) recognized in income on the effective portion of derivatives as a result of reclassification from accumulated other comprehensive income. Additional information regarding previously issued but terminated interest rate contracts, which have related balances within accumulated other comprehensive income being amortized over the underlying life of related debt, can be found in Note 13 in 3M's Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K).

During the first nine months of 2018, the Company entered into forward starting interest rate swaps with a notional amount of \$700 million as hedges against interest rate volatility associated with forecasted issuances of fixed rate debt. Concurrent with the issuance of the medium-term notes in September 2018, 3M terminated \$500 million of these interest rate swaps. The termination resulted in an immaterial gain within accumulated other comprehensive income that will be amortized over the respective lives of the debt.

As of September 30, 2018, the Company had a balance of \$35 million associated with the after-tax net unrealized gain associated with cash flow hedging instruments recorded in accumulated other comprehensive income. This includes a remaining balance of \$7 million (after-tax loss) related to the forward starting interest rate swaps, which will be amortized over the respective lives of the debt. Based on exchange rates as of September 30, 2018, 3M expects to reclassify approximately \$22 million, \$1 million, \$25 million, and \$9 million of the after-tax net unrealized foreign exchange cash flow hedging gains to earnings over the next 12 months, over the remainder of 2018, in 2019, and after 2019, respectively (with the impact offset by earnings/losses from underlying hedged items).

The location in the consolidated statements of income and comprehensive income and amounts of gains and losses related to derivative instruments designated as cash flow hedges are provided in the following table.

	Pretax Gain (Loss) Recognized in Other Comprehensive Income on Effective Portion of Derivative	Pretax Gain (Loss) Recognized in Income on Effective Portion of Derivative as a Result of Reclassification from Accumulated Other Comprehensive Income		Ineffective Portion of Gain (Loss) on Derivative and Amount Excluded from Effectiveness Testing Recognized in Income	
Three months ended September 30, 2018 (Millions)	Amount	Location	Amount	Location	Amount
Foreign currency forward/option contracts	\$ 12	Cost of sales	\$ (37)	Cost of sales	\$ —
Interest rate swap contracts	10	Interest expense	—	Interest expense	—
Total	\$ 22		\$ (37)		\$ —
Nine months ended September 30, 2018 (Millions)	Amount	Location	Amount	Location	Amount
Foreign currency forward/option contracts	\$ 112	Cost of sales	\$ (98)	Cost of sales	\$ —
Interest rate swap contracts	10	Interest expense	(1)	Interest expense	—
Total	\$ 122		\$ (99)		\$ —
Three months ended September 30, 2017 (Millions)	Amount	Location	Amount	Location	Amount
Foreign currency forward/option contracts	\$ (77)	Cost of sales	\$ (2)	Cost of sales	\$ —
Interest rate swap contracts	(3)	Interest expense	—	Interest expense	—
Total	\$ (80)		\$ (2)		\$ —
Nine months ended September 30, 2017 (Millions)	Amount	Location	Amount	Location	Amount
Foreign currency forward/option contracts	\$ (249)	Cost of sales	\$ 21	Cost of sales	\$ —
Interest rate swap contracts	(6)	Interest expense	—	Interest expense	—
Total	\$ (255)		\$ 21		\$ —

Fair Value Hedges:

For derivative instruments that are designated and qualify as fair value hedges, the gain or loss on the derivatives as well as the offsetting loss or gain on the hedged item attributable to the hedged risk are recognized in current earnings.

Fair Value Hedging - Interest Rate Swaps: The Company manages interest expense using a mix of fixed and floating rate debt. To help manage borrowing costs, the Company may enter into interest rate swaps. Under these arrangements, the Company agrees to exchange, at specified intervals, the difference between fixed and floating interest amounts calculated by reference to an agreed-upon notional principal amount. The mark-to-market of these fair value hedges is recorded as gains or losses in interest expense and is offset by the gain or loss of the underlying debt instrument, which also is recorded in interest expense. These fair value hedges are highly effective and, thus, there is no impact on earnings due to hedge ineffectiveness. Additional information regarding designated interest rate swaps can be found in Note 13 in 3M's Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K).

In September 2018, the Company entered into an interest rate swap with a notional amount of \$200 million that converted a portion of the Company's \$400 million aggregate principal amount of fixed rate medium-term notes due 2021 into a floating rate note with an interest rate based on a three-month LIBOR index as a hedge of its exposure to changes in fair value that are attributable to interest rate risk.

The location in the consolidated statements of income and amounts of gains and losses related to derivative instruments designated as fair value hedges and similar information relative to the hedged items are as follows:

Three months ended September 30, 2018 (Millions)	Gain (Loss) on Derivative Recognized in Income		Gain (Loss) on Hedged Item Recognized in Income	
	Location	Amount	Location	Amount
Interest rate swap contracts	Interest expense	\$ —	Interest expense	\$ —
Total		\$ —		\$ —
Nine months ended September 30, 2018 (Millions)	Location		Location	
	Amount		Amount	
Interest rate swap contracts	Interest expense	\$ (12)	Interest expense	\$ 12
Total		\$ (12)		\$ 12
Three months ended September 30, 2017 (Millions)	Location		Location	
	Amount		Amount	
Interest rate swap contracts	Interest expense	\$ —	Interest expense	\$ —
Total		\$ —		\$ —
Nine months ended September 30, 2017 (Millions)	Location		Location	
	Amount		Amount	
Interest rate swap contracts	Interest expense	\$ (4)	Interest expense	\$ 4
Total		\$ (4)		\$ 4

Net Investment Hedges:

The Company may use non-derivative (foreign currency denominated debt) and derivative (foreign exchange forward contracts) instruments to hedge portions of the Company's investment in foreign subsidiaries and manage foreign exchange risk. For instruments that are designated and qualify as hedges of net investments in foreign operations and that meet the effectiveness requirements, the net gains or losses attributable to changes in spot exchange rates are recorded in cumulative translation within other comprehensive income. The remainder of the change in value of such instruments is recorded in earnings. Recognition in earnings of amounts previously recorded in cumulative translation is limited to circumstances such as complete or substantially complete liquidation of the net investment in the hedged foreign operation. To the extent foreign currency denominated debt is not designated in or is dedesignated from a net investment hedge relationship, changes in value of that portion of foreign currency denominated debt due to exchange rate changes are recorded in earnings through their maturity date.

3M's use of foreign exchange forward contracts designated in hedges of the Company's net investment in foreign subsidiaries can vary by time period depending on when foreign currency denominated debt balances designated in such relationships are dedesignated, matured, or are newly issued and designated. Additionally, variation can occur in connection with the extent of the Company's desired foreign exchange risk coverage.

During the first quarter of 2018, the Company dedesignated 300 million Euros of foreign currency denominated debt from a former net investment hedge relationship.

At September 30, 2018, the total notional amount of foreign exchange forward contracts designated in net investment hedges was approximately 150 million Euros and approximately 248 billion South Korean Won, along with a principal amount of long-term debt instruments designated in net investment hedges totaling 4.1 billion Euros. The maturity dates of these derivative and nonderivative instruments designated in net investment hedges range from 2018 to 2031.

The location in the consolidated statements of income and comprehensive income and amounts of gains and losses related to derivative and nonderivative instruments designated as net investment hedges are as follows. There were no reclassifications of the effective portion of net investment hedges out of accumulated other comprehensive income into income for the periods presented in the table below.

	Pretax Gain (Loss) Recognized as Cumulative Translation within Other Comprehensive Income on Effective Portion of Instrument	Ineffective Portion of Gain (Loss) on Instrument and Amount Excluded from Effectiveness Testing Recognized in Income	
		Location	Amount
Three months ended September 30, 2018 (Millions)	Amount		
Foreign currency denominated debt	\$ (14)	Cost of sales	\$ —
Foreign currency forward contracts	(3)	Cost of sales	1
Total	\$ (17)		\$ 1
Nine months ended September 30, 2018 (Millions)	Amount	Location	Amount
Foreign currency denominated debt	\$ 157	Cost of sales	\$ (2)
Foreign currency forward contracts	14	Cost of sales	1
Total	\$ 171		\$ (1)
Three months ended September 30, 2017 (Millions)	Amount	Location	Amount
Foreign currency denominated debt	\$ (179)	Cost of sales	\$ —
Foreign currency forward contracts	(9)	Cost of sales	1
Total	\$ (188)		\$ 1
Nine months ended September 30, 2017 (Millions)	Amount	Location	Amount
Foreign currency denominated debt	\$ (570)	Cost of sales	\$ —
Foreign currency forward contracts	(36)	Cost of sales	6
Total	\$ (606)		\$ 6

Derivatives Not Designated as Hedging Instruments:

Derivatives not designated as hedging instruments include dedesignated foreign currency forward and options contracts that formerly were designated in cash flow hedging relationships (as referenced in the Cash Flow Hedges section above). In addition, 3M enters into foreign exchange forward contracts to offset, in part, the impacts of certain intercompany transactions, foreign currency denominated debt (not otherwise in net investment hedge relationships), and to further mitigate short term currency impacts. In addition, the Company enters into commodity price swaps to offset, in part, fluctuations in costs associated with the use of certain commodities and precious metals. These derivative instruments are not designated in hedging relationships; therefore, fair value gains and losses on these contracts are recorded in earnings. The Company does not hold or issue derivative financial instruments for trading purposes.

The location in the consolidated statements of income and amounts of gains and losses related to derivative instruments not designated as hedging instruments are as follows:

(Millions)	Three months ended September 30, 2018 Gain (Loss) on Derivative Recognized in Income		Nine months ended September 30, 2018 Gain (Loss) on Derivative Recognized in Income	
	Location	Amount	Location	Amount
Foreign currency forward/option contracts	Cost of sales	\$ 11	Cost of sales	\$ 11
Foreign currency forward contracts	Interest expense	(7)	Interest expense	(98)
Total		\$ 4		\$ (87)

(Millions)	Three months ended September 30, 2017 Gain (Loss) on Derivative Recognized in Income		Nine months ended September 30, 2017 Gain (Loss) on Derivative Recognized in Income	
	Location	Amount	Location	Amount
Foreign currency forward/option contracts	Cost of sales	\$ 3	Cost of sales	\$ 8
Foreign currency forward contracts	Interest expense	(67)	Interest expense	(163)
Total		<u>\$ (64)</u>		<u>\$ (155)</u>

Location and Fair Value Amount of Derivative Instruments

The following tables summarize the fair value of 3M's derivative instruments, excluding nonderivative instruments used as hedging instruments, and their location in the consolidated balance sheet. Notional amounts below are presented at period end foreign exchange rates, except interest rate swaps, which are presented using the contract inception date's foreign exchange rate. Additional information with respect to the fair value of derivative instruments is included in Note 13.

September 30, 2018 (Millions)	Gross Notional Amount	Assets		Liabilities	
		Location	Fair Value Amount	Location	Fair Value Amount
Derivatives designated as hedging instruments					
Foreign currency forward/option contracts	\$ 2,107	Other current assets	\$ 51	Other current liabilities	\$ 22
Foreign currency forward/option contracts	1,250	Other assets	41	Other liabilities	11
Interest rate swap contracts	600	Other current assets	—	Other current liabilities	5
Interest rate swap contracts	1,303	Other assets	25	Other liabilities	11
Total derivatives designated as hedging instruments			\$ 117		\$ 49
Derivatives not designated as hedging instruments					
Foreign currency forward/option contracts	\$ 3,201	Other current assets	\$ 21	Other current liabilities	\$ 62
Total derivatives not designated as hedging instruments			\$ 21		\$ 62
Total derivative instruments			\$ 138		\$ 111

December 31, 2017 (Millions)	Gross Notional Amount	Assets		Liabilities	
		Location	Fair Value Amount	Location	Fair Value Amount
Derivatives designated as hedging instruments					
Foreign currency forward/option contracts	\$ 2,204	Other current assets	\$ 7	Other current liabilities	\$ 109
Foreign currency forward/option contracts	1,392	Other assets	20	Other liabilities	56
Interest rate swap contracts	450	Other current assets	—	Other current liabilities	1
Interest rate swap contracts	1,503	Other assets	21	Other liabilities	6
Total derivatives designated as hedging instruments			\$ 48		\$ 172
Derivatives not designated as hedging instruments					
Foreign currency forward/option contracts	\$ 4,974	Other current assets	\$ 30	Other current liabilities	\$ 25
Total derivatives not designated as hedging instruments			\$ 30		\$ 25
Total derivative instruments			\$ 78		\$ 197

Credit Risk and Offsetting of Assets and Liabilities of Derivative Instruments

The Company is exposed to credit loss in the event of nonperformance by counterparties in interest rate swaps, currency swaps, commodity price swaps, and forward and option contracts. However, the Company's risk is limited to the fair value of the instruments. The Company actively monitors its exposure to credit risk through the use of credit approvals and credit limits, and by selecting major international banks and financial institutions as counterparties. 3M enters into master netting arrangements with counterparties when possible to mitigate credit risk in derivative transactions. A master netting arrangement may allow each counterparty to net settle amounts owed between a 3M entity and the counterparty as a result of multiple, separate derivative transactions. As of September 30, 2018, 3M has International Swaps and Derivatives Association (ISDA) agreements with 17 applicable banks and financial institutions which contain netting provisions. In addition to a master agreement with 3M supported by a primary counterparty's parent guarantee, 3M also has associated credit support agreements in place with 16 of its primary derivative counterparties which, among other things, provide the circumstances under which either party is required to post eligible collateral (when the market value of transactions covered by these agreements exceeds specified thresholds or if a counterparty's credit rating has been downgraded to a predetermined rating). The Company does not anticipate nonperformance by any of these counterparties.

3M has elected to present the fair value of derivative assets and liabilities within the Company's consolidated balance sheet on a gross basis even when derivative transactions are subject to master netting arrangements and may otherwise qualify for net presentation. However, the following tables provide information as if the Company had elected to offset the asset and liability balances of derivative instruments, netted in accordance with various criteria in the event of default or termination as stipulated by the terms of netting arrangements with each of the counterparties. For each counterparty, if netted, the Company would offset the asset and liability balances of all derivatives at the end of the reporting period based on the 3M entity that is a party to the transactions. Derivatives not subject to master netting agreements are not eligible for net presentation. As of the applicable dates presented below, no collateral had been received or pledged related to these derivative instruments.

Offsetting of Financial Assets under Master Netting Agreements with Derivative Counterparties

	Gross Amount of Derivative Assets Presented in the Consolidated Balance Sheet	Gross Amounts not Offset in the Consolidated Balance Sheet that are Subject to Master Netting Agreements			Net Amount of Derivative Assets
		Gross Amount of Eligible Offsetting Recognized Derivative Liabilities	Cash Collateral Received		
September 30, 2018 (Millions)					
Derivatives subject to master netting agreements	\$ 138	\$ 45	\$ —		\$ 93
Derivatives not subject to master netting agreements	—				—
Total	\$ 138				\$ 93
December 31, 2017 (Millions)					
Derivatives subject to master netting agreements	\$ 78	\$ 27	\$ —		\$ 51
Derivatives not subject to master netting agreements	—				—
Total	\$ 78				\$ 51

Offsetting of Financial Liabilities under Master Netting Agreements with Derivative Counterparties

	Gross Amount of Derivative Liabilities Presented in the Consolidated Balance Sheet	Gross Amounts not Offset in the Consolidated Balance Sheet that are Subject to Master Netting Agreements			Net Amount of Derivative Liabilities
		Gross Amount of Eligible Offsetting Recognized Derivative Assets	Cash Collateral Pledged		
September 30, 2018 (Millions)					
Derivatives subject to master netting agreements	\$ 111	\$ 45	\$ —		\$ 66
Derivatives not subject to master netting agreements	—				—
Total	\$ 111				\$ 66
December 31, 2017 (Millions)					
Derivatives subject to master netting agreements	\$ 197	\$ 27	\$ —		\$ 170
Derivatives not subject to master netting agreements	—				—
Total	\$ 197				\$ 170

Currency Effects

3M estimates that year-on-year foreign currency transaction effects, including hedging impacts, decreased pre-tax income by approximately \$24 million and \$115 million for the three and nine months ended September 30, 2018. These estimates include transaction gains and losses, including derivative instruments designed to reduce foreign currency exchange rate risks.

NOTE 13. Fair Value Measurements

3M follows ASC 820, *Fair Value Measurements and Disclosures*, with respect to assets and liabilities that are measured at fair value on a recurring basis and nonrecurring basis. Under the standard, fair value is defined as the exit price, or the amount that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants as of the measurement date. The standard also establishes a hierarchy for inputs used in measuring fair value that maximizes the use of observable inputs and minimizes the use of unobservable inputs by requiring that the most observable inputs be used when available. Observable inputs are inputs market participants would use in valuing the asset or liability developed based on market data obtained from sources independent of the Company. Unobservable inputs are inputs that reflect the Company's assumptions about the factors market participants would use in valuing the asset or liability developed based upon the best information available in the circumstances. The hierarchy is broken down into three levels. Level 1 inputs are quoted prices (unadjusted) in active markets for identical assets or liabilities. Level 2 inputs include quoted prices for similar assets or liabilities in active markets, quoted prices for identical or similar assets or liabilities in markets that are not active, and inputs (other than quoted prices) that are observable for the asset or liability, either directly or indirectly. Level 3 inputs are unobservable inputs for the asset or liability. Categorization within the valuation hierarchy is based upon the lowest level of input that is significant to the fair value measurement.

Assets and Liabilities that are Measured at Fair Value on a Recurring Basis:

Refer to Note 14 in the Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K) for a qualitative discussion of the assets and liabilities that are measured at fair value on a recurring and nonrecurring basis, a description of the valuation methodologies used by 3M, and categorization within the valuation framework of ASC 820.

The following tables provide information by level for assets and liabilities that are measured at fair value on a recurring basis.

Description (Millions)	Fair Value at September 30, 2018	Fair Value Measurements Using Inputs Considered as		
		Level 1	Level 2	Level 3
Assets:				
Available-for-sale:				
Marketable securities:				
Commercial paper	\$ 304	\$ —	\$ 304	\$ —
Certificates of deposit/time deposits	31	—	31	—
U.S. municipal securities	30	—	—	30
Derivative instruments — assets:				
Foreign currency forward/option contracts	113	—	113	—
Interest rate swap contracts	25	—	25	—
Liabilities:				
Derivative instruments — liabilities:				
Foreign currency forward/option contracts	95	—	95	—
Interest rate swap contracts	16	—	16	—

Description (Millions)	Fair Value at December 31, 2017	Fair Value Measurements Using Inputs Considered as		
		Level 1	Level 2	Level 3
Assets:				
Available-for-sale:				
Marketable securities:				
Corporate debt securities	\$ 14	\$ —	\$ 14	\$ —
Commercial paper	899	—	899	—
Certificates of deposit/time deposits	76	—	76	—
Asset-backed securities:				
Automobile loan related	16	—	16	—
Credit card related	68	—	68	—
U.S. municipal securities	30	—	—	30
Derivative instruments — assets:				
Foreign currency forward/option contracts	57	—	57	—
Interest rate swap contracts	21	—	21	—
Liabilities:				
Derivative instruments — liabilities:				
Foreign currency forward/option contracts	190	—	190	—
Interest rate swap contracts	7	—	7	—

The following table provides a reconciliation of the beginning and ending balances of items measured at fair value on a recurring basis in the table above that used significant unobservable inputs (Level 3).

Marketable securities — certain U.S. municipal securities only (Millions)	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Beginning balance	\$ 30	\$ 20	\$ 30	\$ 20
Total gains or losses:				
Included in earnings	—	—	—	—
Included in other comprehensive income	—	—	—	—
Purchases and issuances	—	—	—	—
Sales and settlements	—	—	—	—
Transfers in and/or out of level 3	—	—	—	—
Ending balance	\$ 30	\$ 20	\$ 30	\$ 20
Change in unrealized gains or losses for the period included in earnings for securities held at the end of the reporting period	—	—	—	—

In addition, the plan assets of 3M's pension and postretirement benefit plans are measured at fair value on a recurring basis (at least annually). Refer to Note 12 in 3M's Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K).

Assets and Liabilities that are Measured at Fair Value on a Nonrecurring Basis:

Disclosures are required for certain assets and liabilities that are measured at fair value, but are recognized and disclosed at fair value on a nonrecurring basis in periods subsequent to initial recognition. For 3M, such measurements of fair value relate primarily to long-lived asset impairments and adjustment in carrying value of equity securities for which the measurement alternative of cost less impairment plus or minus observable price changes is used. There were no material long-lived asset impairments or adjustments to equity securities using the measurement alternative for the three and nine months ended September 30, 2018 and three months ended September 30, 2017. During the nine months ended September 30, 2017, the Company recognized approximately \$40 million in long-lived asset impairments related to its Electronics and Energy business segment, with the complete carrying amount of such assets written off and included in operating income results.

Fair Value of Financial Instruments:

The Company's financial instruments include cash and cash equivalents, marketable securities, accounts receivable, certain investments, accounts payable, borrowings, and derivative contracts. The fair values of cash equivalents, accounts receivable, accounts payable, and short-term borrowings and current portion of long-term debt approximated carrying values because of the short-term nature of these instruments. Available-for-sale marketable securities, in addition to certain derivative instruments, are recorded at fair values as indicated in the preceding disclosures. For its long-term debt, the Company utilized third-party quotes to estimate fair values (classified as level 2). Information with respect to the carrying amounts and estimated fair values of these financial instruments follow:

(Millions)	September 30, 2018		December 31, 2017	
	Carrying Value	Fair Value	Carrying Value	Fair Value
Long-term debt, excluding current portion	\$ 13,539	\$ 13,666	\$ 12,096	\$ 12,535

The fair values reflected above consider the terms of the related debt absent the impacts of derivative/hedging activity. The carrying amount of long-term debt referenced above is impacted by certain fixed-to-floating interest rate swaps that are designated as fair value hedges and by the designation of certain fixed rate Eurobond securities issued by the Company as hedging instruments of the Company's net investment in its European subsidiaries. A number of 3M's fixed-rate bonds were trading at a premium at September 30, 2018 and December 31, 2017 due to lower interest rates and tighter credit spreads compared to issuance levels.

NOTE 14. Commitments and Contingencies

Legal Proceedings:

The Company and some of its subsidiaries are involved in numerous claims and lawsuits, principally in the United States, and regulatory proceedings worldwide. These include various products liability (involving products that the Company now or formerly manufactured and sold), intellectual property, and commercial claims and lawsuits, including those brought under the antitrust laws, and environmental proceedings. Unless otherwise stated, the Company is vigorously defending all such litigation. Additional information about the Company's process for disclosure and recording of liabilities and insurance receivables related to legal proceedings can be found in Note 15 "Commitments and Contingencies" in the Company's Annual Report on Form 10-K for the year ended December 31, 2017 as updated by the Company's Current Report on Form 8-K dated May 8, 2018.

The following sections first describe the significant legal proceedings in which the Company is involved, and then describe the liabilities and associated insurance receivables the Company has accrued relating to its significant legal proceedings.

Respirator Mask/Asbestos Litigation

As of September 30, 2018, the Company is a named defendant, with multiple co-defendants, in numerous lawsuits in various courts that purport to represent approximately 2,255 individual claimants, compared to approximately 2,230 individual claimants with actions pending at December 31, 2017.

The vast majority of the lawsuits and claims resolved by and currently pending against the Company allege use of some of the Company's mask and respirator products and seek damages from the Company and other defendants for alleged personal injury from workplace exposures to asbestos, silica, coal mine dust or other occupational dusts found in products manufactured by other defendants or generally in the workplace. A minority of the lawsuits and claims resolved by and currently pending against the Company generally allege personal injury from occupational exposure to asbestos from products previously manufactured by the Company, which are often unspecified, as well as products manufactured by other defendants, or occasionally at Company premises.

The Company's current volume of new and pending matters is substantially lower than it experienced at the peak of filings in 2003. The Company expects that filing of claims by unimpaired claimants in the future will continue to be at much lower levels than in the past. Accordingly, the number of claims alleging more serious injuries, including mesothelioma and other malignancies, will represent a greater percentage of total claims than in the past. The Company has prevailed in thirteen of the fourteen cases taken to trial, including eleven of the twelve cases tried to verdict (such trials occurred in 1999, 2000, 2001, 2003, 2004, 2007, 2015, and the cases tried in 2016, 2017, and 2018-described below), and an appellate reversal in 2005 of the 2001 jury verdict adverse to the Company. The remaining case, tried in 2009, was dismissed by the court at the close of plaintiff's evidence, based on the court's legal finding that the plaintiff had not presented sufficient evidence to support a jury verdict. In August 2016, 3M received a unanimous verdict in its favor from a jury in state court in Kentucky, in 3M's first respirator trial involving coal mine dust. The estate of the plaintiff alleged that the 3M 8710 respirator is defective and caused his death because it did not protect him from harmful coal mine dust. The jury rejected plaintiff's claim and returned a verdict finding no liability against 3M. The verdict is final as the plaintiff did not file an appeal. In September 2017, 3M received a unanimous verdict in its favor from a jury in state court in Kentucky in 3M's second respirator trial involving coal mine dust. The jury ultimately determined that the plaintiff's claims were barred by the statute of limitations. In November 2017, the court denied the plaintiff's motion for a new trial. The plaintiff did not file an appeal, thereby ending the litigation. In February 2018, 3M received a verdict in its favor from a jury in state court in California. The plaintiff alleged that the 3M 8710 respirator was defective and caused his mesothelioma because it did not protect him from asbestos fibers. The jury rejected plaintiff's claim and returned a verdict finding no liability against 3M. In April 2018, a jury in state court in Kentucky found 3M's 8710 respirators failed to protect two coal miners from coal mine dust and awarded aggregate compensatory damages of approximately \$2 million and punitive damages totaling \$63 million. In August 2018, the trial court entered judgment and the Company has appealed. The Company believes liability in this case is not probable and estimable. In June 2018, the Company also settled a number of coal mine dust lawsuits for an amount that was not material to the Company.

The Company has demonstrated in these past trial proceedings that its respiratory protection products are effective as claimed when used in the intended manner and in the intended circumstances. Consequently, the Company believes that claimants are unable to establish that their medical conditions, even if significant, are attributable to the Company's respiratory protection products. Nonetheless the Company's litigation experience indicates that claims of persons with malignant conditions are costlier to resolve than

the claims of unimpaired persons, and it therefore believes the average cost of resolving pending and future claims on a per-claim basis will continue to be higher than it experienced in prior periods when the vast majority of claims were asserted by medically unimpaired claimants.

As previously reported, the State of West Virginia, through its Attorney General, filed a complaint in 2003 against the Company and two other manufacturers of respiratory protection products in the Circuit Court of Lincoln County, West Virginia, and amended its complaint in 2005. The amended complaint seeks substantial, but unspecified, compensatory damages primarily for reimbursement of the costs allegedly incurred by the State for worker's compensation and healthcare benefits provided to all workers with occupational pneumoconiosis and unspecified punitive damages. The case was inactive from the fourth quarter of 2007 until late 2013, other than a case management conference in March 2011. In November 2013, the State filed a motion to bifurcate the lawsuit into separate liability and damages proceedings. At the hearing on the motion, the court declined to bifurcate the lawsuit. No liability has been recorded for this matter because the Company believes that liability is not probable and estimable at this time. In addition, the Company is not able to estimate a possible loss or range of loss given the lack of any meaningful discovery responses by the State of West Virginia, the otherwise minimal activity in this case and the fact that the complaint asserts claims against two other manufacturers where a defendant's share of liability may turn on the law of joint and several liability and by the amount of fault, if any, a jury might allocate to each defendant if the case is ultimately tried.

Respirator Mask/Asbestos Liabilities and Insurance Receivables:

The Company regularly conducts a comprehensive legal review of its respirator mask/asbestos liabilities. The Company reviews recent and historical claims data, including without limitation, (i) the number of pending claims filed against the Company, (ii) the nature and mix of those claims (i.e., the proportion of claims asserting usage of the Company's mask or respirator products and alleging exposure to each of asbestos, silica, coal or other occupational dusts, and claims pleading use of asbestos-containing products allegedly manufactured by the Company), (iii) the costs to defend and resolve pending claims, and (iv) trends in filing rates and in costs to defend and resolve claims, (collectively, the "Claims Data"). As part of its comprehensive legal review, the Company regularly provides the Claims Data to a third party with expertise in determining the impact of Claims Data on future filing trends and costs. The third party assists the Company in estimating the costs to defend and resolve pending and future claims. The Company uses these estimates to develop its best estimate of probable liability.

Developments may occur that could affect the Company's estimate of its liabilities. These developments include, but are not limited to, significant changes in (i) the key assumptions underlying the Company's accrual, including, the number of future claims, the nature and mix of those claims, the average cost of defending and resolving claims, and in maintaining trial readiness (ii) trial and appellate outcomes, (iii) the law and procedure applicable to these claims, and (iv) the financial viability of other co-defendants and insurers.

As a result of the Company's review of its respirator mask/asbestos liabilities and as a result of the cost of resolving claims of persons who claim more serious injuries, including mesothelioma and other malignancies, the Company increased its accruals in the first nine months of 2018 for respirator mask/asbestos liabilities by \$56 million. In the first nine months of 2018, the Company made payments for legal fees and settlements of \$64 million related to the respirator mask/asbestos litigation. As of September 30, 2018, the Company had an accrual for respirator mask/asbestos liabilities (excluding Aearo accruals) of \$600 million. This accrual represents the Company's best estimate of probable loss and reflects an estimation period for future claims that may be filed against the Company approaching the year 2050. The Company cannot estimate the amount or upper end of the range of amounts by which the liability may exceed the accrual the Company has established because of the (i) inherent difficulty in projecting the number of claims that have not yet been asserted or the time period in which future claims may be asserted, (ii) the complaints nearly always assert claims against multiple defendants where the damages alleged are typically not attributed to individual defendants so that a defendant's share of liability may turn on the law of joint and several liability, which can vary by state, (iii) the multiple factors described above that the Company considers in estimating its liabilities, and (iv) the several possible developments described above that may occur that could affect the Company's estimate of liabilities.

As of September 30, 2018, the Company's receivable for insurance recoveries related to the respirator mask/asbestos litigation was \$4 million. The Company continues to seek coverage under the policies of certain insolvent and other insurers. Once those claims for coverage are resolved, the Company will have collected substantially all of its remaining insurance coverage for respirator mask/asbestos claims.

Respirator Mask/Asbestos Litigation — Aearo Technologies

On April 1, 2008, a subsidiary of the Company purchased the stock of Aearo Holding Corp., the parent of Aearo Technologies (“Aearo”). Aearo manufactured and sold various products, including personal protection equipment, such as eye, ear, head, face, fall and certain respiratory protection products.

As of September 30, 2018, Aearo and/or other companies that previously owned and operated Aearo’s respirator business (American Optical Corporation, Warner-Lambert LLC, AO Corp. and Cabot Corporation (“Cabot”)) are named defendants, with multiple co-defendants, including the Company, in numerous lawsuits in various courts in which plaintiffs allege use of mask and respirator products and seek damages from Aearo and other defendants for alleged personal injury from workplace exposures to asbestos, silica-related, or other occupational dusts found in products manufactured by other defendants or generally in the workplace.

As of September 30, 2018, the Company, through its Aearo subsidiary, had accruals of \$29 million for product liabilities and defense costs related to current and future Aearo-related asbestos and silica-related claims. This accrual represents the Company’s best estimate of Aearo’s probable loss and reflects an estimation period for future claims that may be filed against Aearo approaching the year 2050. Responsibility for legal costs, as well as for settlements and judgments, is currently shared in an informal arrangement among Aearo, Cabot, American Optical Corporation and a subsidiary of Warner Lambert and their respective insurers (the “Payor Group”). Liability is allocated among the parties based on the number of years each company sold respiratory products under the “AO Safety” brand and/or owned the AO Safety Division of American Optical Corporation and the alleged years of exposure of the individual plaintiff. Aearo’s share of the contingent liability is further limited by an agreement entered into between Aearo and Cabot on July 11, 1995. This agreement provides that, so long as Aearo pays to Cabot a quarterly fee of \$100,000, Cabot will retain responsibility and liability for, and indemnify Aearo against, any product liability claims involving exposure to asbestos, silica, or silica products for respirators sold prior to July 11, 1995. Because of the difficulty in determining how long a particular respirator remains in the stream of commerce after being sold, Aearo and Cabot have applied the agreement to claims arising out of the alleged use of respirators involving exposure to asbestos, silica or silica products prior to January 1, 1997. With these arrangements in place, Aearo’s potential liability is limited to exposures alleged to have arisen from the use of respirators involving exposure to asbestos, silica, or silica products on or after January 1, 1997. To date, Aearo has elected to pay the quarterly fee. Aearo could potentially be exposed to additional claims for some part of the pre-July 11, 1995 period covered by its agreement with Cabot if Aearo elects to discontinue its participation in this arrangement, or if Cabot is no longer able to meet its obligations in these matters.

Developments may occur that could affect the estimate of Aearo’s liabilities. These developments include, but are not limited to: (i) significant changes in the number of future claims, (ii) significant changes in the average cost of resolving claims, (iii) significant changes in the legal costs of defending these claims, (iv) significant changes in the mix and nature of claims received, (v) trial and appellate outcomes, (vi) significant changes in the law and procedure applicable to these claims, (vii) significant changes in the liability allocation among the co-defendants, (viii) the financial viability of members of the Payor Group including exhaustion of available insurance coverage limits, and/or (ix) a determination that the interpretation of the contractual obligations on which Aearo has estimated its share of liability is inaccurate. The Company cannot determine the impact of these potential developments on its current estimate of Aearo’s share of liability for these existing and future claims. If any of the developments described above were to occur, the actual amount of these liabilities for existing and future claims could be significantly larger than the amount accrued.

Because of the inherent difficulty in projecting the number of claims that have not yet been asserted, the complexity of allocating responsibility for future claims among the Payor Group, and the several possible developments that may occur that could affect the estimate of Aearo’s liabilities, the Company cannot estimate the amount or range of amounts by which Aearo’s liability may exceed the accrual the Company has established.

Environmental Matters and Litigation

The Company’s operations are subject to environmental laws and regulations including those pertaining to air emissions, wastewater discharges, toxic substances, and the handling and disposal of solid and hazardous wastes enforceable by national, state, and local authorities around the world, and private parties in the United States and abroad. These laws and regulations provide, under certain circumstances, a basis for the remediation of contamination, for restoration of or compensation for damages to natural resources, and for personal injury and property damage claims. The Company has incurred, and will continue to incur, costs and capital expenditures in complying with these laws and regulations, defending personal injury and property damage claims, and modifying its business operations in light of its environmental responsibilities. In its effort to satisfy its environmental responsibilities and comply with

environmental laws and regulations, the Company has established, and periodically updates, policies relating to environmental standards of performance for its operations worldwide.

Under certain environmental laws, including the United States Comprehensive Environmental Response, Compensation and Liability Act of 1980 (CERCLA) and similar state laws, the Company may be jointly and severally liable, typically with other companies, for the costs of remediation of environmental contamination at current or former facilities and at off-site locations. The Company has identified numerous locations, most of which are in the United States, at which it may have some liability. Please refer to the section entitled “*Environmental Liabilities and Insurance Receivables*” that follows for information on the amount of the accrual.

Environmental Matters

As previously reported, the Company has been voluntarily cooperating with ongoing reviews by local, state, federal (primarily the U.S. Environmental Protection Agency (EPA)), and international agencies of possible environmental and health effects of various perfluorinated compounds, including perfluorooctanoate (“PFOA”), perfluorooctane sulfonate (“PFOS”), perfluorohexane sulfonate (“PFHxS”), or other per- and polyfluoroalkyl substances (collectively “PFAS”). As a result of its phase-out decision in May 2000, the Company no longer manufactures certain PFAS compounds referred to as “perfluorooctanyl compounds,” including PFOA, PFOS, PFHxS, and their pre-cursor compounds. The company ceased manufacturing and using the vast majority of the perfluorooctanyl compounds within approximately two years of the phase-out announcement, and ceased all manufacturing and the last significant use of this chemistry by the end of 2008. Through its ongoing life cycle management and its raw material composition identification processes associated with the Company’s policies covering the use of all persistent and bio-accumulative materials, the Company continues to control or eliminate the presence of certain PFAS in purchased materials or as byproducts in some of 3M’s current fluorochemical manufacturing processes, products, and waste streams.

Regulatory activities concerning PFOA and/or PFOS continue in the United States, Europe and elsewhere, and before certain international bodies. These activities include gathering of exposure and use information, risk assessment, and consideration of regulatory approaches. As the database of studies of both PFOA and PFOS has expanded, the EPA has developed human health effects documents summarizing the available data from these studies. In February 2014, the EPA initiated external peer review of its draft human health effects documents for PFOA and PFOS. The peer review panel met in August 2014. In May 2016, the EPA announced lifetime health advisory levels for PFOA and PFOS at 70 parts per trillion (ppt) (superseding the provisional levels established by the EPA in 2009 of 400 ppt for PFOA and 200 ppt for PFOS). Where PFOA and PFOS are found together, EPA recommends that the concentrations be added together, and the lifetime health advisory for PFOA and PFOS combined is also 70 ppt. Lifetime health advisories, which are non-enforceable and non-regulatory, provide information about concentrations of drinking water contaminants at which adverse health effects are not expected to occur over the specified exposure duration. In an effort to collect exposure information under the Safe Drinking Water Act, the EPA published on May 2, 2012 a list of unregulated substances, including six PFAS, required to be monitored during the period 2013-2015 by public water system suppliers to determine the extent of their occurrence. Through January 2017, the EPA reported results for 4,920 public water supplies nationwide. Based on the 2016 lifetime health advisory, 13 public water supplies exceed the level for PFOA and 46 exceed the level for PFOS (unchanged from the July 2016 EPA summary). A technical advisory issued by EPA in September 2016 on laboratory analysis of drinking water samples stated that 65 public water supplies had exceeded the combined level for PFOA and PFOS. These results are based on one or more samples collected during the period 2012-2015 and do not necessarily reflect current conditions of these public water supplies. EPA reporting does not identify the sources of the PFOA and PFOS in the public water supplies.

The Company is continuing to make progress in its work, under the supervision of state regulators, to address its historic disposal of PFAS-containing waste associated with manufacturing operations at the Decatur, Alabama, Cottage Grove, Minnesota, and Cordova, Illinois plants. As previously reported, the Company entered into a voluntary remedial action agreement with the Alabama Department of Environmental Management (ADEM) to address the presence of PFAS in the soil at the Company’s manufacturing facility in Decatur, Alabama. Pursuant to a permit issued by ADEM, for approximately twenty years, the Company incorporated its wastewater treatment plant sludge containing PFAS in fields at its Decatur facility. After a review of the available options to address the presence of PFAS in the soil, ADEM agreed that the preferred remediation option is to use a multilayer cap over the former sludge incorporation areas on the manufacturing site with subsequent groundwater migration controls and treatment. Implementation of that plan continues and is expected to be completed in 2018.

The Company continues to work with the Minnesota Pollution Control Agency (MPCA) pursuant to the terms of the previously disclosed May 2007 Settlement Agreement and Consent Order to address the presence of certain PFAS in the soil and groundwater at

former disposal sites in Washington County, Minnesota (Oakdale and Woodbury) and at the Company's manufacturing facility at Cottage Grove, Minnesota. Under this agreement, the Company's principal obligations include (i) evaluating releases of certain PFAS from these sites and proposing response actions; (ii) providing treatment or alternative drinking water upon identifying any level exceeding a Health Based Value ("HBV") or Health Risk Limit ("HRL") (i.e., the amount of a chemical in drinking water determined by the Minnesota Department of Health (MDH) to be safe for human consumption over a lifetime) for certain PFAS for which a HBV and/or HRL exists as a result of contamination from these sites; (iii) remediating identified sources of other PFAS at these sites that are not controlled by actions to remediate PFOA and PFOS; and (iv) sharing information with the MPCA about certain perfluorinated compounds. During 2008, the MPCA issued formal decisions adopting remedial options for the former disposal sites in Washington County, Minnesota (Oakdale and Woodbury). In August 2009, the MPCA issued a formal decision adopting remedial options for the Company's Cottage Grove manufacturing facility. During the spring and summer of 2010, 3M began implementing the agreed upon remedial options at the Cottage Grove and Woodbury sites. 3M commenced the remedial option at the Oakdale site in late 2010. At each location the remedial options were recommended by the Company and approved by the MPCA. Remediation work has been completed at the Oakdale and Woodbury sites, and they are in an operational maintenance mode. Remediation will continue at the Cottage Grove site during 2018.

In August 2014, the Illinois EPA approved a request by the Company to establish a groundwater management zone at its manufacturing facility in Cordova, Illinois, which includes ongoing pumping of impacted site groundwater, groundwater monitoring and routine reporting of results.

In May 2017, the MDH issued new HBVs for PFOS and PFOA. The new HBVs are 35 ppt for PFOA and 27 ppt for PFOS. In connection with its announcement the MDH stated that "Drinking water with PFOA and PFOS, even at the levels above the updated values, does not represent an immediate health risk. These values are designed to reduce long-term health risks across the population and are based on multiple safety factors to protect the most vulnerable citizens, which makes them overprotective for most of the residents in our state." In December 2017, the MDH issued a new HBV for perfluorobutane sulfonate (PFBS) of 2 ppb. In February 2018, the MDH published reports finding no unusual rates of certain cancers or adverse birth outcomes (low birth rates or premature births) among residents of Washington and Dakota counties in Minnesota.

The EPA announced a four-step action plan in May 2018 regarding PFAS, which includes evaluating the need to set Safe Drinking Water Act maximum contaminant levels (MCLs) for PFOA and PFOS and beginning the steps necessary to designate PFOA and PFOS as "hazardous substances" under CERCLA.

The U.S. Agency for Toxic Substances and Disease Registry (ATSDR) within the Department of Health and Human Services released a draft Toxicological Profile for PFAS for public review and comment in June 2018. In the draft report, ATSDR proposed draft Minimal Risk Levels (MRLs) for PFOS, PFOA and several other PFAS. An MRL is an estimate of the daily human exposure to a hazardous substance that is likely to be without appreciable risk of adverse non-cancer health effects over a specified duration of exposure. MRLs are not intended to define cleanup or action levels for ATSDR or other agencies. In August 2018, 3M submitted comments on the ATSDR proposal, noting that there are major shortcomings with the current draft, especially with the MRLs, and that the ATSDR's profile must reflect the best science and full weight of evidence known about these chemicals.

In several states, the state legislature or the state environmental agency have been considering actions that would evaluate cleanup standards, groundwater values or drinking water values for PFOS, PFOA and other PFAS.

The Company cannot predict what additional regulatory actions arising from the foregoing proceedings and activities, if any, may be taken regarding such compounds or the consequences of any such actions.

Litigation Related to Historical PFAS Manufacturing Operations in Alabama

As previously reported, a former employee filed a purported class action lawsuit in 2002 in the Circuit Court of Morgan County, Alabama (the "St. John case"), seeking unstated damages and alleging that the plaintiffs suffered fear, increased risk, subclinical injuries, and property damage from exposure to certain perfluorochemicals at or near the Company's Decatur, Alabama, manufacturing facility. The court in 2005 granted the Company's motion to dismiss the named plaintiff's personal injury-related claims on the basis that such claims are barred by the exclusivity provisions of the state's Workers Compensation Act. The plaintiffs' counsel filed an amended complaint in November 2006, limiting the case to property damage claims on behalf of a purported class of residents and property owners in the vicinity of the Decatur plant. In June 2015, the plaintiffs filed an amended complaint adding

additional defendants, including BFI Waste Management Systems of Alabama, LLC; BFI Waste Management of North America, LLC; the City of Decatur, Alabama; Morgan County, Alabama; Municipal Utilities Board of Decatur; and Morgan County, Alabama, d/b/a Decatur Utilities.

In 2005, the judge – in a second purported class action lawsuit filed by three residents of Morgan County, Alabama, seeking unstated compensatory and punitive damages involving alleged damage to their property from emissions of certain perfluorochemical compounds from the Company’s Decatur, Alabama, manufacturing facility that formerly manufactured those compounds (the “Chandler case”) – granted the Company’s motion to abate the case, effectively putting the case on hold pending the resolution of class certification issues in the St. John case. Despite the stay, plaintiffs filed an amended complaint seeking damages for alleged personal injuries and property damage on behalf of the named plaintiffs and the members of a purported class. No further action in the case is expected unless and until the stay is lifted.

In February 2009, a resident of Franklin County, Alabama, filed a purported class action lawsuit in the Circuit Court of Franklin County (the “Stover case”) seeking compensatory damages and injunctive relief based on the application by the Decatur utility’s wastewater treatment plant of wastewater treatment sludge to farmland and grasslands in the state that allegedly contain PFOA, PFOS and other perfluorochemicals. The named plaintiff seeks to represent a class of all persons within the State of Alabama who have had PFOA, PFOS, and other perfluorochemicals released or deposited on their property. In March 2010, the Alabama Supreme Court ordered the case transferred from Franklin County to Morgan County. In May 2010, consistent with its handling of the other matters, the Morgan County Circuit Court abated this case, putting it on hold pending the resolution of the class certification issues in the St. John case.

In October 2015, West Morgan-East Lawrence Water & Sewer Authority (Water Authority) filed an individual complaint against 3M Company, Dyneon, L.L.C., and Daikin America, Inc., in the U.S. District Court for the Northern District of Alabama. The complaint also includes representative plaintiffs who brought the complaint on behalf of themselves, and a class of all owners and possessors of property who use water provided by the Water Authority and five local water works to which the Water Authority supplies water (collectively, the “Water Utilities”). The complaint seeks compensatory and punitive damages and injunctive relief based on allegations that the defendants’ chemicals, including PFOA and PFOS from their manufacturing processes in Decatur, have contaminated the water in the Tennessee River at the water intake, and that the chemicals cannot be removed by the water treatment processes utilized by the Water Authority. In September 2016, the court granted 3M’s motion to dismiss plaintiffs’ trespass claims with prejudice, negligence claims for personal injuries, and private nuisance claims, and denied the motion to dismiss the plaintiffs’ negligence claims for property damage, public nuisance, abatement of nuisance, battery and wantonness.

In June 2016, the Tennessee Riverkeeper, Inc. (Riverkeeper), a non-profit corporation, filed a lawsuit in the U.S. District Court for the Northern District of Alabama against 3M; BFI Waste Systems of Alabama; the City of Decatur, Alabama; and the Municipal Utilities Board of Decatur, Morgan County, Alabama. The complaint alleges that the defendants violated the Resource Conservation and Recovery Act in connection with the disposal of certain PFAS through their ownership and operation of their respective sites. The complaint further alleges such practices may present an imminent and substantial endangerment to health and/or the environment and that Riverkeeper has suffered and will continue to suffer irreparable harm caused by defendants’ failure to abate the endangerment unless the court grants the requested relief, including declaratory and injunctive relief.

In August 2016, a group of over 200 plaintiffs filed a class action against West Morgan-East Lawrence Water and Sewer Authority (Water Authority), 3M, Dyneon, Daikin, BFI, and the City of Decatur in state court in Lawrence County, Alabama. Plaintiffs are residents of Lawrence, Morgan and other counties who are or have been customers of the Water Authority. They contend defendants have released PFAS that contaminate the Tennessee River and, in turn, their drinking water, causing damage to their health and properties. In January 2017, the court in the St. John case, discussed above, stayed this litigation pending resolution of the St. John case.

In January 2017, several hundred plaintiffs sued 3M, its subsidiary Dyneon, and Daikin America in Lawrence and Morgan Counties, Alabama. The plaintiffs are owners of property, residents, and holders of property interests who receive their water from the West Morgan-East Lawrence Water and Sewer Authority (Water Authority). They assert common law claims for negligence, nuisance, trespass, wantonness, and battery, and they seek injunctive relief and punitive damages. The plaintiffs contend that the defendants own and operate manufacturing and disposal facilities in Decatur that have released and continue to release PFOA, PFOS and related chemicals into the groundwater and surface water of their sites, resulting in discharge into the Tennessee River. The plaintiffs also contend that the defendants have discharged into Bakers Creek and the Decatur Utilities Dry Creek Wastewater Treatment Plant,

which, in turn, discharges wastewater containing these chemicals into the Tennessee River. The plaintiffs contend that, as a result of the alleged discharges, the water supplied by the Water Authority to the plaintiffs was, and is, contaminated with PFOA, PFOS, and related chemicals at a level dangerous to humans.

In November 2017, a purported class action (the “King” case) was filed against 3M, its subsidiary Dyneon, Daikin America, and the West Morgan-East Lawrence Water and Sewer Authority (Water Authority) in the U.S. District Court for the Northern District of Alabama. The plaintiffs are residents of Lawrence and Morgan County, Alabama who receive their water from the Water Authority. They assert various common law claims, including negligence, nuisance, wantonness, and fraudulent concealment, and they seek injunctive relief, attorneys’ fees, compensatory and punitive damages for their alleged personal injuries. The plaintiffs contend that the defendants own and operate manufacturing and disposal facilities in Decatur that have released and continue to release PFOA, PFOS and related chemicals into the groundwater and surface water of their sites, resulting in discharge into the Tennessee River. The plaintiffs also contend that the defendants have discharged chemicals into the Decatur Utilities Dry Creek Wastewater Treatment Plant, which, in turn, discharged wastewater containing these chemicals into the Tennessee River. The plaintiffs contend that, as a result of the alleged discharges, the water supplied by the Water Authority to the plaintiffs was, and is, contaminated with PFOA, PFOS, and related chemicals at a level dangerous to humans.

In January 2018, certain property owners in Trinity, Alabama filed a lawsuit against 3M, Dyneon, and three unnamed defendants in the U.S. District Court for the Northern District of Alabama. The plaintiffs assert claims for negligence, strict liability, trespass, nuisance, wanton and reckless conduct, and citizen suit claims for violation of the Resource Conservation and Recovery Act. They allege these claims arise from the defendants’ contamination of their property by disposal of PFAS in a landfill located on their property. The plaintiffs seek compensatory and punitive damages and a court order directing the defendants to remediate all PFAS contamination on their property. In September 2018, the case was dismissed by stipulation of the parties.

In September 2018, an individual plaintiff filed a lawsuit in the U.S. District Court for the Northern District of Alabama raising allegations and claims substantially similar to those asserted by plaintiffs in the King case. 3M has not yet been served in this case.

Litigation Related to Historical PFAS Manufacturing Operations in Minnesota

In July 2016, the City of Lake Elmo filed a lawsuit in the U.S. District Court for the District of Minnesota against 3M alleging that the City suffered damages from drinking water supplies contaminated with PFAS, including costs to construct alternative sources of drinking water. Trial is scheduled to begin in September 2019.

State Attorneys General Litigation related to PFAS

In December 2010, the State of Minnesota, by its Attorney General, filed a lawsuit in Hennepin County District Court against 3M to recover damages (including unspecified assessment costs and reasonable attorney’s fees) for alleged injury to, destruction of, and loss of use of certain of the State’s natural resources under the Minnesota Environmental Response and Liability Act (MERLA) and the Minnesota Water Pollution Control Act (MWPCA), as well as statutory nuisance and common law claims of trespass, nuisance, and negligence with respect to the presence of PFAS in the groundwater, surface water, fish or other aquatic life, and sediments (the “NRD Lawsuit”). The State also sought declarations under MERLA that 3M is responsible for all damages the State may suffer in the future for injuries to natural resources from releases of PFAS into the environment, and that 3M is responsible for compensation for future loss or destruction of fish, aquatic life, and other damages under the MWPCA. In September 2017, the State’s damages expert submitted a report that contended the State incurred \$5 billion in damages. In November 2017, the State of Minnesota filed a motion for leave to amend its complaint to seek punitive damages from 3M, and 3M filed a motion for summary judgment contending, among other things, that the State’s claims were barred by the applicable statute of limitations. In December 2017, the court urged the parties to attempt to resolve the litigation before trial, and in January 2018, the court appointed a mediator to facilitate that process. In February 2018, 3M and the State of Minnesota reached a resolution of the NRD Lawsuit. Under the terms of the settlement, 3M agreed to provide an \$850 million grant to the State for a special “3M Water Quality and Sustainability Fund.” This Fund will enable projects that support water sustainability in the Twin Cities East Metro region, such as continued delivery of water to residents and enhancing groundwater recharge to support sustainable growth. The projects will also result in habitat and recreation improvements, such as fishing piers, trails, and open space preservation. 3M recorded a pre-tax charge of \$897 million, inclusive of legal fees and other related obligations, in the first quarter of 2018 associated with the resolution of this matter.

In June 2018, the State of New York, by its Attorney General, filed a lawsuit in Albany County Supreme Court against 3M, Tyco Fire Products LP, Chemguard, Inc., Buckeye Fire Equipment Co., National Foam, Inc., and Kidde-Fenwal, Inc., seeking to recover (1) the costs incurred in responding to the contamination caused by Aqueous Film Forming Foam (AFFF) manufactured by 3M and others; (2) damages for injury to, destruction of, and loss of the State's natural resources and related recreational series; and (3) property damage.

In July 2018, the governor of Michigan requested that the Michigan Attorney General file a lawsuit against 3M and others related to PFAS in a public letter. The Michigan Attorney General has not yet determined whether he will do so.

Aqueous Film Forming Foam (AFFF) Environmental Litigation

3M manufactured and marketed AFFF for use in firefighting at airports and military bases from approximately 1963 to 2000. As of September 30, 2018, 75 putative class action and other lawsuits have been filed against 3M and other defendants in various state and federal courts in Colorado, Delaware, Florida, Massachusetts, New York, Pennsylvania, and Washington where current or former airports, military bases, or fire training facilities are or were located. In these cases, plaintiffs typically allege that certain PFAS used in AFFF contaminated the soil and groundwater where AFFF was used and seek damages for loss of use and enjoyment of properties, diminished property values, investigation costs, remediation costs, and in some cases, personal injury and funds for medical monitoring. Several companies have been sued along with 3M, including Ansul Co. (acquired by Tyco, Inc.), Angus Fire, Buckeye Fire Protection Co., Chemguard, National Foam, Inc., and United Technologies Corp.

Other PFAS-related Environmental Litigation

3M manufactured and marketed certain products containing PFAS for use by customers in their manufacturing process. As of September 30, 2018, the following cases have been filed against 3M, another manufacturer, and their customers in connection with the use and disposal of these products.

In September 2017, three complaints were filed in the U.S. District Court for the Northern District of New York against 3M, Saint-Gobain Performance Plastics Corp. ("Saint-Gobain"), Honeywell International Inc. ("Honeywell") and E.I. DuPont De Nemours and Company. Plaintiffs allege that 3M manufactured and sold PFOA that was used for manufacturing purposes at Saint-Gobain's and Honeywell's facilities located in the Village of Hoosick Falls and the Town of Hoosick. Plaintiffs claim that the drinking water around Hoosick Falls became contaminated with unsafe levels of PFOA due to the activities of the defendants, and allege that they suffered bodily injury due to the ingestion and inhalation of PFOA. Plaintiffs seek unstated compensatory, consequential, and punitive damages, as well as attorneys' fees and costs.

In December 2017, plaintiffs filed a punitive class action in federal court against 3M, Wolverine World Wide and Waste Management, Inc., alleging negligence, trespass, intentional and negligent infliction of emotional distress, battery, products liability, public and private nuisance, fraudulent concealment, and unjust enrichment. Each count was filed against each defendant. The action arises from Wolverine's allegedly improper disposal of materials and wastes related to their shoe manufacturing operations. Plaintiffs allege Wolverine used 3M Scotchgard in its manufacturing process and that chemicals from 3M's product have contaminated the environment after being disposed of near drinking water sources. In addition to the federal court class action, as of September 30, 2018, 3M has been named a defendant in 122 private individual actions in Michigan state court based on similar allegations. 3M is aware of additional cases filed against Wolverine in Michigan state court in which 3M is likely to be added as a defendant in the future.

In Alabama, 3M is defending two lawsuits filed in state court by local public water suppliers relating to 3M's sale of PFAS-containing products to carpet manufacturers in Georgia. The plaintiffs in these cases allege that the carpet manufacturers improperly discharged PFAS into the surface water and groundwater, contaminating drinking water supplies of cities located downstream along the Coosa River.

Other Environmental Litigation

In July 2018, the Company, along with more than 120 other companies, was served with a complaint seeking cost recovery and contribution towards the cleaning up of approximately eight miles of the Lower Passaic River in New Jersey. The plaintiff, Occidental Chemical Corporation, alleges that it agreed to design and pay the estimated \$165 million cost to remove and cap sediment containing

eight chemicals of concern, including PCBs and dioxins. The complaint seeks to spread those costs among the defendants, including the Company. The Company's involvement in the case relates to its past use of two commercial drum conditioning facilities in New Jersey. Whether, and to what extent, the Company may be required to contribute to the costs at issue in the case remains to be determined.

For environmental litigation matters described above, no liability has been recorded as the Company believes liability in those matters is not probable and estimable and the Company is not able to estimate a possible loss or range of loss at this time.

Environmental Liabilities and Insurance Receivables

As of September 30, 2018, the Company had recorded liabilities of \$29 million for estimated "environmental remediation" costs based upon an evaluation of currently available facts with respect to each individual site and also recorded related insurance receivables of \$8 million. The Company records liabilities for remediation costs on an undiscounted basis when they are probable and reasonably estimable, generally no later than the completion of feasibility studies or the Company's commitment to a plan of action. Liabilities for estimated costs of environmental remediation, depending on the site, are based primarily upon internal or third-party environmental studies, and estimates as to the number, participation level and financial viability of any other potentially responsible parties, the extent of the contamination and the nature of required remedial actions. The Company adjusts recorded liabilities as further information develops or circumstances change. The Company expects that it will pay the amounts recorded over the periods of remediation for the applicable sites, currently ranging up to 20 years.

As of September 30, 2018, the Company had recorded liabilities of \$69 million for "other environmental liabilities" based upon an evaluation of currently available facts to implement the Settlement Agreement and Consent Order with the MPCA (including the best estimate of the probable liability under the settlement of the NRD Lawsuit for interim treatment of municipal and private wells), the remedial action agreement with ADEM, as well as presence in the soil and groundwater at the Company's manufacturing facilities in Decatur, Alabama, and Cottage Grove, Minnesota, and at two former disposal sites in Washington County, Minnesota (Oakdale and Woodbury). The Company expects that most of the spending will occur over the next four years.

It is difficult to estimate the cost of environmental compliance and remediation given the uncertainties regarding the interpretation and enforcement of applicable environmental laws and regulations, the extent of environmental contamination and the existence of alternative cleanup methods. Developments may occur that could affect the Company's current assessment, including, but not limited to: (i) changes in the information available regarding the environmental impact of the Company's operations and products; (ii) changes in environmental regulations, changes in permissible levels of specific compounds in drinking water sources, or changes in enforcement theories and policies, including efforts to recover natural resource damages; (iii) new and evolving analytical and remediation techniques; (iv) success in allocating liability to other potentially responsible parties; and (v) the financial viability of other potentially responsible parties and third-party indemnitors. For sites included in both "environmental remediation liabilities" and "other environmental liabilities," at which remediation activity is largely complete and remaining activity relates primarily to operation and maintenance of the remedy, including required post-remediation monitoring, the Company believes the exposure to loss in excess of the amount accrued would not be material to the Company's consolidated results of operations or financial condition. However, for locations at which remediation activity is largely ongoing, the Company cannot estimate a possible loss or range of loss in excess of the associated established accruals for the reasons described above.

Other Matters

Department of Labor Investigation

The U.S. Department of Labor (DOL) notified 3M in April 2015 that it had commenced an investigation of 3M's pension plan pursuant to the federal Employee Retirement Income Security Act of 1974, as amended (ERISA). The DOL has stated its investigation relates to certain private equity investments, plan expenses, securities lending, and distributions of plan benefits. In response to certain DOL requests, 3M produced documents and made employees available for interviews. In December 2016, the DOL issued certain subpoenas to 3M and 3M Investment Management Corp. relating to this investigation. 3M has produced additional responsive documents and is cooperating with the DOL in its investigation. In June 2018, the DOL issued a letter indicating that it did not intend to take further action.

Product Liability Litigation

As of September 30, 2018, the Company is a named defendant in lawsuits involving approximately 4,955 plaintiffs (compared to approximately 4,270 plaintiffs at December 31, 2017) who allege the Bair Hugger™ patient warming system caused a surgical site infection. Nearly all of the lawsuits are pending in federal court in Minnesota. The plaintiffs claim they underwent various joint arthroplasty, cardiovascular, and other surgeries and later developed surgical site infections due to the use of the Bair Hugger™ patient warming system (the Bair Hugger™ product line was acquired by 3M as part of the 2010 acquisition of Arizant, Inc., a leading manufacturer of patient warming solutions designed to prevent hypothermia and maintain normal body temperature in surgical settings). The complaints seek damages and other relief based on theories of strict liability, negligence, breach of express and implied warranties, failure to warn, design and manufacturing defect, fraudulent and/or negligent misrepresentation/concealment, unjust enrichment, and violations of various state consumer fraud, deceptive or unlawful trade practices and/or false advertising acts. One case, from the U.S. District Court for the Western District of Tennessee is a putative nationwide class action. The U.S. Judicial Panel on Multidistrict Litigation (MDL) granted the plaintiffs' motion to transfer and consolidate all cases pending in federal courts to the U.S. District Court for the District of Minnesota to be managed in a multi-district proceeding during the pre-trial phase of the litigation. In 2017, the U.S. District Court and the Minnesota state courts denied the plaintiffs' motions to amend their complaints to add claims for punitive damages. At a joint hearing before the U.S. District Court and the Minnesota State court, on the parties' motion to exclude each other's experts, and 3M's motion for summary judgment with respect to general causation, the federal court did not exclude the plaintiffs' experts and denied 3M's motion for summary judgment on general causation. In January 2018, the state court, in hearing the same arguments, excluded plaintiffs' experts and granted 3M's motion for summary judgment on general causation, dismissing all 61 cases pending before the state court in Minnesota. Plaintiffs have appealed that ruling and the state court's punitive damages ruling, and oral argument before the Minnesota Court of Appeals is scheduled in November 2018. In April 2018, the federal court partially granted 3M's motion for summary judgment in the first bellwether case, leaving for trial a claim for strict liability based upon design defect. The court dismissed the plaintiff's claims for negligence, failure to warn, and common law and statutory fraud. In the trial of the first bellwether case in May 2018, the jury returned a unanimous verdict in 3M's favor finding that the Bair Hugger™ patient warming system was not defective and was not the cause of the plaintiff's injury.

In June 2016, the Company was served with a putative class action filed in the Ontario Superior Court of Justice for all Canadian residents who underwent various joint arthroplasty, cardiovascular, and other surgeries and later developed surgical site infections due to the use of the Bair Hugger™ patient warming system. The representative plaintiff seeks relief (including punitive damages) under Canadian law based on theories similar to those asserted in the MDL. No liability has been recorded for the Bair Hugger™ litigation because the Company believes that any such liability is not probable and estimable at this time.

In September 2011, 3M Oral Care launched Lava Ultimate CAD/CAM dental restorative material. The product was originally indicated for inlay, onlay, veneer, and crown applications. In June 2015, 3M Oral Care voluntarily removed crown applications from the product's instructions for use, following reports from dentists of patients' crowns debonding, requiring additional treatment. The product remains on the market for other applications. 3M communicated with the U.S. Food and Drug Administration, as well as regulators outside the United States. 3M also informed customers and distributors of its action, offered to accept return of unused materials and provide refunds. In May 2018, 3M reached a preliminary settlement for an amount that did not have a material impact to the Company of the lawsuit pending in the U.S. District Court for the District of Minnesota that sought certification of a class of dentists in the United States and its territories. The settlement is subject to the court's approval and certification of the settlement class, with a right of class members to opt-out of the settlement and bring individual claims against the Company.

For product liability litigation matters described in this section for which a liability has been recorded, the Company believes the amount recorded is not material to the Company's consolidated results of operations or financial condition. In addition, the Company is not able to estimate a possible loss or range of loss in excess of the established accruals at this time.

NOTE 15. Stock-Based Compensation

The 3M 2016 Long-Term Incentive Plan provides for the issuance or delivery of up to 123,965,000 shares of 3M common stock pursuant to awards granted under the plan. Awards may be issued in the form of incentive stock options, nonqualified stock options, progressive stock options, stock appreciation rights, restricted stock, restricted stock units, other stock awards, and performance units and performance shares. The remaining shares available for grant under the LTIP Program are 26.3 million as of September 30, 2018.

The Company's annual stock option and restricted stock unit grant is made in February to provide a strong and immediate link between the performance of individuals during the preceding year and the size of their annual stock compensation grants. The grant to eligible employees uses the closing stock price on the grant date. Accounting rules require recognition of expense under a non-substantive vesting period approach, requiring compensation expense recognition when an employee is eligible to retire. Employees are considered eligible to retire at age 55 and after having completed ten years of service. This retiree-eligible population represents 38 percent of the annual grant stock-based compensation award expense dollars; therefore, higher stock-based compensation expense is recognized in the first quarter.

In addition to the annual grants, the Company makes other minor grants of stock options, restricted stock units and other stock-based grants. The Company issues cash settled restricted stock units and stock appreciation rights in certain countries. These grants do not result in the issuance of common stock and are considered immaterial by the Company.

Amounts recognized in the financial statements with respect to stock-based compensation programs, which include stock options, restricted stock, restricted stock units, performance shares and the General Employees' Stock Purchase Plan (GESPP), are provided in the following table. Capitalized stock-based compensation amounts were not material for the three and nine months ended September 30, 2018 and 2017.

Stock-Based Compensation Expense

(Millions)	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Cost of sales	\$ 8	\$ 9	\$ 40	\$ 41
Selling, general and administrative expenses	35	45	177	186
Research, development and related expenses	7	6	41	39
Stock-based compensation expenses	\$ 50	\$ 60	\$ 258	\$ 266
Income tax benefits	\$ (20)	\$ (35)	\$ (137)	\$ (257)
Stock-based compensation expenses (benefits), net of tax	\$ 30	\$ 25	\$ 121	\$ 9

Stock Option Program

The following table summarizes stock option activity during the nine months ended September 30, 2018:

(Options in thousands)	Number of Options	Weighted Average Exercise Price	Weighted Average Remaining Contractual Life (months)	Aggregate Intrinsic Value (millions)
Under option —				
January 1	34,965	\$ 125.73		
Granted:				
Annual	3,211	233.19		
Exercised	(2,974)	91.16		
Forfeited	(106)	187.42		
September 30	35,096	\$ 138.30	69	\$ 2,613
Options exercisable				
September 30	26,624	\$ 121.37	58	\$ 2,379

Stock options vest over a period from one year to three years with the expiration date at 10 years from date of grant. As of September 30, 2018, there was \$84 million of compensation expense that has yet to be recognized related to non-vested stock option based awards. This expense is expected to be recognized over the remaining weighted-average vesting period of 22 months. The total intrinsic values of stock options exercised were \$411 million and \$526 million during the nine months ended September 30, 2018 and 2017, respectively. Cash received from options exercised was \$270 million and \$468 million for the nine months ended September 30, 2018 and 2017, respectively. The Company's actual tax benefits realized for the tax deductions related to the exercise of employee stock options were \$87 million and \$179 million for the nine months ended September 30, 2018 and 2017, respectively.

For the primary 2018 annual stock option grant, the weighted average fair value at the date of grant was calculated using the Black-Scholes option-pricing model and the assumptions that follow.

Stock Option Assumptions

	Annual 2018
Exercise price	\$ 233.63
Risk-free interest rate	2.7 %
Dividend yield	2.4 %
Expected volatility	21.0 %
Expected life (months)	78
Black-Scholes fair value	\$ 41.59

Expected volatility is a statistical measure of the amount by which a stock price is expected to fluctuate during a period. For the 2018 annual grant date, the Company estimated the expected volatility based upon the following three volatilities of 3M stock: the median of the term of the expected life rolling volatility; the median of the most recent term of the expected life volatility; and the implied volatility on the grant date. The expected term assumption is based on the weighted average of historical grants.

Restricted Stock and Restricted Stock Units

The following table summarizes restricted stock and restricted stock unit activity during the nine months ended September 30, 2018:

(Shares in thousands)	Number of Shares	Weighted Average Grant Date Fair Value
Nonvested balance —		
As of January 1	1,994	\$ 162.60
Granted		
Annual	467	233.61
Other	6	208.36
Vested	(633)	164.40
Forfeited	(34)	186.02
As of September 30	1,800	\$ 180.11

As of September 30, 2018, there was \$92 million of compensation expense that has yet to be recognized related to non-vested restricted stock units and restricted stock. This expense is expected to be recognized over the remaining weighted-average vesting period of 23 months. The intrinsic value of restricted stock units and restricted stock that vested during the nine months ended September 30, 2018 and 2017 was \$154 million and \$135 million, respectively. The Company's actual tax benefits realized for the tax deductions related to the vesting of restricted stock units and restricted stock was \$29 million and \$44 million for the nine months ended September 30, 2018 and 2017, respectively.

Restricted stock units granted generally vest three years following the grant date assuming continued employment. Dividend equivalents equal to the dividends payable on the same number of shares of 3M common stock accrue on these restricted stock units during the vesting period, although no dividend equivalents are paid on any of these restricted stock units that are forfeited prior to the

vesting date. Dividends are paid out in cash at the vest date on restricted stock units. Since the rights to dividends are forfeitable, there is no impact on basic earnings per share calculations. Weighted average restricted stock unit shares outstanding are included in the computation of diluted earnings per share.

Performance Shares

Instead of restricted stock units, the Company makes annual grants of performance shares to members of its executive management. The 2018 performance criteria for these performance shares (organic volume growth, return on invested capital, free cash flow conversion, and earning per share growth) were selected because the Company believes that they are important drivers of long-term stockholder value. The number of shares of 3M common stock that could actually be delivered at the end of the three-year performance period may be anywhere from 0% to 200% of each performance share granted, depending on the performance of the Company during such performance period. When granted, these performance shares are awarded at 100% of the estimated number of shares at the end of the three-year performance period and are reflected under “Granted” in the table below. Non-substantive vesting requires that expense for the performance shares be recognized over one or three years depending on when each individual became a 3M executive. The 2018 performance share grant accrues dividends, therefore the grant date fair value is equal to the closing stock price on the date of grant. Since the rights to dividends are forfeitable, there is no impact on basic earnings per share calculations. Weighted average performance shares whose performance period is complete are included in computation of diluted earnings per share.

The following table summarizes performance share activity during the nine months ended September 30, 2018:

(Shares in thousands)	Number of Shares	Weighted Average Grant Date Fair Value
Undistributed balance —		
As of January 1	686	\$ 171.90
Granted	166	229.13
Distributed	(206)	159.82
Performance change	(49)	196.85
Forfeited	(28)	204.09
As of September 30	569	\$ 189.19

As of September 30, 2018, there was \$31 million of compensation expense that has yet to be recognized related to performance shares. This expense is expected to be recognized over the remaining weighted-average earnings period of 10 months. The total fair values of performance shares that were distributed were \$48 million and \$55 million for the nine months ended September 30, 2018 and 2017, respectively. The Company’s actual tax benefits realized for the tax deductions related to the distribution of performance shares were \$11 million and \$15 million for the nine months ended September 30, 2018 and 2017, respectively.

NOTE 16. Business Segments

3M's businesses are organized, managed and internally grouped into segments based on differences in markets, products, technologies and services. 3M manages its operations in five business segments: Industrial; Safety and Graphics; Health Care; Electronics and Energy; and Consumer. 3M's five business segments bring together common or related 3M technologies, enhancing the development of innovative products and services and providing for efficient sharing of business resources. Transactions among reportable segments are recorded at cost. 3M is an integrated enterprise characterized by substantial intersegment cooperation, cost allocations and inventory transfers. Therefore, management does not represent that these segments, if operated independently, would report the operating income information shown. The difference between operating income and pre-tax income relates to interest income and interest expense, which are not allocated to business segments, along with non-service cost components of pension and postretirement net periodic benefit costs.

As part of 3M's continuing effort to improve the alignment of its businesses around markets and customers, the Company made the following changes, effective in the first quarter of 2018, and other revisions impacting business segment reporting:

Consolidation of customer account activity within international countries – expanding dual credit reporting

- The Company consolidated its customer account activity in each country into centralized sales districts for certain countries that make up approximately 70 percent of 3M's 2017 international net sales. Expansion of these initiatives, which previously had been deployed only in the U.S., reduces the complexity for customers when interacting with multiple 3M businesses. 3M business segment reporting measures include dual credit to business segments for certain sales and related operating income. This dual credit is based on which business segment provides customer account activity with respect to a particular product sold in a specific country. The expansion of alignment of customer accounts within additional countries increased the attribution of dual credit across 3M's business segments. Additionally, certain sales and operating income results for electronic bonding product lines that were previously equally divided between the Electronics and Energy business segment and the Industrial business segment are now reported similarly to dual credit.

Centralization of manufacturing and supply technology platforms

- Certain shared film manufacturing and supply technology platform resources formerly reflected within the Electronics and Energy business segment were combined with other shared and centrally managed material resource centers of expertise within Corporate and Unallocated.

In addition, as discussed in Note 1, 3M adopted ASU NO. 2017-07, *Improving the Presentation of Net Periodic Pension Cost and Net Periodic Postretirement Benefit Cost*, effective January 1, 2018 on a retrospective basis. As a result, operating income for 3M's business segments has been revised to reflect non-service cost components of pension and postretirement net periodic benefit costs within other expense (income), net.

The financial information presented herein reflects the impact of the preceding changes for all periods presented.

Business Segment Information

(Millions)	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Net Sales				
Industrial	\$ 3,023	\$ 3,023	\$ 9,315	\$ 8,905
Safety and Graphics	1,660	1,551	5,258	4,670
Health Care	1,445	1,485	4,501	4,369
Electronics and Energy	1,443	1,515	4,130	4,096
Consumer	1,235	1,279	3,585	3,521
Corporate and Unallocated	35	3	47	6
Elimination of Dual Credit	(689)	(684)	(2,016)	(1,900)
Total Company	\$ 8,152	\$ 8,172	\$ 24,820	\$ 23,667
Operating Income				
Industrial	\$ 667	\$ 672	\$ 2,110	\$ 1,910
Safety and Graphics	412	411	1,375	1,661
Health Care	446	467	1,341	1,304
Electronics and Energy	457	430	1,659	1,011
Consumer	291	311	770	732
Corporate and Unallocated	(77)	(112)	(1,329)	(256)
Elimination of Dual Credit	(180)	(171)	(502)	(459)
Total Company	\$ 2,016	\$ 2,008	\$ 5,424	\$ 5,903

Corporate and unallocated operating income includes a variety of miscellaneous items, such as corporate investment gains and losses, certain derivative gains and losses, certain insurance-related gains and losses, certain litigation and environmental expenses, corporate restructuring charges and certain under- or over-absorbed costs (e.g. pension, stock-based compensation) that the Company may choose not to allocate directly to its business segments. Corporate and Unallocated also includes sales, costs, and income from contract manufacturing, transition services and other arrangements with the acquirer of substantially all of the Communication Markets Division following its divestiture in June 2018. Because this category includes a variety of miscellaneous items, it is subject to fluctuation on a quarterly and annual basis.

3M business segment reporting measures include dual credit to business segments for certain sales and related operating income. Management evaluates each of its five business segments based on net sales and operating income performance, including dual credit reporting to further incentivize sales growth. As a result, 3M reflects additional (“dual”) credit to another business segment when the customer account activity (“sales district”) with respect to the particular product sold to the external customer is provided by a different business segment. This additional dual credit is largely reflected at the division level. For example, certain respirators are primarily sold by the Personal Safety Division within the Safety and Graphics business segment; however, a sales district within the Industrial business segment provides the contact for sales of the product to particular customers. In this example, the non-primary selling segment (Industrial) would also receive credit for the associated net sales initiated through its sales district and the related approximate operating income. The assigned operating income related to dual credit activity may differ from operating income that would result from actual costs associated with such sales. The offset to the dual credit business segment reporting is reflected as a reconciling item entitled “Elimination of Dual Credit,” such that sales and operating income in total are unchanged.

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM*

To the Shareholders and Board of Directors of 3M Company

Results of Review of Financial Statements

We have reviewed the accompanying consolidated balance sheet of 3M Company and its subsidiaries as of September 30, 2018 and the related consolidated statements of income and comprehensive income for the three-month and nine-month periods ended September 30, 2018 and 2017 and the consolidated statements of cash flows for the nine-month periods ended September 30, 2018 and 2017, including the related notes (collectively referred to as the “interim financial statements”). Based on our reviews, we are not aware of any material modifications that should be made to the accompanying interim financial statements for them to be in conformity with accounting principles generally accepted in the United States of America.

We have previously audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States), the consolidated balance sheet of the Company as of December 31, 2017, and the related consolidated statements of income and comprehensive income, of changes in equity, and of cash flows for the year then ended (not presented herein), and in our report dated February 8, 2018, except for the change in the manner in which the Company presents and classifies certain pension and postretirement net periodic benefit costs in the statement of income discussed in Notes 1 and 5 and in the composition of reportable segments discussed in Notes 3 and 17, as to which the date is May 8, 2018, we expressed an unqualified opinion on those consolidated financial statements. In our opinion, the information set forth in the accompanying consolidated balance sheet information as of December 31, 2017, is fairly stated, in all material respects, in relation to the consolidated balance sheet from which it has been derived.

Basis for Review Results

These interim financial statements are the responsibility of the Company’s management. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (“PCAOB”) and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB. We conducted our review in accordance with the standards of the PCAOB. A review of interim financial information consists principally of applying analytical procedures and making inquiries of persons responsible for financial and accounting matters. It is substantially less in scope than an audit conducted in accordance with the standards of the PCAOB, the objective of which is the expression of an opinion regarding the financial statements taken as a whole. Accordingly, we do not express such an opinion.

/s/ PricewaterhouseCoopers LLP

Minneapolis, Minnesota
October 25, 2018

* Pursuant to Rule 436(c) of the Securities Act of 1933 (“Act”) this should not be considered a “report” within the meaning of Sections 7 and 11 of the Act and the independent registered public accounting firm liability under Section 11 does not extend to it.

Item 2. Management’s Discussion and Analysis of Financial Condition and Results of Operations.

Management’s Discussion and Analysis of Financial Condition and Results of Operations (MD&A) is designed to provide a reader of 3M’s financial statements with a narrative from the perspective of management. 3M’s MD&A is presented in the following sections:

- Overview
- Results of Operations
- Performance by Business Segment
- Financial Condition and Liquidity
- Cautionary Note Concerning Factors That May Affect Future Results

Forward-looking statements in Part I, Item 2 may involve risks and uncertainties that could cause results to differ materially from those projected (refer to the section entitled “Cautionary Note Concerning Factors That May Affect Future Results” in Part I, Item 2 and the risk factors provided in Part II, Item 1A for discussion of these risks and uncertainties).

OVERVIEW

3M is a diversified global manufacturer, technology innovator and marketer of a wide variety of products and services. As more fully described in both the Performance by Business Segment section in MD&A and in Note 16, effective in the first quarter of 2018, the Company changed its business segment reporting in its continuing effort to improve the alignment of businesses around markets and customers. These changes included consolidation of customer account activity into centralized sales districts within certain countries that make up approximately 70 percent of 3M’s 2017 international net sales, impacting dual credit reporting, in addition to the centralization of manufacturing and supply technology platforms within Corporate and Unallocated. Business segment information presented herein reflects the impact of these changes for all periods presented.

3M manages its operations in five operating business segments: Industrial; Safety and Graphics; Health Care; Electronics and Energy; and Consumer. From a geographic perspective, any references to EMEA refer to Europe, Middle East and Africa on a combined basis.

Earnings per share attributable to 3M common shareholders – diluted:

The following table provides the increase (decrease) in diluted earnings per share for the three and nine months ended September 30, 2018 compared to 2017.

(Earnings per diluted share)	Three months ended September 30, 2018	Nine months ended September 30, 2018
Same period last year	\$ 2.33	\$ 7.08
Increase/(decrease) in earnings per share - diluted, due to:		
2017 divestiture of identity management business	—	(0.54)
Organic growth/productivity and other	0.12	0.71
Acquisitions/other divestiture gains	0.01	0.02
Communication Markets Division stranded costs and prior year lost income	(0.03)	(0.04)
Foreign exchange impacts	(0.08)	(0.02)
Legal-related charges	—	(0.07)
Other expense	(0.05)	(0.14)
Income tax rate, excluding Tax Cuts and Jobs Act (TCJA) measurement period adjustment	0.22	0.56
Shares of common stock outstanding	0.06	0.09
2018 divestiture of Communication Markets Division, net of related restructuring actions	—	0.48
Current period, excluding MN Natural Resource Damages (NRD) resolution and TCJA measurement period adjustment	\$ 2.58	\$ 8.13
TCJA measurement period adjustment	—	(0.36)
MN NRD resolution	—	(1.16)
Current period	\$ 2.58	\$ 6.61

For the third quarter of 2018, net income attributable to 3M was \$1.543 billion, or \$2.58 per diluted share compared to \$1.429 billion or \$2.33 per diluted share in the same period last year, an increase of 11 percent on a per diluted share basis. For the first nine months of 2018, net income attributable to 3M was \$4.002 billion, or \$6.61 per diluted share compared to \$4.335 billion or \$7.08 per diluted share in the same period last year, a decrease of 7 percent on a per diluted share basis. Excluding the first quarter 2018 pre-tax impact of \$897 million related to the resolution of the Minnesota natural resource damages (NRD) matter and the \$217 million measurement period adjustment relative to the accounting for the 2017 enactment of the Tax Cuts and Jobs Act (TCJA), net income was \$4.929 billion, or \$8.13 per diluted share, an increase of 15 percent on a per diluted share basis compared to the first nine months of 2017. The Company refers to various measures excluding the Minnesota NRD resolution and TCJA measurement period adjustment. These non-GAAP measures are further described and reconciled to the most directly comparable GAAP financial measures in the section that follows.

Additional discussion related to the components of the year-on-year change in earnings per diluted share follows:

2017 divestiture of identity management business:

- In May 2017, 3M completed the related sale or transfer of control, as applicable of its identity management business and reflected a pre-tax gain of \$457 million, which was reported within the Company's Safety and Graphics business. The earnings per share impact reflects the specific income tax rate used for this divestiture.

Organic growth/productivity and other:

- Third quarter and first nine months year-on-year benefits include higher organic local-currency sales, selling price increases, and business transformation, which is having a positive impact on 3M's productivity efforts. Higher raw material costs partially offset these year-on-year benefits.
- Lower year-on-year restructuring (other than activity related to the Communication Markets Division divestiture) and portfolio and footprint actions increased pre-tax earnings by approximately \$23 million and \$249 million in the third quarter and first nine months of 2018, respectively. These charges included \$24 million related to exit activities and \$80 million in asset charges, accelerated depreciation and other costs taken in the first quarter of 2017, \$99 million in restructuring actions and \$51 million in asset charges, accelerated depreciation and other costs taken in the second quarter of 2017, in addition to \$35 million in asset charges, accelerated depreciation and other costs taken in the third quarter of 2017.
- Lost operating income (loss) from divested businesses other than Communication Markets Division was neutral year-on-year to earnings per diluted share for the third quarter of 2018 and increased earnings per diluted share by 2 cents for the first nine months of 2018.

Acquisitions/other divestiture gains:

- In aggregate, acquisitions and year-on-year divestitures gains (other than the sale of the Communication Markets Division and identity management business) increased earnings per diluted share by 1 cent year-on-year for the third quarter of 2018 and increased earnings per diluted share by 2 cents for the first nine months of 2018.

Communication Markets Division stranded costs and prior year lost income:

- Remaining stranded costs in addition to lost operating income from the divestiture of the Communication Markets Division decreased earnings per diluted share by 3 cents and 4 cents year-on-year for the third quarter of 2018 and first nine months of 2018, respectively.

Foreign exchange impacts:

- Foreign currency impacts (net of hedging) decreased pre-tax earnings by approximately \$65 million and approximately \$15 million, or the equivalent of 8 cents and 2 cents per diluted share, for the third quarter and first nine months of 2018, respectively, excluding the impact of foreign currency changes on tax rates.

Legal-related charges

- In the second quarter of 2018, 3M reached agreements in principle on a number of respiratory mask/asbestos claims and an oral care product liability matter, the implications of which resulted in an increase in certain legal accruals. Refer to Note 14 for further details.

Other expense:

- Third quarter and first nine month's interest expense (net of interest income) increased \$26 million and \$80 million year-on-year, respectively, as a result of higher U.S. average debt balances and higher borrowing costs.
- On a combined basis, higher defined benefit pension and postretirement service cost expense and defined contribution expense, in addition to lower income related to non-service cost components of pension and postretirement expense, increased expense year-on-year.

Income tax rate, excluding Tax Cuts and Jobs Act (TCJA) measurement period adjustment:

- The effective tax rate for the third quarter of 2018 was 21.3 percent, a decrease of 7.0 percentage points versus 2017. The effective tax rate for the first nine months of 2018 was 24.0 percent, an decrease of 2.1 percentage points versus 2017.
- The effect of income taxes on items that had specific tax rates are reflected within their respective diluted earnings per share impacts in the table above for both the three and nine months ended September 30, 2018. Additionally, as discussed in the section below, excluding the Minnesota NRD Resolution and measurement period adjustment related to TCJA, the effective income tax rate was 20.0 percent in first nine months of 2018, a decrease of 6.1 percentage points versus 2017.
- Factors that decreased the effective tax rate for the third quarter of 2018 primarily related to the favorable aspects of the TCJA such as the decrease in the U.S. income tax rate and provisions incentivizing foreign-derived intangible income (FDII), in addition to impacts associated with composition of geographic mix of income before taxes. This decrease was partially offset by the impacts of TCJA, such as the elimination of the domestic manufacturing deduction and the global intangible low-taxed income (GILTI) provision; as well as unfavorable changes related to lower excess tax benefits related to employee share-based payments. Refer to Note 8 for additional details.

Shares of common stock outstanding:

- Lower shares outstanding increased earnings per share by 6 cents and 9 cents per diluted share for the third quarter and first nine months of 2018, respectively. Weighted-average diluted shares outstanding in the third quarter and first nine months of 2018 declined 2.3 percent and 1.2 percent year-on-year, respectively, which benefited earnings per share. The decrease in the outstanding weighted-average diluted shares relates to the Company's purchase of \$1.1 billion and \$3.6 billion of its own stock in the third quarter and first nine months of 2018, respectively.

2018 divestiture of Communication Markets Division, net of related restructuring actions and exit activities:

- In June 2018, 3M completed the sale of substantially all of its Communication Markets Division and reflected a pre-tax gain of \$494 million as a result of this divestiture, which was reported within the Company's Electronics and Energy business. During the second quarter of 2018, management approved and committed to undertake certain restructuring actions related to addressing corporate functional costs following the Communication Markets Division divestiture. These actions affected approximately 900 positions worldwide and resulted in a second quarter 2018 pre-tax charge of \$105 million. The aggregate net impact of the gain on sale and related restructuring actions increased earnings per share by 48 cents per diluted share for the first nine months of 2018 and reflects the specific income tax rate associated with these items.

Operating income, operating income margin, income before taxes, net income, earnings per share, and effective tax rate adjusted for impacts of the Minnesota NRD resolution and the measurement period adjustment to the impact of the enactment of the Tax Cuts and Jobs Act (TCJA) - (non-GAAP measures):

As further discussed in Note 14, in February 2018, 3M reached an agreement with the State of Minnesota that resolved the previously disclosed Natural Resource Damages (NRD) lawsuit filed by the State against the Company related to certain PFAS present in the environment. Under the terms of the settlement, 3M agreed to provide an \$850 million grant to the State for a special "3M Water Quality and Sustainability Fund." This Fund will enable projects that support water sustainability in the Twin Cities East Metro region, such as continued delivery of water to residents and enhancing groundwater recharge to support sustainable growth. The projects will also result in habitat and recreation improvements, such as fishing piers, trails, and open space preservation. 3M recorded

a pre-tax charge of \$897 million, inclusive of legal fees and other related obligations, in the first quarter of 2018 associated with the resolution of this matter.

As further described in Note 8, during the first quarter of 2018, 3M recorded a tax expense of \$217 related to a measurement period adjustment to the provisional amounts recorded in December 2017, from the enactment of the Tax Cuts and Jobs Act (TCJA). 3M's provisional accounting continues to be subject to adjustment during the measurement period of up to one year following the December 2017 enactment of TCJA.

In addition to reporting financial results in accordance with U.S. GAAP, the Company also provides non-GAAP measures that adjust for the impacts of the NRD resolution and measurement period adjustment to the impact of enactment of the TCJA. These items represent significant charges that impacted the Company's financial results. Operating income, income before taxes, net income, earnings per share, and the effective tax rate are all measures for which 3M provides the reported GAAP measure and an adjusted measure. The adjusted measures are not in accordance with, nor are they a substitute for, GAAP measures. The Company considers these non-GAAP measures in evaluating and managing the Company's operations. The Company believes that discussion of results adjusted for these items is meaningful to investors as it provides a useful analysis of ongoing underlying operating trends. The determination of these items may not be comparable to similarly titled measures used by other companies.

Adjusted income, operating income margin, earnings per share, & effective tax rate (non-GAAP measures) (Dollars in millions, except per share amounts)	Nine months ended September 30,				
	2017	2018			
	Reported GAAP Measure	Reported GAAP Measure	Adjustment for Measurement Period Accounting of TCJA	Adjustment for MN NRD Resolution	Adjusted Non-GAAP Measure
Net sales	\$ 23,667	\$ 24,820	\$ —	\$ —	\$ 24,820
Operating income	\$ 5,903	\$ 5,424	\$ —	\$ 897	\$ 6,321
Operating income margin	24.9 %	21.9 %			25.5 %
Income before taxes	\$ 5,876	\$ 5,280	\$ —	\$ 897	\$ 6,177
Provision for income taxes	\$ 1,532	\$ 1,266	\$ (217)	\$ 187	\$ 1,236
Effective tax rate	26.1 %	24.0 %			20.0 %
Net income attributable to 3M	\$ 4,335	\$ 4,002	\$ 217	\$ 710	\$ 4,929
Earnings per diluted share	\$ 7.08	\$ 6.61	\$ 0.36	\$ 1.16	\$ 8.13
Earnings per diluted share percent change		(6.6)%			14.8 %

Sales and operating income by business segment:

The following tables contain sales and operating income results by business segment for the three and nine months ended September 30, 2018 and 2017. Refer to the section entitled "Performance by Business Segment" later in MD&A for additional discussion concerning 2018 versus 2017 results, including Corporate and Unallocated. Refer to Note 16 for additional information on business segments, including Elimination of Dual Credit.

(Dollars in millions)	Three months ended September 30,				% change	
	2018		2017		Net Sales	Oper. Income
	Net Sales	Oper. Income	Net Sales	Oper. Income		
Business Segments						
Industrial	\$ 3,023	\$ 667	\$ 3,023	\$ 672	— %	(0.7)%
Safety and Graphics	1,660	412	1,551	411	7.0	0.4
Health Care	1,445	446	1,485	467	(2.8)	(4.4)
Electronics and Energy	1,443	457	1,515	430	(4.8)	6.2
Consumer	1,235	291	1,279	311	(3.4)	(6.7)
Corporate and Unallocated	35	(77)	3	(112)	—	—
Elimination of Dual Credit	(689)	(180)	(684)	(171)	—	—
Total Company	\$ 8,152	\$ 2,016	\$ 8,172	\$ 2,008	(0.2)%	0.4 %

(Dollars in millions)	Nine months ended September 30,				% change	
	2018		2017		Net Sales	Oper. Income
	Net Sales	Oper. Income	Net Sales	Oper. Income		
Business Segments						
Industrial	\$ 9,315	\$ 2,110	\$ 8,905	\$ 1,910	4.6 %	10.5 %
Safety and Graphics	5,258	1,375	4,670	1,661	12.6 %	(17.2) %
Health Care	4,501	1,341	4,369	1,304	3.0 %	2.8 %
Electronics and Energy	4,130	1,659	4,096	1,011	0.8 %	64.1 %
Consumer	3,585	770	3,521	732	1.8 %	5.1 %
Corporate and Unallocated	47	(1,329)	6	(256)	—	—
Elimination of Dual Credit	(2,016)	(502)	(1,900)	(459)	—	—
Total Company	\$ 24,820	\$ 5,424	\$ 23,667	\$ 5,903	4.9 %	(8.1) %

Worldwide Sales Change By Business Segment	Three months ended September 30, 2018				
	Organic local-currency sales	Acquisitions	Divestitures	Translation	Total sales change
Industrial	2.2 %	— %	(0.1)%	(2.1)%	— %
Safety and Graphics	2.2	8.9	(1.9)	(2.2)	7.0
Health Care	(1.1)	—	—	(1.7)	(2.8)
Electronics and Energy	2.3	—	(6.1)	(1.0)	(4.8)
Consumer	(2.0)	—	—	(1.4)	(3.4)
Total Company	1.3 %	1.7 %	(1.5)%	(1.7)%	(0.2)%

Worldwide Sales Change By Business Segment	Nine months ended September 30, 2018				
	Organic local-currency sales	Acquisitions	Divestitures	Translation	Total sales change
Industrial	3.4 %	— %	(0.1)%	1.3 %	4.6 %
Safety and Graphics	5.8	9.8	(4.2)	1.2	12.6
Health Care	1.7	0.1	—	1.2	3.0
Electronics and Energy	3.0	—	(3.2)	1.0	0.8
Consumer	1.3	—	—	0.5	1.8
Total Company	3.3 %	1.9 %	(1.4)%	1.1 %	4.9 %

Sales by geographic area:

Percent change information compares the third quarter and first nine months of 2018 with the same period last year, unless otherwise indicated. From a geographic perspective, any references to EMEA refer to Europe, Middle East and Africa on a combined basis.

	Three months ended September 30, 2018					
	United States	Asia Pacific	Europe, Middle East & Africa	Latin America/Canada	Other Unallocated	Worldwide
Net sales (millions)	\$ 3,265	\$ 2,621	\$ 1,527	\$ 741	\$ (2)	\$ 8,152
% of worldwide sales	40.1 %	32.1 %	18.7 %	9.1 %	—	100.0 %
Components of net sales change:						
Volume — organic	(0.9)%	3.0 %	(2.5)%	(0.1)%	—	0.1 %
Price	1.4	0.2	1.6	2.2	—	1.2
Organic local-currency sales	0.5	3.2	(0.9)	2.1	—	1.3
Acquisitions	2.3	0.5	2.9	0.8	—	1.7
Divestitures	(1.5)	(0.6)	(2.9)	(1.5)	—	(1.5)
Translation	—	(1.5)	(3.0)	(6.9)	—	(1.7)
Total sales change	1.3 %	1.6 %	(3.9)%	(5.5)%	—	(0.2)%
Total sales change:						
Industrial	2.3 %	1.6 %	(2.5)%	(7.0)%	—	— %
Safety and Graphics	10.6 %	4.6 %	8.0 %	(1.9)%	—	7.0 %
Health Care	(5.9)%	7.4 %	(3.8)%	(3.0)%	—	(2.8)%
Electronics and Energy	(9.3)%	0.9 %	(32.7)%	(19.4)%	—	(4.8)%
Consumer	(0.9)%	(8.7)%	(7.8)%	(4.6)%	—	(3.4)%
Organic local-currency sales change:						
Industrial	2.3 %	3.4 %	1.1 %	1.4 %	—	2.2 %
Safety and Graphics	1.4 %	4.9 %	0.7 %	1.3 %	—	2.2 %
Health Care	(6.0)%	10.1 %	(1.4)%	3.9 %	—	(1.1)%
Electronics and Energy	4.8 %	2.7 %	(5.2)%	1.3 %	—	2.3 %
Consumer	(0.9)%	(7.1)%	(5.4)%	4.7 %	—	(2.0)%

Additional information beyond what is included in the preceding table is as follows:

- In the Asia Pacific geographic area, China/Hong Kong total sales increased 8 percent and organic local-currency sales increased 10 percent. In Japan, total sales decreased 9 percent and organic local-currency sales decreased 7 percent. Excluding electronics, Japan had a total sales decrease of 1 percent, while organic local-currency sales increased 1 percent.
- In the EMEA geographic area, Central/East Europe and Middle East/Africa total sales decreased by 6 percent due in part to foreign currency translation impacts, while organic local-currency sales increased 4 percent. West Europe total sales declined 3 percent and organic local-currency sales decreased 2 percent.
- In the Latin America/Canada geographic area, total sales decreased 1 percent in Mexico, while organic local-currency sales increased 3 percent. In Canada, total sales decreased 3 percent, driven by foreign currency translation impacts, while organic local-currency sales growth was flat. In Brazil, total sales decreased 15 percent, as organic local-currency sales growth of 5 percent was more than offset by foreign currency translation impacts.

Nine months ended September 30, 2018						
	United States	Asia Pacific	Europe, Middle East & Africa	Latin America/Canada	Other Unallocated	Worldwide
Net sales (millions)	\$ 9,657	\$ 7,801	\$ 5,077	\$ 2,289	\$ (4)	\$ 24,820
% of worldwide sales	38.9 %	31.4 %	20.5 %	9.2 %	—	100.0 %
Components of net sales change:						
Volume — organic	1.6 %	4.4 %	0.2 %	2.1 %	—	2.3 %
Price	1.1	0.2	1.6	1.7	—	1.0
Organic local-currency sales	2.7	4.6	1.8	3.8	—	3.3
Acquisitions	2.6	0.6	3.0	0.9	—	1.9
Divestitures	(1.4)	(0.6)	(2.4)	(1.4)	—	(1.4)
Translation	—	1.8	4.0	(2.8)	—	1.1
Total sales change	3.9 %	6.4 %	6.4 %	0.5 %	—	4.9 %
Total sales change:						
Industrial	3.4 %	6.5 %	6.6 %	(0.9)%	—	4.6 %
Safety and Graphics	13.3 %	11.1 %	18.1 %	2.9 %	—	12.6 %
Health Care	(1.2)%	11.3 %	5.2 %	3.0 %	—	3.0 %
Electronics and Energy	(1.8)%	4.1 %	(11.8)%	(8.5)%	—	0.8 %
Consumer	3.1 %	(0.4)%	0.9 %	(0.4)%	—	1.8 %
Organic local-currency sales change:						
Industrial	3.4 %	4.1 %	2.6 %	3.3 %	—	3.4 %
Safety and Graphics	5.5 %	7.4 %	5.6 %	3.8 %	—	5.8 %
Health Care	(1.3)%	9.2 %	1.0 %	5.6 %	—	1.7 %
Electronics and Energy	4.3 %	3.6 %	(2.3)%	0.9 %	—	3.0 %
Consumer	3.1 %	(2.4)%	(2.9)%	3.9 %	—	1.3 %

Additional information beyond what is included in the preceding table is as follows:

- In the Asia Pacific geographic area, China/Hong Kong total sales increased 14 percent and organic local-currency sales increased 11 percent. In Japan, total sales decreased 2 percent and organic local-currency sales decreased 3 percent. Excluding electronics, Japan had a total sales increase of 4 percent and organic local-currency sales increased 3 percent.
- In the EMEA geographic area, Central/East Europe and Middle East/Africa total sales increased 2 percent and organic local-currency sales increased 5 percent. West Europe total sales grew 8 percent, driven by foreign currency translation impacts, while organic local-currency sales increased 1 percent.
- In the Latin America/Canada geographic area, total sales increased 2 percent in Mexico, as organic local-currency sales increased 3 percent. In Canada, total sales increased 6 percent, driven by organic local-currency sales growth of 5 percent in addition to foreign currency translation impacts. In Brazil, total sales decreased 8 percent, as organic local-currency sales growth of 5 percent was more than offset by foreign currency translation impacts.

Managing currency risks:

The weaker U.S. dollar had a positive impact on sales and earnings in the first nine months of 2018 compared to the same period last year, which was partially offset by the negative impact of the strengthening dollar in the third quarter of 2018 compared to the same period last year. 3M utilizes a number of tools to hedge currency risk related to earnings. 3M uses natural hedges such as pricing, productivity, hard currency and hard currency-indexed billings, and localizing source of supply. 3M also uses financial hedges to mitigate currency risk. In the case of more liquid currencies, 3M hedges a portion of its aggregate exposure, using a 12, 24 or 36 month horizon, depending on the currency in question. For less liquid currencies, financial hedging is frequently more expensive with more limitations on tenor. Thus this risk is largely managed via local operational actions using natural hedging tools as discussed above. In either case, 3M's hedging approach is designed to mitigate a portion of foreign currency risk and reduce volatility, ultimately allowing time for 3M's businesses to respond to changes in the marketplace.

Financial condition:

3M generated \$4.181 billion of operating cash flows in the first nine months of 2018, a decrease of \$199 million when compared to the first nine months of 2017, with this decrease primarily due to the Minnesota NRD resolution. Refer to the section entitled “Financial Condition and Liquidity” later in MD&A for a discussion of items impacting cash flows.

In February 2016, 3M’s Board of Directors authorized the repurchase of up to \$10 billion of 3M’s outstanding common stock, with no pre-established end date. In the first nine months of 2018, the Company purchased \$3.6 billion of its own stock, compared to \$1.6 billion of stock purchases in the first nine months of 2017. As of September 30, 2018, approximately \$1.5 billion remained available under the February 2016 authorization. The Company expects to purchase \$4.0 billion to \$5.0 billion of its own stock in 2018. In January 2018, 3M’s Board of Directors declared a first-quarter 2018 dividend of \$1.36 per share, an increase of 16 percent. This marked the 60th consecutive year of dividend increases for 3M. In May 2018, 3M’s Board of Directors declared a second quarter 2018 dividend of \$1.36 per share. In August 2018, 3M’s Board of Directors declared a third quarter 2018 dividend of \$1.36 per share.

3M currently has an AA- credit rating with a stable outlook from Standard & Poor’s and has an A1 credit rating with a stable outlook from Moody’s Investors Service. The Company generates significant ongoing cash flow and has proven access to capital markets funding throughout business cycles.

3M expects to contribute up to \$400 million of cash to its global defined benefit pension and postretirement plans in 2018. The Company does not have a required minimum cash pension contribution obligation for its U.S. plans in 2018.

RESULTS OF OPERATIONS

Net Sales:

Refer to the preceding “Overview” section and the “Performance by Business Segment” section later in MD&A for additional discussion of sales change.

Operating Expenses:

(Percent of net sales)	Three months ended September 30,			Nine months ended September 30,		
	2018	2017	Change	2018	2017	Change
Cost of sales	51.0 %	49.7 %	1.3 %	50.7 %	50.6 %	0.1 %
Selling, general and administrative expenses	19.0	20.0	(1.0)	23.9	20.6	3.3
Research, development and related expenses	5.3	5.7	(0.4)	5.6	6.0	(0.4)
Gain on sale of businesses	—	—	—	(2.1)	(2.1)	—
Operating income margin	24.7 %	24.6 %	0.1 %	21.9 %	24.9 %	(3.0)%

3M expects global defined benefit pension and postretirement service cost expense in 2018 to increase by approximately \$26 million pre-tax when compared to 2017, which impacts cost of sales; selling, general and administrative expenses (SG&A); and research, development and related expenses (R&D). The year-on-year increase in defined benefit pension and postretirement service cost expense and defined contribution expense for the third quarter and first nine months of 2018 was approximately \$4 million and \$29 million, respectively.

The Company is investing in an initiative called business transformation, with these investments impacting cost of sales, SG&A, and R&D. Business transformation encompasses the ongoing multi-year phased implementation of an enterprise resource planning (ERP) system on a worldwide basis, as well as changes in processes and internal/external service delivery across 3M.

Cost of Sales:

Cost of sales includes manufacturing, engineering and freight costs.

Cost of sales, measured as a percent of sales, increased in the third quarter and first nine months of 2018 primarily due to foreign currency effects (net of hedge losses). Additionally, cost of sales for the first nine months of 2018 were increased by the second quarter 2018 Communication Markets Division related restructuring charges as discussed in Note 5. This increase was partially offset by 2017 portfolio and supply chain footprint optimization charges that did not repeat in 2018, and selling price increases. Selling prices increased net sales year-on-year by 1.2 percent and 1.0 percent in third quarter and first nine months of 2018, respectively. These were partially offset by raw material cost increases and higher defined benefit pension and postretirement service cost expense and defined contribution expense.

Selling, General and Administrative Expenses:

SG&A in dollars decreased 5.5 percent and increased 21.5 percent in the third quarter and first nine months of 2018, respectively, when compared to the same period last year. The decrease in the third quarter of 2018 primarily relates to 2017 portfolio and supply chain footprint optimization charges that did not repeat in 2018. The increase in the first nine months is primarily associated with the Communication Markets Division-related restructuring charges (as discussed in Note 5) and the charge related to the Minnesota NRD resolution (as discussed earlier in the “*Operating income, operating income margin, income before taxes, net income, earnings per share, and effective tax rate adjusted for impacts of the Minnesota NRD resolution and the measurement period adjustment to the impact of the enactment of the Tax Cuts and Jobs Act (TCJA) - (non-GAAP measures)*” section and further in Note 14).

Research, Development and Related Expenses:

R&D in dollars decreased \$38 million in both the third quarter of 2018 and first nine months of 2018, respectively, when compared to the same period last year. The decrease primarily relates to R&D no longer incurred related to the Communication Markets Division, which was divested in the second quarter of 2018. 3M continued to invest in its key initiatives, including R&D aimed at disruptive innovation programs with the potential to create entirely new markets and disrupt existing markets.

Gain on Sale of Businesses:

In the first quarter of 2018, 3M closed on the sale of certain personal safety product offerings primarily focused on noise, environmental, and heat stress monitoring. In addition, 3M divested a polymer additives compounding business, formerly part of the Company’s Industrial business, and reflected a gain on final closing adjustments from a prior divestiture which, in aggregate, were not material. These divestitures resulted in a gain on sale of businesses of \$24 million.

In the second quarter of 2018, 3M completed the sale of substantially all of its Communication Markets Division to Corning Incorporated. This business consists of optical fiber and copper passive connectivity solutions for the telecommunications industry including 3M’s xDSL, FTTx, and structured cabling solutions and, in certain countries, telecommunications system integration services. This divestiture resulted in a gain on sale of \$494 million, which was reported within the Company’s Electronics and Energy business. 3M also divested an abrasives glass products business, formerly part of the Company’s Industrial business, which resulted in a gain on sale of less than \$15 million. Refer to Note 3 for additional detail on these divestitures.

Operating Income:

3M uses operating income as one of its primary business segment performance measurement tools. Refer to the table below for a reconciliation of operating income margins for the three and nine months ended September 30, 2018 versus 2017.

(Percent of net sales)	Three months ended September 30, 2018	Nine months ended September 30, 2018
Same period last year	24.6 %	24.9 %
Increase/(decrease) in operating income margin, due to:		
2017 divestiture of identity management business	—	(1.9)
Organic volume/productivity and other	0.7	1.5
Acquisitions/other divestiture gains	(0.3)	(0.3)
Communication Markets Division stranded costs	(0.1)	—
Selling price and raw material impact	0.3	0.3
Foreign exchange impacts	(0.5)	(0.4)
Legal-related charges	—	(0.2)
2018 divestiture of Communication Markets Division, net of related restructuring actions	—	1.6
Current period, excluding MN Natural Resource Damages (NRD) resolution	24.7 %	25.5 %
MN NRD resolution	—	(3.6)
Current period	24.7 %	21.9 %

Operating income margins increased 0.1 percentage point in the third quarter of 2018 when compared to the third quarter of 2017, and declined 3.0 percentage points in the first nine months of 2018 when compared to the first nine months of 2017.

Additional discussion related to the components of the year-on-year change in operating income margins follows:

2017 divestiture of identity management business:

- Operating income margins decreased year-on-year due to the gain on the May 2017 divestiture of the Company's former identity management business.

Organic volume/productivity and other:

- Operating income margins increased year-on-year due to benefits from organic local-currency growth and productivity, in addition to lower year-on-year portfolio and supply chain footprint optimization charges.
- Operating income margins decreased year-on-year due to higher defined benefit pension and postretirement service cost expense and defined contribution expense.
- Operating income margins increased year-on-year due to the lost lower-margin operating income from divested businesses.

Acquisitions/other divestiture gains:

- Acquisition impacts (primarily related to Scott Safety) in addition to lower year-on-year divestiture gains other than the identity management business and Communication Markets decreased operating margins year-on-year.

Communication Markets Division stranded costs:

- Remaining stranded costs to be addressed from the divestiture of the Communication Markets Division reduced operating margins year-on-year.

Selling price and raw material impact:

- Higher selling prices, partially offset by raw material cost increases, benefited operating income margins year-on-year.

Foreign exchange impacts:

- Foreign currency effects (net of hedge gains) decreased operating income margins year-on-year.

Legal-related charges:

- In the second quarter of 2018, 3M reached agreements in principle on a number of respiratory mask/asbestos claims and an oral care product liability matter, the implications of which resulted in an increase in certain legal accruals. Refer to Note 14 for further details.

2018 divestiture of Communication Markets Division, net of related restructuring actions:

- In June 2018, 3M completed the sale of substantially all of its Communication Markets Division and reflected a pre-tax gain of \$494 million as a result of this divestiture, which was reported within the Company's Electronics and Energy business. During the second quarter of 2018, management approved and committed to undertake certain restructuring actions related to addressing corporate functional costs following the Communication Markets Division divestiture. These actions resulted in a second quarter 2018 pre-tax charge of \$105 million.

MN NRD Resolution:

- Operating income margins for the first nine months of 2018 decreased 3.0 percentage points year-on-year. Excluding the first quarter 2018 impact of the Minnesota NRD resolution (as discussed earlier in the "*Operating income, operating income margin, income before taxes, net income, earnings per share, and effective tax rate adjusted for impacts of the Minnesota NRD resolution and the measurement period adjustment to the impact of the enactment of the Tax Cuts and Jobs Act (TCJA) - (non-GAAP measures)*" section and further in Note 14), operating margins increased 0.6 percentage points for the first nine months of 2018.

Other Expense (Income), Net:

See Note 6 for a detailed breakout of this line item.

Interest expense increased in the third quarter and first nine months of 2018 compared to the same periods in 2017 due to higher U.S. average debt balances and higher borrowing costs. Interest income also increased year-on-year for the third quarter and first nine months of 2018 due to higher average interest rates and investment balances.

Effective January 1, 2018, in conjunction with 3M's adoption of ASU No. 2017-07, all pension and postretirement net periodic benefit cost components (except the service cost component) are reported within other expense (income), net. For additional details, refer to Note 1 (Significant Accounting Policies). Year-on-year pension and postretirement net periodic benefit non-service costs increased \$14 million and \$37 million for the third quarter and first nine months of 2018, respectively. The year-on-year increases were primarily due to an increase in the net actuarial amortization expense.

Provision for Income Taxes:

(Percent of pre-tax income)	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Effective tax rate	21.3 %	28.3 %	24.0 %	26.1 %

The effective tax rate for the third quarter of 2018 was 21.3 percent, compared to 28.3 percent in 2017, a decrease of 7.0 percentage points. The effective tax rate for the first nine months of 2018 was 24.0 percent, compared to 26.1 percent in 2017, an decrease of 2.1 percentage points. The changes in the tax rates between years were impacted by many factors, including the enactment of the Tax Cuts and Jobs Act (TCJA) in December 2017 and measurement period adjustments to the Company's provisional accounting thereof as further described in the Overview, "*Operating income, operating income margin, income before taxes, net income, earnings per share, and effective tax rate adjusted for impacts of the Minnesota NRD resolution and the measurement period adjustment to the impact of the enactment of the Tax Cuts and Jobs Act (TCJA) - (non-GAAP measures)*" section and in Note 8. 3M's provisional accounting is subject to adjustment during the measurement period of up to one year following the December 2017 enactment of the TCJA, as provided by recent SEC guidance.

Considering the impacts of the TCJA (including the Company's measurement period adjustments through the third quarter of 2018 to the Company's provisional accounting for the TCJA's enactment) and other factors, 3M currently estimates its effective tax rate for

2018 will be approximately 23 to 25 percent. The tax rate can vary from quarter to quarter due to discrete items, such as the settlement of income tax audits, changes in tax laws, measurement period adjustment effects on the provisional items such as the remaining analyses regarding the impact of the TCJA and employee share-based payment accounting; as well as recurring factors, such as the geographic mix of income before taxes.

Refer to Note 8 for further discussion of income taxes.

Net Income Attributable to Noncontrolling Interest:

(Millions)	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Net income attributable to noncontrolling interest	\$ 3	\$ 4	\$ 12	\$ 9

Net income attributable to noncontrolling interest represents the elimination of the income or loss attributable to non-3M ownership interests in 3M consolidated entities. The primary noncontrolling interest relates to 3M India Limited, of which 3M's effective ownership is 75 percent.

Currency Effects:

3M estimates that year-on-year currency effects, including hedging impacts, decreased pre-tax income by approximately \$65 million and approximately \$15 million, respectively, for the third quarter and first nine months of 2018 compared to 2017. This estimate includes the effect of translating profits from local currencies into U.S. dollars; the impact of currency fluctuations on the transfer of goods between 3M operations in the United States and abroad; and transaction gains and losses, including derivative instruments designed to reduce foreign currency exchange rate risks. 3M estimates that year-on-year foreign currency transaction effects, including hedging impacts, decreased pre-tax income by approximately \$24 million and \$115 million for the three and nine months ended September 30, 2018. These estimates include transaction gains and losses, including derivative instruments designed to reduce foreign currency exchange rate risks.

Significant Accounting Policies:

Information regarding new accounting standards is included in Note 1 to the Consolidated Financial Statements.

PERFORMANCE BY BUSINESS SEGMENT

Disclosures relating to 3M's business segments are provided in Note 16. As part of 3M's continuing effort to improve the alignment of its businesses around markets and customers, the Company made the following changes, effective in the first quarter of 2018, and other revisions impacting business segment reporting:

Consolidation of customer account activity within international countries – expanding dual credit reporting

- The Company consolidated its customer account activity in each country into centralized sales districts for certain countries that make up approximately 70 percent of 3M's 2017 international net sales. Expansion of these initiatives, which previously had been deployed only in the U.S., reduces the complexity for customers when interacting with multiple 3M businesses. 3M business segment reporting measures include dual credit to business segments for certain sales and related operating income. This dual credit is based on which business segment provides customer account activity with respect to a particular product sold in a specific country. The expansion of alignment of customer accounts within additional countries increased the attribution of dual credit across 3M's business segments. Additionally, certain sales and operating income results for electronic bonding product lines that were previously equally divided between the Electronics and Energy business segment and the Industrial business segment are now reported similarly to dual credit. As a result, previously reported aggregate business segment net sales and operating income for total year 2017 increased \$1.568 billion and \$402 million, respectively, offset by similar increases in the elimination of dual credit net sales and operating income amounts.

Centralization of manufacturing and supply technology platforms

- Certain shared film manufacturing and supply technology platform resources formerly reflected within the Electronics and Energy business segment were combined with other shared and centrally managed material resource centers of expertise within Corporate and Unallocated. This change resulted in a decrease in previously reported net sales and an increase in operating income for total year 2017 of \$1 million and \$42 million, respectively, in the Electronics and Energy segment, offset by a corresponding increase in net sales and decrease in operating income within Corporate and Unallocated.

In addition, as discussed in Note 1, 3M adopted ASU No. 2017-07, *Improving the Presentation of Net Periodic Pension Cost and Net Periodic Postretirement Benefit Cost*, effective January 1, 2018 on a retrospective basis. As a result, operating income for 3M's business segments has been revised to reflect non-service components of pension and postretirement net periodic benefit costs within other expense (income), net.

Business segment information presented herein reflects the impact of these changes for all periods presented. 3M manages its operations in five business segments. The reportable segments are Industrial; Safety and Graphics; Electronics and Energy; Health Care; and Consumer.

Corporate and Unallocated:

In addition to these five business segments, 3M assigns certain costs to "Corporate and Unallocated," which is presented separately in the preceding business segments table and in Note 16. Corporate and Unallocated includes a variety of miscellaneous items, such as corporate investment gains and losses, certain derivative gains and losses, certain insurance-related gains and losses, certain litigation and environmental expenses, corporate restructuring charges and certain under- or over-absorbed costs (e.g. pension, stock-based compensation) that the Company determines not to allocate directly to its business segments. Corporate and Unallocated also includes sales, costs, and income from contract manufacturing, transition services and other arrangements with the acquirer of substantially all of the Communication Markets Division following its divestiture in June 2018. Because this category includes a variety of miscellaneous items, it is subject to fluctuation on a quarterly and annual basis.

Corporate and Unallocated operating expenses were largely flat and increased by \$1.1 billion in the third quarter and first nine months of 2018, respectively, when compared to the same period last year. In the second quarter, operating expenses included the restructuring charge of \$105 million discussed in Note 5 related to addressing corporate functional costs following the Communication Markets Division divestiture, net of operating income from contractual manufacturing and other arrangements described in the paragraph above. In the first quarter of 2018, the Minnesota NRD resolution (\$897 million), inclusive of legal fees and other related obligations, was reflected in Corporate and Unallocated. In addition, 3M's defined benefit pension and postretirement service-cost expense allocation to Corporate and Unallocated increased year-on-year.

Operating Business Segments:

Information related to 3M's business segments for the third quarter and first nine months of both 2018 and 2017 are presented in the tables that follow. Organic local-currency sales include both organic volume impacts plus selling price impacts. Acquisition impacts, if any, are measured separately for the first twelve months post-transaction. The divestiture impacts, if any, foreign currency translation impacts and total sales change are also provided for each business segment. Any references to EMEA relate to Europe, Middle East and Africa on a combined basis.

Refer to the preceding "Sales and operating income by geographic area" section for organic local-currency sales growth by business segment within major geographic areas.

Refer to 3M's Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K), Item 1, Business, for discussion of 3M products that are included in each business segment.

Industrial Business:

	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Sales (millions)	\$ 3,023	\$ 3,023	\$ 9,315	\$ 8,905
Sales change analysis:				
Organic local-currency	2.2 %	7.1 %	3.4 %	6.0 %
Divestitures	(0.1)	(0.6)	(0.1)	(0.6)
Translation	(2.1)	0.6	1.3	(0.3)
Total sales change	— %	7.1 %	4.6 %	5.1 %
Operating income (millions)	\$ 667	\$ 672	\$ 2,110	\$ 1,910
Percent change	(0.7)%	6.0 %	10.5 %	(1.5)%
Percent of sales	22.1 %	22.2 %	22.7 %	21.5 %

Third Quarter 2018 results:

Sales in Industrial totaled \$3.0 billion, which was flat compared to the same period last year. Organic local-currency sales increased 2.2 percent, divestitures decreased sales by 0.1 percent, and foreign currency translation decreased sales by 2.1 percent.

On an organic local-currency sales basis:

- Sales growth was led by advanced materials, automotive and aerospace, and industrial adhesives and tapes. Benefits included 3M's automotive OEM's increased penetration across applications such as structural tapes, adhesives, acoustics, light weighting and electronics solutions.
- Automotive aftermarket declined, as the industry saw a softer collision repair market versus prior year.

Acquisitions and divestitures:

- In October 2016, 3M sold the assets of its temporary protective films business.
- In the first quarter of 2018, 3M completed the sale of its polymer additives compounding business.
- In May 2018, 3M divested an abrasives glass products business.

Operating income:

- Operating income margins decreased 0.1 percentage points. Third quarter 2017 included a 0.3 percentage point impact from charges related to portfolio and footprint actions.

First Nine Months 2018 results:

Sales in Industrial totaled \$9.3 billion, up 4.6 percent in U.S. dollars. Organic local-currency sales increased 3.4 percent, divestitures decreased sales by 0.1 percent, and foreign currency translation increased sales by 1.3 percent.

On an organic local-currency sales basis:

- Sales growth increased in advanced materials, automotive and aerospace, abrasives, separation and purification, and industrial adhesives and tapes. Benefits included 3M's automotive OEM's increased penetration across applications such as structural tapes, adhesives, acoustics, light weighting and electronics solutions.
- Automotive aftermarket sales declined, as growth in products and solutions for retail car care was more than offset by softening of the collision repair market.

Acquisitions and divestitures:

- In January 2016, 3M completed its sale of its pressurized polyurethane foam adhesives business (formerly known as Polyfoam).
- In October 2016, 3M sold the assets of its temporary protective films business.
- In the first quarter of 2018, 3M completed the sale of its polymer additives compounding business.
- In May 2018, 3M divested an abrasives glass products business.

Operating income:

Operating income margins increased 1.2 percentage points, helped by organic sales growth across most of the portfolio in addition to benefiting from expenses related to portfolio and footprint actions taken in the first nine months of 2017 that were not repeated in the first nine months of 2018.

Safety and Graphics Business:

	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Sales (millions)	\$ 1,660	\$ 1,551	\$ 5,258	\$ 4,670
Sales change analysis:				
Organic local-currency	2.2 %	6.5 %	5.8 %	5.0 %
Acquisitions	8.9	—	9.8	—
Divestitures	(1.9)	(4.9)	(4.2)	(3.0)
Translation	(2.2)	0.8	1.2	(0.2)
Total sales change	7.0 %	2.4 %	12.6 %	1.8 %
Operating income (millions)	\$ 412	\$ 411	\$ 1,375	\$ 1,661
Percent change	0.4 %	12.0 %	(17.2)%	45.9 %
Percent of sales	24.8 %	26.4 %	26.1 %	35.6 %

Third Quarter 2018 results:

Sales in Safety and Graphics totaled \$1.7 billion, up 7.0 percent in U.S. dollars. Organic local-currency sales increased 2.2 percent, acquisitions increased sales by 8.9 percent, divestitures reduced sales by 1.9 percent, and foreign currency translation decreased sales by 2.2 percent.

On an organic local-currency sales basis:

- Sales increased in personal safety, transportation safety, and commercial solutions.
- Sales declined in roofing granules, as production slowed within the shingle manufacturing industry when compared to prior year.

Acquisitions and divestitures:

- Acquisition sales growth in 2018 reflects the acquisition of Scott Safety in October 2017. Scott Safety is a premier manufacturer of innovative products, including self-contained breathing apparatus systems, gas and flame detection instruments and other safety devices.
- 2017 divestitures include the sale of its safety prescription eyewear business (first quarter 2017), the sale of 3M's identity management business and tolling and automated license/number plate business (both in second quarter 2017) and electronic monitoring business (fourth quarter 2017).
- In February 2018, 3M closed on the sale of certain personal safety product offerings primarily focused on noise, environmental, and heat stress monitoring.

Operating income:

- Operating income margins decreased 1.6 percentage points, which included a 1.5 percentage point impact related to the impact on operating margins from the Scott Safety acquisition.

First Nine Months 2018 results:

Sales in Safety and Graphics totaled \$5.3 billion, up 12.6 percent in U.S. dollars. Organic local-currency sales increased 5.8 percent, acquisitions increased sales by 9.8 percent, divestitures reduced sales by 4.2 percent, and foreign currency translation increased sales by 1.2 percent.

On an organic local-currency sales basis:

- Sales increased in personal safety, commercial solutions, and transportation safety.
- Sales declined in roofing granules due to the slower production within the shingle manufacturing industry during the third quarter.

Acquisitions and divestitures:

- Acquisition sales growth in 2018 reflects the acquisition of Scott Safety in October 2017. Scott Safety is a premier manufacturer of innovative products, including self-contained breathing apparatus systems, gas and flame detection instruments and other safety devices.
- 2017 divestitures include the sale of its safety prescription eyewear business (first quarter 2017), the sale of 3M's identity management business and tolling and automated license/number plate business (both in second quarter 2017) and electronic monitoring business (fourth quarter 2017).
- In February 2018, 3M closed on the sale of certain personal safety product offerings primarily focused on noise, environmental, and heat stress monitoring.

Operating income:

- Operating income margins decreased 9.5 percentage points, primarily related to the year-on-year impact of 2017 divestiture gains.

Health Care Business:

	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Sales (millions)	\$ 1,445	\$ 1,485	\$ 4,501	\$ 4,369
Sales change analysis:				
Organic local-currency	(1.1) %	6.8 %	1.7 %	4.1 %
Acquisitions	—	—	0.1	—
Translation	(1.7)	0.8	1.2	(0.2)
Total sales change	(2.8) %	7.6 %	3.0 %	3.9 %
Operating income (millions)	\$ 446	\$ 467	\$ 1,341	\$ 1,304
Percent change	(4.4) %	10.3 %	2.8 %	(1.6) %
Percent of sales	30.9 %	31.4 %	29.8 %	29.8 %

Third Quarter 2018 results:

Sales in Health Care totaled \$1.4 billion, down 2.8 percent in U.S. dollars. Organic local-currency sales decreased 1.1 percent, and foreign currency translation decreased sales by 1.7 percent.

On an organic local-currency sales basis:

- Sales increased in food safety, health information systems, and oral care.
- Sales declined in medical solutions, as strong growth in vascular access and advanced wound care solutions were more than offset by weaknesses in other areas of the portfolio.
- Drug delivery systems also declined due to softness in project-based drug delivery business, which is highly dependent on the timing of regulatory approval processes and customer R&D budgets. However, the pipeline continues to strengthen and 3M expects the business to stabilize in 2019.
- In developing markets, Health Care organic local-currency sales growth was led by China/Hong Kong.

Acquisitions:

- In September 2017, 3M acquired Elution Technologies, LLC, a manufacturer of food safety test kits.

Operating income:

- Operating income margins decreased 0.5 percentage points year-on-year, largely due to lower organic sales growth within the portfolio.

First Nine Months 2018 results:

Sales in Health Care totaled \$4.5 billion, up 3.0 percent in U.S. dollars. Organic local-currency sales increased 1.7 percent, acquisitions increased sales by 0.1 percent, and foreign currency translation increased sales by 1.2 percent.

On an organic local-currency sales basis:

- Sales growth was led by food safety, health information systems, and medical solutions.
- Oral care sales also increased, with continued positive growth internationally, particularly in developing economies.
- Sales declined in drug delivery systems.

Acquisitions:

- In September 2017, 3M acquired Elution Technologies, LLC, a manufacturer of food safety test kits.

Operating income:

- Operating income margins were flat year-on-year.

Electronics and Energy Business:

	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Sales (millions)	\$ 1,443	\$ 1,515	\$ 4,130	\$ 4,096
Sales change analysis:				
Organic local-currency	2.3 %	14.1 %	3.0 %	11.9 %
Divestitures	(6.1)	(0.1)	(3.2)	(0.2)
Translation	(1.0)	—	1.0	(0.2)
Total sales change	(4.8)%	14.0 %	0.8 %	11.5 %
Operating income (millions)	\$ 457	\$ 430	\$ 1,659	\$ 1,011
Percent change	6.2 %	29.6 %	64.1 %	27.8 %
Percent of sales	31.7 %	28.4 %	40.2 %	24.7 %

Third Quarter 2018 results:

Sales in Electronics and Energy totaled \$1.4 billion, down 4.8 percent in U.S. dollars. Organic local-currency sales increased 2.3 percent, divestitures reduced sales by 6.1 percent, and foreign currency translation decreased sales by 1.0 percent.

Total sales within the electronics-related and energy-related businesses increased 1 percent and decreased 19 percent, respectively. Total sales increased 1 percent in Asia Pacific.

On an organic local-currency sales basis:

- Sales increased 1 percent in 3M's electronics-related businesses, driven by strong demand for specialty fluids and abrasives for the semiconductor and data center markets within electronics materials solutions. This growth was partially offset by a decline in display materials and systems due to softness in consumer electronics.
- Sales increased 6 percent in 3M's energy-related businesses, driven by strong growth in renewable energy and pipe coating solutions within electrical markets.
- Sales increased 3 percent in Asia Pacific, where 3M's electronics business is concentrated.

Divestitures:

- In December 2016, 3M sold the assets of its cathode battery technology out-licensing business.
- In the fourth quarter of 2017, 3M sold the assets of its electrical marking/labeling business.
- In June 2018, 3M completed the sale of substantially all of its Communication Markets Division and recorded a pre-tax gain of approximately \$494 million. Refer to Note 3 for additional details.

Operating income:

- Operating income margins increased 3.3 percentage points, driven by underlying productivity gains related to organic local-currency growth. Operating margins also increased 0.8 percentage points from the lost lower-margin operating income of the former Communication Markets Division.

First Nine Months 2018 results:

Sales in Electronics and Energy totaled \$4.1 billion, up 0.8 percent in U.S. dollars. Organic local-currency sales increased 3.0 percent, divestitures reduced sales by 3.2 percent, and foreign currency translation increased sales by 1.0 percent.

Total sales within the electronics-related and energy-related businesses were increased 4 percent and decreased 5 percent, respectively. Total sales increased 4 percent in Asia Pacific.

On an organic local-currency sales basis:

- Sales increased 3 percent in 3M's electronics-related businesses, driven by increases in electronics materials solutions. This growth was partially offset by a decline in display materials and systems due to softness in consumer electronics.

- Sales increased 5 percent in 3M's energy-related businesses, driven by electrical markets.
- Sales increased 4 percent in Asia Pacific, where 3M's electronics business is concentrated.

Divestitures:

- In December 2016, 3M sold the assets of its cathode battery technology out-licensing business.
- In the fourth quarter of 2017, 3M sold the assets of its electrical marking/labeling business.
- In June 2018, 3M completed the sale of substantially all of its Communication Markets Division and recorded a pre-tax gain of approximately \$494 million. Refer to Note 3 for additional details.

Operating income:

- Operating income margins increased 15.5 percentage points, driven by the Communication Markets Division divestiture gain.

Consumer Business:

	Three months ended September 30,		Nine months ended September 30,	
	2018	2017	2018	2017
Sales (millions)	\$ 1,235	\$ 1,279	\$ 3,585	\$ 3,521
Sales change analysis:				
Organic local-currency	(2.0)%	3.3 %	1.3 %	1.5 %
Translation	(1.4)	0.3	0.5	0.2
Total sales change	(3.4)%	3.6 %	1.8 %	1.7 %
Operating income (millions)	\$ 291	\$ 311	\$ 770	\$ 732
Percent change	(6.7)%	(1.1)%	5.1 %	(11.6)%
Percent of sales	23.5 %	24.3 %	21.5 %	20.8 %

Third Quarter 2018 results:

Sales in Consumer totaled \$1.2 billion, a decrease of 3.4 percent in U.S. dollars. Organic local-currency sales decreased 2.0 percent and foreign currency translation decreased sales by 1.4 percent.

On an organic local-currency sales basis:

- Sales grew in home improvement, while stationery and office, home care, and consumer health care declined.
- Third quarter sales were impacted by stronger than expected second quarter sales in anticipation of the Company's roll out of its ERP system in the U.S.

Operating income:

- Operating income margins decreased 0.8 percentage points year-on-year, impacted by the ERP deployment in the U.S, partially offset by 0.7 percentage points of portfolio and footprint actions taken in the third quarter of 2017 that were not repeated in the third quarter of 2018.

First Nine Months 2018 results:

Sales in Consumer totaled \$3.6 billion, an increase of 1.8 percent in U.S. dollars. Organic local-currency sales increased 1.3 percent and foreign currency translation increased sales by 0.5 percent.

On an organic local-currency sales basis:

- Sales grew in home improvement, building on a track record of strong performance over the past several years.
- Home care and stationery and office supplies were flat, while consumer health care declined.

Operating income:

- Operating income margins increased 0.7 percentage points year-on-year, benefiting from expenses related to portfolio and footprint actions taken in the first nine months of 2017 that were not repeated in the first nine months of 2018.

FINANCIAL CONDITION AND LIQUIDITY

The strength and stability of 3M's business model and strong free cash flow capability, together with proven capital markets access, positions the Company to be able to add further leverage to its capital structure. Investing in 3M's businesses to drive organic growth remains the first priority for capital deployment, including research and development, capital expenditures, and commercialization capability. Investment in organic growth will be supplemented by complementary acquisitions. 3M will also continue to return cash to shareholders through dividends and share repurchases. Sources for cash availability in the United States, such as ongoing cash flow from operations and access to capital markets, have historically been sufficient to fund dividend payments to shareholders, as well as funding U.S. acquisitions and other items as needed. The TCJA creates additional repatriation opportunities for 3M to access international cash positions on a continual and on-going basis and will help support U.S. capital deployments needs. For those international earnings still considered to be reinvested indefinitely, the Company currently has no plans or intentions to repatriate these funds for U.S. operations. See Note 9 in 3M's Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K) for further information on earnings considered to be reinvested indefinitely.

3M's primary short-term liquidity needs are met through cash on hand and U.S. commercial paper issuances. 3M believes it will have continuous access to the commercial paper market. 3M's commercial paper program permits the Company to have a maximum of \$5 billion outstanding with a maximum maturity of 397 days from date of issuance. At September 30, 2018, there was no commercial paper issued and outstanding.

Total debt:

The strength of 3M's capital structure and significant ongoing cash flows provide 3M proven access to capital markets. Additionally, the Company's maturity profile is staggered to help ensure refinancing needs in any given year are reasonable in proportion to the total portfolio. 3M currently has an AA- credit rating with a stable outlook from Standard & Poor's and has an A1 credit rating with a stable outlook from Moody's Investors Service.

The Company's total debt was \$897 million higher at September 30, 2018 when compared to December 31, 2017. Increases in debt related to the third quarter 2018 issuance of \$2.25 billion of medium-term notes, net of \$450 million repayment of maturing medium-term notes and the net impact of repayments and borrowings of international subsidiaries along with foreign currency effects, partially offset by having zero commercial paper outstanding at September 30, 2018, as compared to \$745 million at December 31, 2017. For discussion of repayments of and proceeds from debt refer to the following "Cash Flows from Financing Activities" section.

Effective February 24, 2017, the Company updated its "well-known seasoned issuer" (WKSI) shelf registration statement, which registers an indeterminate amount of debt or equity securities for future issuance and sale. This replaced 3M's previous shelf registration dated May 16, 2014. In May 2016, in connection with the WKSI shelf, 3M entered into an amended and restated distribution agreement relating to the future issuance and sale (from time to time) of the Company's medium-term notes program (Series F), up to the aggregate principal amount of \$18 billion, which was an increase from the previous aggregate principal amount up to \$9 billion of the same Series.

As of September 30, 2018, the total amount of debt issued as part of the medium-term notes program (Series F), inclusive of debt issued in September 2018 and prior years is approximately \$15.3 billion (utilizing the foreign exchange rates applicable at the time of issuance for the Euro denominated debt). Information with respect to long-term debt issuances and maturities for the periods presented is included in Note 11 of 3M's Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K).

In March 2016, 3M amended and restated its existing \$2.25 billion five-year revolving credit facility expiring in August 2019 to a \$3.75 billion five-year revolving credit facility expiring in March 2021. This credit agreement includes a provision under which 3M may request an increase of up to \$1.25 billion (at lenders' discretion), bringing the total facility up to \$5.0 billion. This revolving credit facility is undrawn at September 30, 2018. Under the \$3.75 billion credit agreement, the Company is required to maintain its EBITDA to Interest Ratio as of the end of each fiscal quarter at not less than 3.0 to 1. This is calculated (as defined in the agreement) as the ratio of consolidated total EBITDA for the four consecutive quarters then ended to total interest expense on all funded debt for the same period. At September 30, 2018, this ratio was approximately 21 to 1. Debt covenants do not restrict the payment of

dividends. Apart from the committed facilities, an additional \$281 million in stand-alone letters of credit and bank guarantees were also issued and outstanding at September 30, 2018. These instruments are utilized in connection with normal business activities.

Cash, cash equivalents and marketable securities:

At September 30, 2018, 3M had \$3.6 billion of cash, cash equivalents and marketable securities, of which approximately \$2.985 billion was held by the Company's foreign subsidiaries and approximately \$566 million was held by the United States. These balances are invested in bank instruments and other high-quality fixed income securities. At December 31, 2017, cash, cash equivalents and marketable securities held by the Company's foreign subsidiaries and by the United States totaled approximately \$3.975 billion and \$180 million, respectively.

Net Debt (non-GAAP measure):

Net debt is not defined under U.S. GAAP and may not be computed the same as similarly titled measures used by other companies. The Company defines net debt as total debt less the total of cash, cash equivalents and current and long-term marketable securities. 3M believes net debt is meaningful to investors as 3M considers net debt and its components to be important indicators of liquidity and financial position. The following table provides net debt as of September 30, 2018 and December 31, 2017.

(Millions)	September 30, 2018	December 31, 2017	Change
Total debt	\$ 14,846	\$ 13,949	\$ 897
Less: Cash, cash equivalents and marketable securities	3,550	4,156	(606)
Net debt (non-GAAP measure)	\$ 11,296	\$ 9,793	\$ 1,503

Refer to the preceding "Total Debt" and "Cash, Cash Equivalents and Marketable Securities" sections for additional details.

Balance Sheet:

3M's strong balance sheet and liquidity provide the Company with significant flexibility to fund its numerous opportunities going forward. The Company will continue to invest in its operations to drive growth, including continual review of acquisition opportunities.

The Company uses working capital measures that place emphasis and focus on certain working capital assets, such as accounts receivable and inventory activity.

Working capital (non-GAAP measure):

(Millions)	September 30, 2018	December 31, 2017	Change
Current assets	\$ 14,419	\$ 14,277	\$ 142
Less: Current liabilities	7,336	7,687	(351)
Working capital (non-GAAP measure)	\$ 7,083	\$ 6,590	\$ 493

Various assets and liabilities, including cash and short-term debt, can fluctuate significantly from month to month depending on short-term liquidity needs. Working capital is not defined under U.S. generally accepted accounting principles and may not be computed the same as similarly titled measures used by other companies. The Company defines working capital as current assets minus current liabilities. 3M believes working capital is meaningful to investors as a measure of operational efficiency and short-term financial health.

Working capital increased \$493 million compared with December 31, 2017. Current asset balance changes increased working capital by \$142 million, driven by increases in accounts receivable and inventories (discussed further below), partially offset by decreases in cash and cash equivalents and marketable securities. Current liability balance changes increased working capital by \$351 million, primarily due to decreases in short-term debt.

Accounts receivable increased \$418 million from December 31, 2017, primarily due to increased sales. Foreign currency impacts decreased September 30, 2018 accounts receivable by \$156 million and acquisitions, net of divestitures, decreased accounts receivable by \$22 million. Inventory increased \$403 million from December 31, 2017, impacted by maintenance of additional inventory during

the deployment in the U.S. of the Company's ERP system. Foreign currency impacts decreased September 30, 2018 inventory by \$139 million and acquisitions, net of divestitures, decreased inventory by \$21 million.

Cash Flows:

Cash flows from operating, investing and financing activities are provided in the tables that follow. Individual amounts in the Consolidated Statement of Cash Flows exclude the effects of acquisitions, divestitures and exchange rate impacts on cash and cash equivalents, which are presented separately in the cash flows. Thus, the amounts presented in the following operating, investing and financing activities tables reflect changes in balances from period to period adjusted for these effects.

Cash Flows from Operating Activities:

(Millions)	Nine months ended September 30,	
	2018	2017
Net income including noncontrolling interest	\$ 4,014	\$ 4,344
Depreciation and amortization	1,117	1,195
Company pension and postretirement contributions	(303)	(314)
Company pension and postretirement expense	306	243
Stock-based compensation expense	258	266
Gain on sale of businesses	(530)	(490)
Income taxes (deferred and accrued income taxes)	49	144
Accounts receivable	(596)	(595)
Inventories	(562)	(436)
Accounts payable	148	(25)
Other — net	280	48
Net cash provided by operating activities	\$ 4,181	\$ 4,380

Cash flows from operating activities can fluctuate significantly from period to period, as pension funding decisions, tax timing differences and other items can significantly impact cash flows.

In the first nine months of 2018, cash flows provided by operating activities decreased \$199 million compared to the same period last year, with this decrease primarily due to the Minnesota NRD resolution. Additional factors that decreased operating cash flows were increases in inventory and accounts receivable. The combination of accounts receivable, inventories and accounts payable increased working capital by \$1.010 billion in the first nine months of 2018, compared to the working capital increases of \$1.056 billion in the first nine months of 2017. Additional discussion on working capital changes is provided earlier in the "Financial Condition and Liquidity" section.

Cash Flows from Investing Activities:

(Millions)	Nine months ended September 30,	
	2018	2017
Purchases of property, plant and equipment (PP&E)	\$ (1,046)	\$ (914)
Proceeds from sale of PP&E and other assets	143	18
Acquisitions, net of cash acquired	13	(12)
Purchases and proceeds from maturities and sale of marketable securities and investments, net	714	(310)
Proceeds from sale of businesses, net of cash sold	806	862
Other — net	8	2
Net cash provided by (used in) investing activities	\$ 638	\$ (354)

Investments in property, plant and equipment enable growth across many diverse markets, helping to meet product demand and increasing manufacturing efficiency. The Company expects 2018 capital spending to be approximately \$1.6 billion as 3M continues to invest in its businesses.

3M invests in renewal and maintenance programs, which pertain to cost reduction, cycle time, maintaining and renewing current capacity, eliminating pollution, and compliance. Costs related to maintenance, ordinary repairs, and certain other items are expensed. 3M also invests in growth, which adds to capacity, driven by new products, both through expansion of current facilities and new facilities. Finally, 3M also invests in other initiatives, such as information technology (IT) and laboratory facilities.

Refer to Note 3 for information on acquisitions and divestitures. The Company is actively considering additional acquisitions, investments and strategic alliances, and from time to time may also divest certain businesses. Proceeds from sale of businesses in the first nine months of 2018 primarily relate to the sale of substantially all of 3M's Communication Markets Division and the sale of certain personal safety product offerings primarily focused on noise, environmental, and heat stress monitoring within the Safety and Graphics business segment.

Purchases of marketable securities and investments and proceeds from maturities and sale of marketable securities and investments are primarily attributable to asset-backed securities, certificates of deposit/time deposits, commercial paper, and other securities, which are classified as available-for-sale. Refer to Note 9 for more details about 3M's diversified marketable securities portfolio. Purchases of investments include additional survivor benefit insurance, plus investments in equity securities.

Cash Flows from Financing Activities:

(Millions)	Nine months ended September 30,	
	2018	2017
Change in short-term debt — net	\$ (698)	\$ 60
Repayment of debt (maturities greater than 90 days)	(456)	(650)
Proceeds from debt (maturities greater than 90 days)	2,247	—
Total cash change in debt	\$ 1,093	\$ (590)
Purchases of treasury stock	(3,601)	(1,564)
Proceeds from issuances of treasury stock pursuant to stock option and benefit plans	401	582
Dividends paid to shareholders	(2,406)	(2,104)
Other — net	(36)	(23)
Net cash used in financing activities	\$ (4,549)	\$ (3,699)

Total debt was approximately \$14.8 billion at September 30, 2018 and \$13.9 billion at December 31, 2017. Increases in debt related to the third quarter 2018 issuance of \$2.25 billion of medium-term notes, which was partially offset by the \$450 million repayment of maturing medium-term notes, the net impact of repayments and borrowings of international subsidiaries along with foreign currency effects, and the repayment of commercial paper. Outstanding commercial paper was zero at September 30, 2018, as compared to \$745 million at December 31, 2017. Net commercial paper issuances and repayments and borrowings by international subsidiaries are largely reflected in "Change in short-term debt — net" in the preceding table. 3M's primary short-term liquidity needs are met through cash on hand and U.S. commercial paper issuances.

Repurchases of common stock are made to support the Company's stock-based employee compensation plans and for other corporate purposes. In February 2016, 3M's Board of Directors authorized the repurchase of up to \$10 billion of 3M's outstanding common stock, with no pre-established end date. The Company expects full-year 2018 gross share repurchases will be in the range of \$4.0 billion to \$5.0 billion. For more information, refer to the table titled "Issuer Purchases of Equity Securities" in Part II, Item 2. The Company does not utilize derivative instruments linked to the Company's stock.

3M has paid dividends each year since 1916. In January 2018, 3M's Board of Directors declared a first-quarter 2018 dividend of \$1.36 per share, an increase of 16 percent. This is equivalent to an annual dividend of \$5.44 per share and marked the 60th consecutive year of dividend increases for 3M. In May 2018, 3M's Board of Directors declared a third quarter 2018 dividend of \$1.36 per share. In August 2018, 3M's Board of Directors declared a third quarter 2018 dividend of \$1.36 per share.

Other cash flows from financing activities may include various other items, such as distributions to or sales of noncontrolling interests, changes in cash overdraft balances, and principal payments for capital leases.

Free Cash Flow (non-GAAP measure):

Free cash flow and free cash flow conversion are not defined under U.S. generally accepted accounting principles (GAAP). Therefore, they should not be considered a substitute for income or cash flow data prepared in accordance with U.S. GAAP and may not be comparable to similarly titled measures used by other companies. The Company defines free cash flow as net cash provided by operating activities less purchases of property, plant and equipment. It should not be inferred that the entire free cash flow amount is available for discretionary expenditures. The Company defines free cash flow conversion as free cash flow divided by net income attributable to 3M. The Company believes free cash flow and free cash flow conversion are meaningful to investors as they are useful measures of performance and the Company uses these measures as an indication of the strength of the company and its ability to generate cash. The first quarter of each year is typically 3M's seasonal low for free cash flow and free cash flow conversion. In the table below details the components of free cash flow for the nine months ended September 30, 2018 and 2017.

In the first nine months of 2018, free cash flow conversion was impacted by the Minnesota NRD resolution, the measurement period adjustment relative to the accounting for the 2017 enactment of the TCJA, and the divestiture of the Communication Markets Division, net of related restructuring actions. Refer to the preceding "Cash Flows from Operating Activities" section for discussion of additional items that impacted operating cash flow. Refer to the preceding "Cash Flows from Investing Activities" section for discussion on capital spending for property, plant and equipment.

(Millions)	Nine months ended September 30,	
	2018	2017
Major GAAP Cash Flow Categories		
Net cash provided by operating activities	\$ 4,181	\$ 4,380
Net cash provided by investing activities	638	(354)
Net cash used in financing activities	(4,549)	(3,699)
Free Cash Flow (non-GAAP measure)		
Net cash provided by operating activities	\$ 4,181	\$ 4,380
Purchases of property, plant and equipment (PP&E)	(1,046)	(914)
Free cash flow	\$ 3,135	\$ 3,466
Net income attributable to 3M	\$ 4,002	\$ 4,335
Free cash flow conversion	78 %	80 %

CAUTIONARY NOTE CONCERNING FACTORS THAT MAY AFFECT FUTURE RESULTS

This Quarterly Report on Form 10-Q, including “Management’s Discussion and Analysis of Financial Condition and Results of Operations” in Part I, Item 2, contains forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. The Company may also make forward-looking statements in other reports filed with the Securities and Exchange Commission, in materials delivered to shareholders and in press releases. In addition, the Company’s representatives may from time to time make oral forward-looking statements.

Forward-looking statements relate to future events and typically address the Company’s expected future business and financial performance. Words such as “plan,” “expect,” “aim,” “believe,” “project,” “target,” “anticipate,” “intend,” “estimate,” “will,” “should,” “could,” “forecast” and other words and terms of similar meaning, typically identify such forward-looking statements. In particular, these include, among others, statements relating to:

- the Company’s strategy for growth, future revenues, earnings, cash flow, uses of cash and other measures of financial performance, and market position,
- worldwide economic, political, and capital markets conditions, such as interest rates, foreign currency exchange rates, financial conditions of our suppliers and customers, and natural and other disasters or climate change affecting the operations of the Company or our suppliers and customers,
- new business opportunities, product development, and future performance or results of current or anticipated products,
- the scope, nature or impact of acquisition, strategic alliance and divestiture activities,
- the outcome of contingencies, such as legal and regulatory proceedings,
- future levels of indebtedness, common stock repurchases and capital spending,
- future availability of and access to credit markets,
- pension and postretirement obligation assumptions and future contributions,
- asset impairments,
- tax liabilities,
- information technology security, and
- the effects of changes in tax (including the Tax Cuts and Jobs Act), environmental and other laws and regulations in the United States and other countries in which we operate.

The Company assumes no obligation to update or revise any forward-looking statements.

Forward-looking statements are based on certain assumptions and expectations of future events and trends that are subject to risks and uncertainties. Actual future results and trends may differ materially from historical results or those reflected in any such forward-looking statements depending on a variety of factors. Important information as to these factors can be found in this document, including, among others, “Management’s Discussion and Analysis of Financial Condition and Results of Operations” under the headings of “Overview,” “Financial Condition and Liquidity” and annually in “Critical Accounting Estimates.” Discussion of these factors is incorporated by reference from Part I, Item 1A, “Risk Factors,” of this document, and should be considered an integral part of Part II, Item 7, “Management’s Discussion and Analysis of Financial Condition and Results of Operations.” For additional information concerning factors that may cause actual results to vary materially from those stated in the forward-looking statements, see our reports on Form 10-K, 10-Q and 8-K filed with the SEC from time to time.

Item 3. Quantitative and Qualitative Disclosures About Market Risk.

In the context of Item 3, 3M is exposed to market risk due to the risk of loss arising from adverse changes in foreign currency exchange rates, interest rates and commodity prices. Changes in those factors could impact the Company’s results of operations and financial condition. For a discussion of sensitivity analysis related to these types of market risks, refer to Part II, Item 7A, Quantitative and Qualitative Disclosures About Market Risk, in 3M’s Current Report on Form 8-K dated May 8, 2018 (which updated 3M’s 2017 Annual Report on Form 10-K). There have been no material changes in information that would have been provided in the context of Item 3 from the end of the preceding year until September 30, 2018. However, the Company does provide risk management discussion in various places in this Quarterly Report on Form 10-Q, primarily in the Derivatives note.

Item 4. Controls and Procedures.

a. The Company carried out an evaluation, under the supervision and with the participation of its management, including the Chief Executive Officer and Chief Financial Officer, of the effectiveness of the design and operation of the Company's "disclosure controls and procedures" (as defined in the Exchange Act Rule 13a-15(e)) as of the end of the period covered by this report. Based upon that evaluation, the Chief Executive Officer and Chief Financial Officer concluded that the Company's disclosure controls and procedures are effective.

b. There was no change in the Company's internal control over financial reporting that occurred during the Company's most recently completed fiscal quarter that has materially affected, or is reasonably likely to materially affect, the Company's internal control over financial reporting.

The Company is implementing an enterprise resource planning ("ERP") system on a worldwide basis, which is expected to improve the efficiency of certain financial and related transaction processes. The gradual implementation is expected to occur in phases over the next several years. The implementation of a worldwide ERP system will likely affect the processes that constitute our internal control over financial reporting and will require testing for effectiveness.

The Company completed implementation with respect to various processes/sub-processes in certain subsidiaries/locations, including aspects relative to the United States, and will continue to roll out the ERP system over the next several years. As with any new information technology application we implement, this application, along with the internal controls over financial reporting included in this process, was appropriately considered within the testing for effectiveness with respect to the implementation in these instances. We concluded, as part of its evaluation described in the above paragraphs, that the implementation of the ERP system in these circumstances has not materially affected our internal control over financial reporting.

3M COMPANY
FORM 10-Q
For the Quarterly Period Ended September 30, 2018
PART II. Other Information

Item 1. Legal Proceedings.

Discussion of legal matters is incorporated by reference from Part I, Item 1, Note 14, “Commitments and Contingencies” of this document, and should be considered an integral part of Part II, Item 1, “Legal Proceedings.”

Item 1A. Risk Factors.

Provided below is a cautionary discussion of what we believe to be the most important risk factors applicable to the Company. Discussion of these factors is incorporated by reference into and considered an integral part of Part I, Item 2, “Management’s Discussion and Analysis of Financial Conditions and Results of Operations.”

** Results are impacted by the effects of, and changes in, worldwide economic, political, and capital markets conditions.* The Company operates in more than 70 countries and derives approximately 60 percent of its revenues from outside the United States. The Company’s business is subject to global competition and geopolitical risks and may be adversely affected by factors in the United States and other countries that are beyond its control, such as slower economic growth, disruptions in financial markets, economic downturns in the form of either contained or widespread recessionary conditions, inflation, elevated unemployment levels, sluggish or uneven recovery, government actions impacting international trade agreements, imposing trade restrictions such as tariffs, and retaliatory counter measures, government deficit reduction and other austerity measures in specific countries or regions, or in the various industries in which the Company operates; social, political or labor conditions in specific countries or regions; natural and other disasters or climate change affecting the operations of the Company or its customers and suppliers; or adverse changes in the availability and cost of capital, interest rates, tax rates, tax laws, or exchange control, ability to expatriate earnings and other regulations in the jurisdictions in which the Company operates.

** Change in the Company’s credit ratings could increase cost of funding.* The Company’s credit ratings are important to 3M’s cost of capital. The major rating agencies routinely evaluate the Company’s credit profile and assign debt ratings to 3M. This evaluation is based on a number of factors, which include financial strength, business and financial risk, as well as transparency with rating agencies and timeliness of financial reporting. 3M currently has an AA- credit rating with a stable outlook from Standard & Poor’s and has an A1 credit rating with a stable outlook from Moody’s Investors Service. The Company’s credit ratings have served to lower 3M’s borrowing costs and facilitate access to a variety of lenders. The addition of further leverage to the Company’s capital structure could impact 3M’s credit ratings in the future. Failure to maintain strong investment grade ratings would adversely affect the Company’s cost of funding and could adversely affect liquidity and access to capital markets.

** The Company’s results are affected by competitive conditions and customer preferences.* Demand for the Company’s products, which impacts revenue and profit margins, is affected by (i) the development and timing of the introduction of competitive products; (ii) the Company’s response to downward pricing to stay competitive; (iii) changes in customer order patterns, such as changes in the levels of inventory maintained by customers and the timing of customer purchases which may be affected by announced price changes, changes in the Company’s incentive programs, or the customer’s ability to achieve incentive goals; and (iv) changes in customers’ preferences for our products, including the success of products offered by our competitors, and changes in customer designs for their products that can affect the demand for some of the Company’s products.

** Foreign currency exchange rates and fluctuations in those rates may affect the Company’s ability to realize projected growth rates in its sales and earnings.* Because the Company’s financial statements are denominated in U.S. dollars and approximately 60 percent of the Company’s revenues are derived from outside the United States, the Company’s results of operations and its ability to realize projected growth rates in sales and earnings could be adversely affected if the U.S. dollar strengthens significantly against foreign currencies.

** The Company’s growth objectives are largely dependent on the timing and market acceptance of its new product offerings, including its ability to continually renew its pipeline of new products and to bring those products to market.* This ability may be adversely affected by difficulties or delays in product development, such as the inability to identify viable new products, obtain adequate

intellectual property protection, or gain market acceptance of new products. There are no guarantees that new products will prove to be commercially successful.

** The Company's future results are subject to fluctuations in the costs and availability of purchased components, compounds, raw materials and energy, including oil and natural gas and their derivatives, due to shortages, increased demand, supply interruptions, currency exchange risks, natural disasters and other factors.* The Company depends on various components, compounds, raw materials, and energy (including oil and natural gas and their derivatives) supplied by others for the manufacturing of its products. It is possible that any of its supplier relationships could be interrupted due to natural and other disasters and other events, or be terminated in the future. Any sustained interruption in the Company's receipt of adequate supplies could have a material adverse effect on the Company. In addition, while the Company has a process to minimize volatility in component and material pricing, no assurance can be given that the Company will be able to successfully manage price fluctuations or that future price fluctuations or shortages will not have a material adverse effect on the Company.

** Acquisitions, strategic alliances, divestitures, and other unusual events resulting from portfolio management actions and other evolving business strategies, and possible organizational restructuring could affect future results.* The Company monitors its business portfolio and organizational structure and has made and may continue to make acquisitions, strategic alliances, divestitures and changes to its organizational structure. With respect to acquisitions, future results will be affected by the Company's ability to integrate acquired businesses quickly and obtain the anticipated synergies.

** The Company's future results may be affected if the Company generates fewer productivity improvements than estimated.* The Company utilizes various tools, such as Lean Six Sigma, and engages in ongoing global business transformation. Business transformation is defined as changes in processes and internal/external service delivery across 3M to move to more efficient business models to improve operational efficiency and productivity, while allowing 3M to serve customers with greater speed and efficiency. This is enabled by the ongoing multi-year phased implementation of an enterprise resource planning (ERP) system on a worldwide basis. There can be no assurance that all of the projected productivity improvements will be realized.

** The Company employs information technology systems to support its business, including ongoing phased implementation of an ERP system as part of business transformation on a worldwide basis over the next several years. Security breaches and other disruptions to the Company's information technology infrastructure could interfere with the Company's operations, compromise information belonging to the Company or its customers, suppliers, and employees, exposing the Company to liability which could adversely impact the Company's business and reputation.* In the ordinary course of business, the Company relies on centralized and local information technology networks and systems, some of which are managed by third parties, to process, transmit and store electronic information, and to manage or support a variety of businesses. Additionally, the Company collects and stores certain data, including proprietary business information, and may have access to confidential or personal information in certain of our businesses that is subject to privacy and security laws, regulations and customer-imposed controls. Despite our cybersecurity measures (including employee and third-party training, monitoring of networks and systems, patching, maintenance, and backup of systems and data), the Company's information technology networks and infrastructure may still be vulnerable to damage, disruptions or shutdowns due to attacks by hackers, breaches, employee error or malfeasance, power outages, computer viruses, ransomware, telecommunication or utility failures, systems failures, service or cloud provider breaches, natural disasters or other catastrophic events. It is possible for such vulnerabilities to remain undetected for an extended period, up to and including several years. While we have experienced, and expect to continue to experience, these types of vulnerabilities to the Company's information technology networks and infrastructure, none of them to date has had a material impact to the Company. There may be other challenges and risks as the Company upgrades and standardizes its ERP system on a worldwide basis. Any such events could result in legal claims or proceedings, liability or penalties under privacy laws, disruptions or shutdowns, and damage to the Company's reputation, which could adversely affect the Company's business. Although the Company maintains insurance coverage for various cybersecurity and business continuity risks, there can be no guarantee that all costs or losses incurred will be fully insured.

** The Company's defined benefit pension and postretirement plans are subject to financial market risks that could adversely impact our results.* The performance of financial markets and discount rates impact the Company's funding obligations under its defined benefit plans. Significant changes in market interest rates, decreases in the fair value of plan assets and investment losses on plan assets, and relevant legislative or regulatory changes relating to defined benefit plan funding may increase the Company's funding obligations and adversely impact its results of operations and cash flows.

* The Company's future results may be affected by various legal and regulatory proceedings and legal compliance risks, including those involving product liability, antitrust, intellectual property, environmental, the U.S. Foreign Corrupt Practices Act and other anti-bribery, anti-corruption, or other matters. The outcome of these legal proceedings may differ from the Company's expectations because the outcomes of litigation, including regulatory matters, are often difficult to reliably predict. Various factors or developments can lead the Company to change current estimates of liabilities and related insurance receivables where applicable, or make such estimates for matters previously not susceptible of reasonable estimates, such as a significant judicial ruling or judgment, a significant settlement, significant regulatory developments or changes in applicable law. A future adverse ruling, settlement or unfavorable development could result in future charges that could have a material adverse effect on the Company's results of operations or cash flows in any particular period. For a more detailed discussion of the legal proceedings involving the Company and the associated accounting estimates, see the discussion in Note 14 "Commitments and Contingencies" within the Notes to Consolidated Financial Statements.

Item 2. Unregistered Sales of Equity Securities and Use of Proceeds.

Issuer Purchases of Equity Securities

Repurchases of 3M common stock are made to support the Company's stock-based employee compensation plans and for other corporate purposes. In February 2016, 3M's Board of Directors authorized the repurchase of up to \$10 billion of 3M's outstanding common stock, with no pre-established end date.

Issuer Purchases of Equity Securities (registered pursuant to Section 12 of the Exchange Act)

Period	Total Number of Shares Purchased (1)	Average Price Paid per Share	Total Number of Shares Purchased as Part of Publicly Announced Plans or Programs (2)	Maximum Approximate Dollar Value of Shares that May Yet Be Purchased under the Plans or Programs (Millions)
January 1-31, 2018	714,575	\$ 245.98	714,138	\$ 4,894
February 1-28, 2018	1,420,634	\$ 233.78	1,420,599	\$ 4,562
March 1-31, 2018	1,791,496	\$ 228.82	1,791,496	\$ 4,152
Total January 1-March 31, 2018	3,926,705	\$ 233.74	3,926,233	\$ 4,152
April 1-30, 2018	2,135,968	\$ 213.63	2,135,968	\$ 3,696
May 1-31, 2018	3,283,170	\$ 201.64	3,282,339	\$ 3,034
June 1-30, 2018	2,358,619	\$ 200.31	2,358,619	\$ 2,562
Total April 1-June 30, 2018	7,777,757	\$ 204.53	7,776,926	\$ 2,562
July 1-31, 2018	1,851,663	\$ 201.17	1,851,663	\$ 2,189
August 1-31, 2018	1,813,661	\$ 205.37	1,813,661	\$ 1,817
September 1-30, 2018	1,476,649	\$ 211.62	1,476,649	\$ 1,504
Total July 1-September 30, 2018	5,141,973	\$ 205.65	5,141,973	\$ 1,504
Total January 1-September 30, 2018	16,846,435	\$ 211.68	16,845,132	\$ 1,504

- (1) The total number of shares purchased includes: (i) shares purchased under the Board's authorizations described above, and (ii) shares purchased in connection with the exercise of stock options.
- (2) The total number of shares purchased as part of publicly announced plans or programs includes shares purchased under the Board's authorizations described above.

Item 3. Defaults Upon Senior Securities. — No matters require disclosure.

Item 4. Mine Safety Disclosures. Pursuant to Section 1503 of the Dodd-Frank Wall Street Reform and Consumer Protection Act (the “Act”), the Company is required to disclose, in connection with the mines it operates, information concerning mine safety violations or other regulatory matters in its periodic reports filed with the SEC. The information concerning mine safety violations or other regulatory matters required by Section 1503(a) of the Act is included in Exhibit 95 to this quarterly report.

Item 5. Other Information. — No matters require disclosure.

Item 6. Exhibits.

- (12) [Calculation of ratio of earnings to fixed charges.](#)
- (15) [A letter from the Company’s independent registered public accounting firm regarding unaudited interim consolidated financial statements.](#)
- (31.1) [Certification of the Chief Executive Officer pursuant to Section 302 of the Sarbanes-Oxley Act of 2002, 18 U.S.C. Section 1350.](#)
- (31.2) [Certification of the Chief Financial Officer pursuant to Section 302 of the Sarbanes-Oxley Act of 2002, 18 U.S.C. Section 1350.](#)
- (32.1) [Certification of the Chief Executive Officer pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, 18 U.S.C. Section 1350.](#)
- (32.2) [Certification of the Chief Financial Officer pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, 18 U.S.C. Section 1350.](#)
- (95) [Mine Safety Disclosures.](#)
- (101.INS) XBRL Instance Document.
- (101.SCH) XBRL Taxonomy Extension Schema Document.
- (101.CAL) XBRL Taxonomy Extension Calculation Linkbase Document.
- (101.DEF) XBRL Taxonomy Extension Definition Linkbase Document.
- (101.LAB) XBRL Taxonomy Extension Label Linkbase Document.
- (101.PRE) XBRL Taxonomy Extension Presentation Linkbase Document.

SIGNATURES

Pursuant to the requirements of the Securities Exchange Act of 1934, the Registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

3M COMPANY
(Registrant)

Date: October 25, 2018

By /s/ Nicholas C. Gangestad
Nicholas C. Gangestad,

Senior Vice President and Chief Financial Officer
(Mr. Gangestad is the Principal Financial Officer and has
been duly authorized to sign on behalf of the Registrant.)